KLayout-PEX Documentation

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1 Introduction

1.1 Motivation

In Electronic Design and Automation (EDA) for Integrated Circuits (ICs), a schematic presents an abstraction in comparison to the layout that will eventually be taped-out and fabricated by the semiconductor foundry.

While in the schematic, a connection between device terminals is seen as an equipotential, the stacked geometries in a specific layout introduce parasitic effects, which can be thought of additional resistors, capacitors (and inductors), not modeled by and missing in the original schematic.

To be able to simulate these effects, a parasitic extraction tool (PEX) is used, to extract a netlist from the layout, which represents the original schematic (created from the layout active and passive elements) augmented with the additional parasitic devices.

1.2 Acknowledgements

This project is funded by the JKU/SAL IWS Lab, a collaboration of Johannes Kepler University and Silicon Austria Labs.

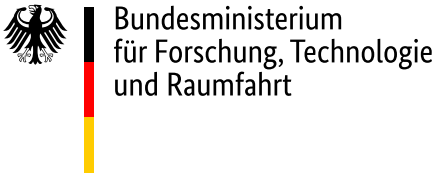
This project is further funded by the German project FMD-QNC (16ME0831).



(a) Johannes Kepler University: Institute for Integrated Circuits and Quantum Computing



(a) Silicon Austria Labs



(a) Bundesministerium für Forschung, Technologie und Raumfahrt

1.3 About KLayout-PEX

KLayout is an open source VLSI layout viewer and editor.

KLayout-PEX (short `KPEX`) is a PEX tool, well integrated with `KLayout` by using its API.

There are multiple PEX engines supported, currently:

- FasterCap integration (field solver engine)
- MAGIC integration (wrapper calling `magic`)
- Analytical 2.5D engine (parasitic concepts and formulas of MAGIC, implemented using KLayout methods)

Tip

KPEX *tool* source code itself is made publicly available on GitHub (follow this link) and shared under the GPL-3.0 license.
KPEX *documentation* source code is made publicly available on GitHub (follow this link) and shared under the Apache-2.0 license.
Please feel free to create issues and/or submit pull requests on GitHub to fix errors and omissions!
The production of the tool and this document would be impossible without these (and many more) great open-source software products: `KLayout`, `FasterCap`, `MAGIC`, `protobuf`, `Quarto`, `Python`, `ngspice`, `Numpy`, `Scipy`, `Matplotlib`, `Git`, `Docker`, `Ubuntu`, `Linux`…

1.4 Status

> **Caution**
>
> Currently, KPEX is developed as a Python prototype, using the KLayout Python API. This allows for a faster development cycle during the current prototyping phase.
> Eventually, critical parts will be re-implemented (in C++, and parallelized), to improve performance. As we're already using the KLayout API (which is pretty similar between Python, Ruby and C++), this will be relatively straight-forward.

> **Warning**
>
> Please keep in mind that this software is early stage, and not yet intended for production use.

Engine	PEX Type	Status	Description
KPEX/MAGIC	CC, RC	Usable	Wrapper engine, using installed magic tool
KPEX/FasterCap	CC	Usable, **pending QA**	Field solver engine using FasterCap
KPEX/FastHenry2	R, L	Planned	Field solver engine using FastHenry2
KPEX/2.5D	CC	Usable, **pending QA**	Prototype engine implementing MAGIC concepts/formulas with KLayout means
KPEX/2.5D	R	Usable, **pending QA**	Prototype engine implementing resistance extraction with KLayout means
KPEX/2.5D	RC,RCC	Planned	

1.5 Installation

Generally, KPEX is deployed using PyPi (Python Package Index), install via:

```
pip3 install --upgrade klayout-pex

kpex --version   # check the installed version
kpex --help      # this will help with command line arguments
```

As for the dependencies, there are multiple options available.

1.5.1 Option 1: Using IIC-OSIC-TOOLS Docker Image

We provide a comprehensive, low entry barrier Docker image that comes pre-installed with most relevant open source ASIC tools, as well as the open PDKs. This is a pre-compiled Docker image which allows to do circuit design on a virtual machine on virtually any type of computing equipment (personal PC, Raspberry Pi, cloud server) on various operating systems (Windows, macOS, Linux).

For further information please look at the Docker Hub page and for detailed instructions at the IIC-OSIC-TOOLS GitHub page.

> **Linux**
>
> In this document, we assume that users have a basic knowledge of Linux and how to operate it using the terminal (shell). If you are not yet familiar with Linux (which is basically a must when doing integrated circuit design as many tools are only available on Linux), then please check out a Linux introductory course or tutorial online, there are many resources available.
> A summary of important Linux shell commands is provided in IIC-JKU Linux Cheatsheet.

1.5.2 Option 2: Standalone Installation

- KLayout layout tool:
  - is mandatory for all engines (besides the MAGIC-wrapper)
  - get the latest pre-built package version
  - or follow the build instructions
- FasterCap engine:
  - optional, required to run the FasterCap engine
  - either compile your own version from the GitHub repository
  - or use precompiled versions available at https://github.com/martinjankoehler/FasterCap/releases
- MAGIC-wrapper engine:
  - optional, required to run the MAGIC-wrapper engine
  - `MAGIC 8.3.679` or newer is required: older revisions report twice the sidewall capacitance (see Section 5.4 and https://github.com/martinjankoehler/magic/issues/6#issuecomment-5371056429)
  - Follow the installation instructions at the GitHub repository
- Skywater sky130A PDK:
  - optional, for now, KPEX technology specific files are deployed within the `klayout-pex` Python package
  - `pip3 install --upgrade ciel` (install PDK package manager)
  - `ciel ls-remote --pdk sky130A` (retrieve available PDK releases
    - for example `PRE-RELEASE 0c1df35fd535299ea1ef74d1e9e15dedaeb34c32 (2024.12.11)`)
  - `ciel enable --pdk sky130A 0c1df35fd535299ea1ef74d1e9e15dedaeb34c32` (install a PDK version)
  - PDK files now have been installed under `$HOME/.volare/sky130A`
- IHP SG13G2 PDK:
  - optional, for now, KPEX technology specific files are deployed within the `klayout-pex` Python package
  - `pip3 install --upgrade ciel` (install PDK package manager)
  - `ciel ls-remote --pdk ihp-sg13g2` (retrieve available PDK releases
    - for example `PRE-RELEASE cb7daaa8901016cf7c5d272dfa322c41f024931f (2025.07.18)`)
  - `ciel enable --pdk ihp-sg13g2 cb7daaa8901016cf7c5d272dfa322c41f024931f` (install a PDK version)
  - PDK files now have been installed under `$HOME/.volare/ihp-sg13g2`

1.5.3 Useful tools: `meshlab`

For previewing generated 3D geometries, representing the input to `FasterCap`, we recommend installing MeshLab. The generated `STL`-files are located at `output/<design>/Geometries/*.stl`.

2 First Steps

- The command line tool `kpex` is used to trigger the parasitic extraction flow from the terminal.
- Get help calling `kpex --help`.

2.1 Example Layouts

Example layouts are included in the `testdata/designs` subdirectory of the KLayout-PEX source code:

```
git clone https://github.com/iic-jku/klayout-pex.git

# for sky130A
find testdata/designs/sky130A -name "*.gds.gz"

# for IHP SG13G2
find testdata/designs/ihp_sg13g2 -name "*.gds.gz"
```

2.2 Generating PEX25D

`kpex pex25d` generates a PEX25D geometry and material description without running a solver. It is separate from the analytical KPEX/2.5D engine. See Section 9.1 for the format specification, benefits, and standalone tools.

From the KLayout-PEX source directory:

```
kpex pex25d --pdk sky130A --blackbox y \
  --gds testdata/designs/sky130A/nmos_diode2/nmos_diode2.gds.gz \
  --output_file nmos_diode2.pex25d \
  --output_scene nmos_diode2.pex25d.scene.pb
```

The `.pex25d` file contains the text representation. The `.pex25d.scene.pb` file contains the resolved protobuf scene, including absolute layer heights and terminal intersections. Generation runs the reference validator by default; diagnostics determine the exit code even when output files have been written.

Use the standalone tool to inspect or validate an artifact:

```
pex25d show nmos_diode2.pex25d
pex25d validate nmos_diode2.pex25d --strict
```

For a complete, hand-written file that can be validated and exported without layout generation, use the downloadable example in Section 9.1.4. Terminal semantics and the current state of KLayout-generated terminals are explained in Section 9.1.5.7.

> **Current Sky130 stack validation**
>
> The current default Sky130 dielectric stack can report `PEX25D-E0260` for `nild6`/`capild` anchoring, including with the layout above. Generated files should not be treated as validated solver input when the command fails. See Section 9.1.6.1 for the diagnostic and a limited terminal-testing workaround.

2.3 Running the KPEX/FasterCap engine

Preconditions:

- `klayout-pex` was installed, see Section 1.5
- `FasterCap` was installed, see Section 1.5

> **Note**
>
> Normally, devices with SPICE (Nagel 1975) simulation models (e.g. like MOM-capacitors[1] in the sky130A PDK) are ignored ("blackboxed") during parasitic extraction.
> `kpex` has an option `--blacklist n` to allow extraction of those devices (whiteboxing), which can be useful during development (during the prototype phase, whiteboxing is actually the default setting, so please use `--blacklist y` to explicitly configure blackboxing).

Let's try the following:

```
kpex --pdk sky130A --blackbox n --gds \
  testdata/designs/sky130A/*/cap_vpp_04p4x04p6_l1m1m2_noshield.gds.gz
```

> **Note**
>
> This will report an error that we have not activated one or more engines, and list the available engines:
>
> | Argument | Description |
> |---|---|
> | `--fastercap` | Run kpex/FasterCap engine |
> | `--2.5D` | Run kpex/2.5D engine |
> | `--magic` | Run MAGIC engine |

Now, to run the FasterCap engine (might take a couple of minutes):

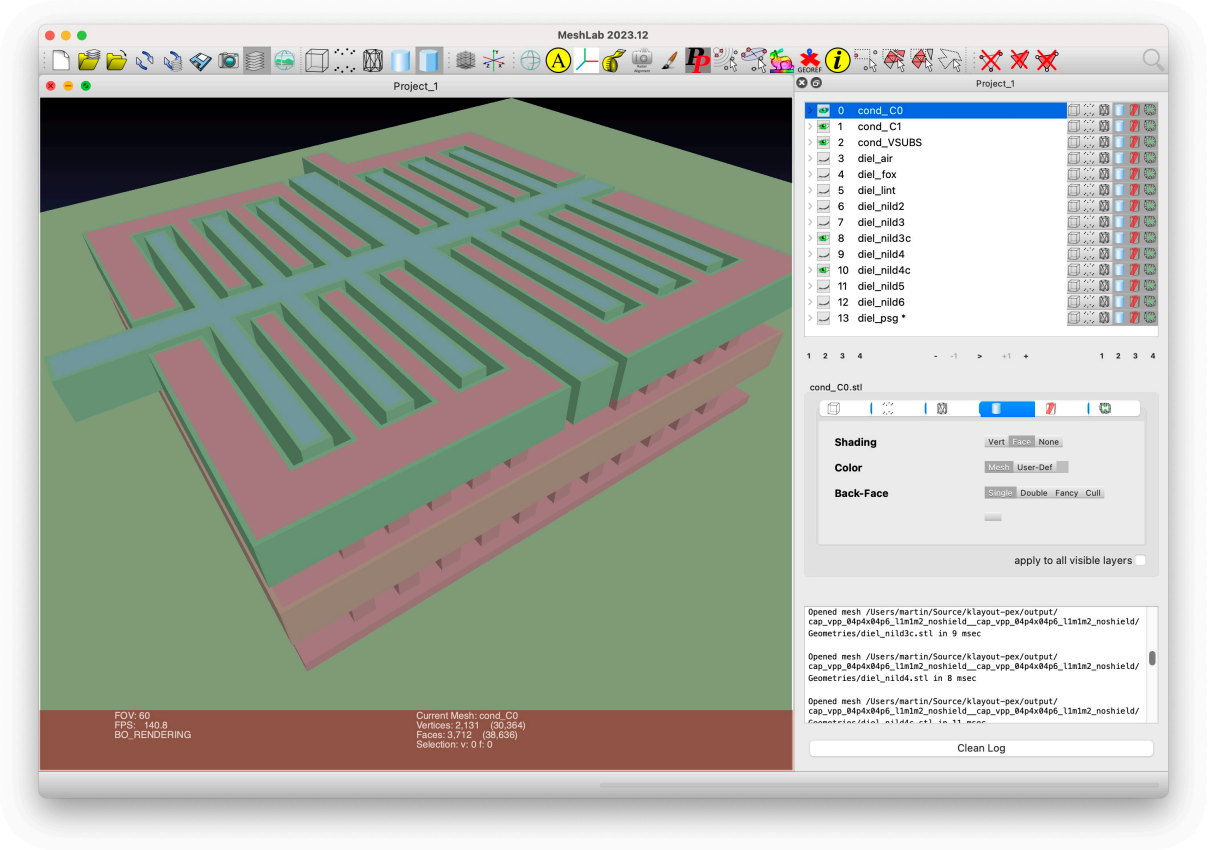
```
kpex --pdk sky130A --blackbox n --fastercap --gds \
  testdata/designs/sky130A/*/cap_vpp_04p4x04p6_l1m1m2_noshield.gds.gz
```

Within the output directory (defaults to `output`), `KPEX` creates a subdirectory `Geometries`, containing STL-files that provide a preview of the `FasterCap` input geometries. Use `MeshLab` (see Section 1.5.3) to open and preview those files:

```
ls -d output/cap_vpp_04*/Geometries/*.stl
```

> **Tip**
>
> - Open the `*.stl` files in `MeshLab`
> - Use the eye buttons to hide and show each file/mesh
> - Use the align tool ("A" in the toolbar) to assign different colors
> - Start by showing only on the conductors (files named `cond_*.stl`)
> - Then try showing different dielectrics (files named `diel_*.stl`), to see how they surround the conductors.

[1] Metal-Oxide-Metal capacitors



In the log file, we see the output of FasterCap including the Maxwell capacitance matrix:

```
Capacitance matrix is:
Dimension 3 x 3
g1_VSUBS  5.2959e-09 -4.46971e-10 -1.67304e-09
g2_C1  -5.56106e-10 1.5383e-08 -1.47213e-08
g3_C0  -1.69838e-09 -1.48846e-08 1.64502e-08
```

`KPEX` interprets this matrix and prints a CSV netlist, which can be pasted into a spreadsheet application:

```
Device;Net1;Net2;Capacitance [fF]
Cext_0_1;VSUBS;C1;0.5
Cext_0_2;VSUBS;C0;1.69
Cext_1_2;C1;C0;14.8
Cext_1_1;C1;VSUBS;0.08
```

In addition, a SPICE netlist is generated.

2.4 Running the `KPEX/MAGIC` engine

Preconditions:

- `klayout-pex` was installed, see Section 1.5
- `magic` was installed, see Section 1.5

The magic section of `kpex --help` describes the arguments and their defaults. Important arguments:

- `--magicrc`: specify location of the `magicrc` file
- `--gds`: path to the GDS input layout
- `--magic`: enable magic engine

```
kpex --pdk sky130A --magic --gds \
  testdata/designs/sky130A/*/cap_vpp_04p4x04p6_l1m1m2_noshield.gds.gz
```

3 Supporting new PDKs

For every supported PDK[2], a `KPEX` technology definition is required, as well as customized PEX-"LVS" scripts.

3.1 Customized PEX-"LVS" scripts

`KLayout` has built-in support for Layout-Versus-Schematic (LVS) scripts, based on its Ruby API. Customized "LVS" scripts are ("ab")used in `KPEX`, not with the intent of comparing Layout-Versus-Schematic, but rather to extract the connectivity/net information for all polygons across multiple layers. The resulting net information is stored in a `KLayout LVS Database` ("LVSDB") within the run directory.

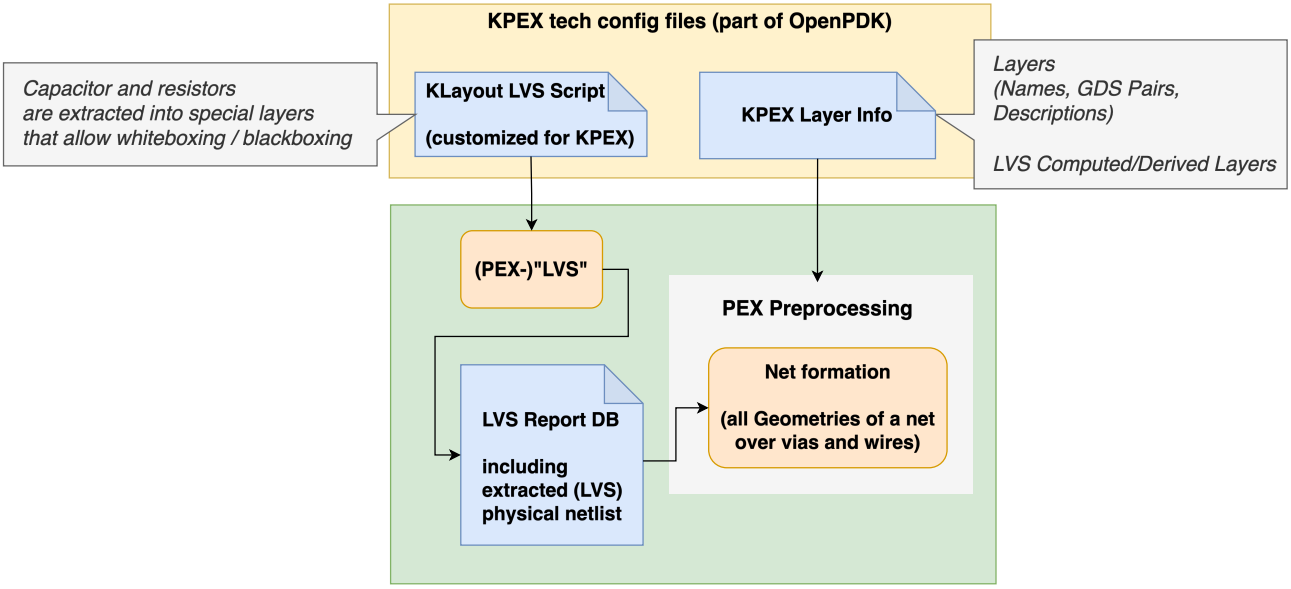


Figure 4: `KPEX` Net Formation

These customized "LVS" scripts are stored in:

- Skywater sky130A: `pdk/sky130A/libs.tech/kpex/sky130.lvs`
- IHP SG13G2: `pdk/ihp_sg13gs/libs.tech/kpex/sg13g2.lvs`

What's specific about this customization:

- Layers names must be assigned, using KLayout's `(name(layer, name))` function
- MOM[3] capacitors, MIM[4] capacitors and resistors should be extracted to separate layers, to enable blackboxing / whiteboxing.

The layer names in the script must correspond with the names configured in the tech JSON file.

[2]Process Design Kit
[3]Metal-Oxide-Metal capacitors
[4]Metal-Insulator-Metal capacitors

3.2 Technology Definition Files

The KPEX technology definition format uses Google Protocol Buffers, so there is:

- formal schema files, defining the structure and data types involved
  - protos/tech.proto: main schema / entry point, includes the others
  - protos/process_stack.proto: describes details of the process stack, such as dielectrics and heights of layers
  - protos/process_parasitics.proto: parasitic tables, used to parametrize the 2.5D engine
- multiple concrete instantiations, that adhere to this schema (called *messages* in the protobuf lingo)
  - in the form of JSON files
  - Skywater 130A: klayout_pex_protobuf/sky130A_tech.pb.json
  - IHP SG13G2: klayout_pex_protobuf/ihp_sg13g2_tech.pb.json

3.2.1 JSON tech files for supported PDKs

> **Note**
>
> The built-in JSON tech files are programmatically generated during the build process[5]. Therefore they not part of the repository source code, but of course part of the deployed Python wheels. To review those, look into your Python site-packages[6]/klayout_pex_protobuf.

3.2.2 Process Stackup Definition

For 3D solvers, like the KPEX/FasterCap engine, 3D information about the dimensions (e.g. z-offset and thickness) of metal layers is required, as well as the dielectrics in-between.

The Skywater sky130A process includes more intricate types of dielectrics (e.g., compared to IHP sg13g2), therefore I'll use this PDK as an example here.

We can derive the following data from Figure 5:

- Metal layers = {poly, li, metal1, metal2, metal3, metal4, metal5, capm, cap2m}
  - thickness
  - z-offset (relative to substrate)
- Contact layers = {licon}
- Via layers = {mcon, via1, via2, via3, via4}
- Dielectrics
  - dielectric constant k, the relative permittivity of the dielectric material
  - Different types:
    - Sidewall dielectrics = {IOX, SPNIT, NILD3_C, NILD4_C}
      - width of the sidewall around metals
      - height above metal
    - Conformal dielectrics = {LINT, TOPNIT, capild}
      - thickness above metal
      - thickness where no metal

[6] C++ generator scripts the built-in tech files are located in cxx/gen_tech_pb/pdk/*.cpp.
[6] To find the site-packages directory for the klayout-pex package, call pip3 show klayout-pex.

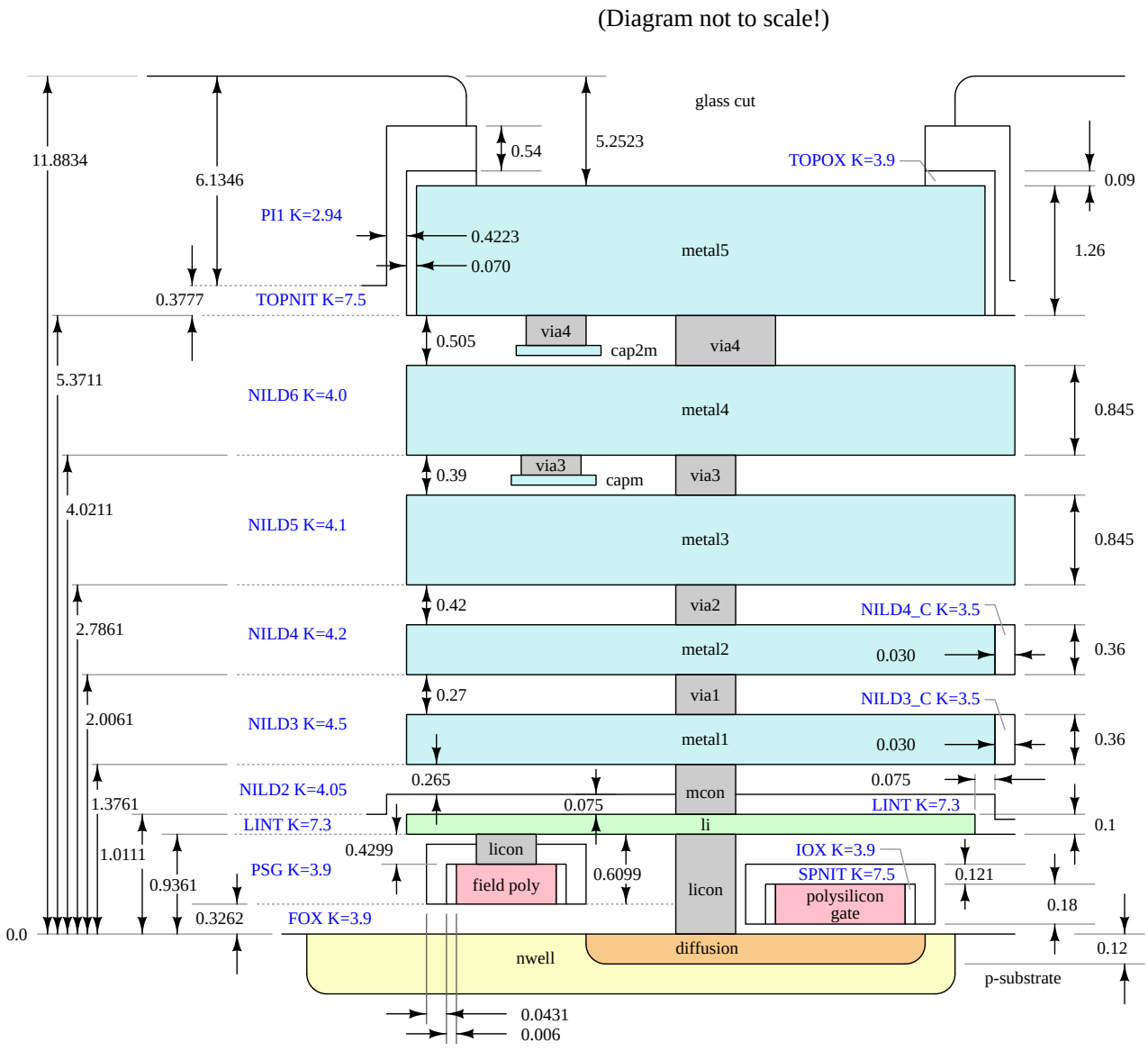


Figure 5: Skywater `sky130A` Process Stackup (Source: (Authors 2020))

        - thickness of the sidewall
    - Simple dielectrics = {PSG, NILD2, NILD3, NILD4, NILD5, NILD6}
        - embracing everything between 2 layers, including the sidewall and conformal dielectrics

An except of the `process_stack.proto` schema is shown in the code listing below, note:

- Line 3: `ProcessStackInfo.LayerType` is an enumeration type for the possible types of layers in the process stack
- Line 58: `ProcessStackInfo.layers` is a list of `LayerInfo`
  - the order of this list defines the orders of layers
- Line 41: `MetalLayer` describes a metal layer
- Line 44: `MetalLayer.contact_above` points to the via connecting to the metal layer above (omitted for the top metal layer)
- Line 31, 38: `ConformalDielectricLayer.reference` and `SidewallDielectricLayer.reference`
  - these refer to a dielectric or metal layer that they wrap around
  - `ConformalDielectricLayer` and `ConformalDielectricLayer` can be wrapped, e.g. `SPNIT` wraps `IOX`, which wraps `poly`

```
1  message ProcessStackInfo {
2
3      enum LayerType { …
4          LAYER_TYPE_SIMPLE_DIELECTRIC = 50;
5          LAYER_TYPE_CONFORMAL_DIELECTRIC = 60;
6          LAYER_TYPE_SIDEWALL_DIELECTRIC = 70;
7          LAYER_TYPE_METAL = 80;
8      }
9
10     message Contact { // Contact/Via
11         string name = 1;
12         string layer_below = 9;
13         string metal_above = 10;
14         double thickness = 20;
15
16         double width = 30;
17         double spacing = 31;
18         double border = 32;
19     }
20     …
21     message SimpleDielectricLayer { // Simple dielectric
22         double dielectric_k = 10;
23         string reference = 30;
24     }
25
26     message ConformalDielectricLayer { // Conformal dielectric
27         double dielectric_k = 10;
28         double thickness_over_metal = 20;
29         double thickness_where_no_metal = 21;
30         double thickness_sidewall = 22;
31         string reference = 30;
32     }
33
```

```
    message SidewallDielectricLayer { // Sidewall dielectric
        double dielectric_k = 10;
        double height_above_metal = 20; // might be 0 if none
        double width_outside_sidewall = 21;
        string reference = 30;
    }

    message MetalLayer { // Metal
        double z = 1;          // z-offset in µm (of layer bottom), above substrate
        double thickness = 2;  // thickness in µm
        Contact contact_above = 40;
    }

    message LayerInfo {
        string name = 1;
        LayerType layer_type = 2;
        oneof parameters { …
            SimpleDielectricLayer simple_dielectric_layer = 12;
            ConformalDielectricLayer conformal_dielectric_layer = 13;
            SidewallDielectricLayer sidewall_dielectric_layer = 14;
            MetalLayer metal_layer = 15;
        }
    }

    repeated LayerInfo layers = 100;
}
```

4 KPEX/FasterCap Engine

FasterCap is a 3D and 2D parallel capacitance field solver, inspired by FastCap2. https://www.fastfieldsolvers.com/fastercap.htm

Starting from an input layout (e.g. GDS file) and a process stack-up (part of the Section 3.2), KPEX creates input geometries for FasterCap. After running FasterCap, the Maxwell capacitance matrix is parsed and interpreted to obtain the parasitic capacitances.

See Section 2.3 to get started with a first extraction example.

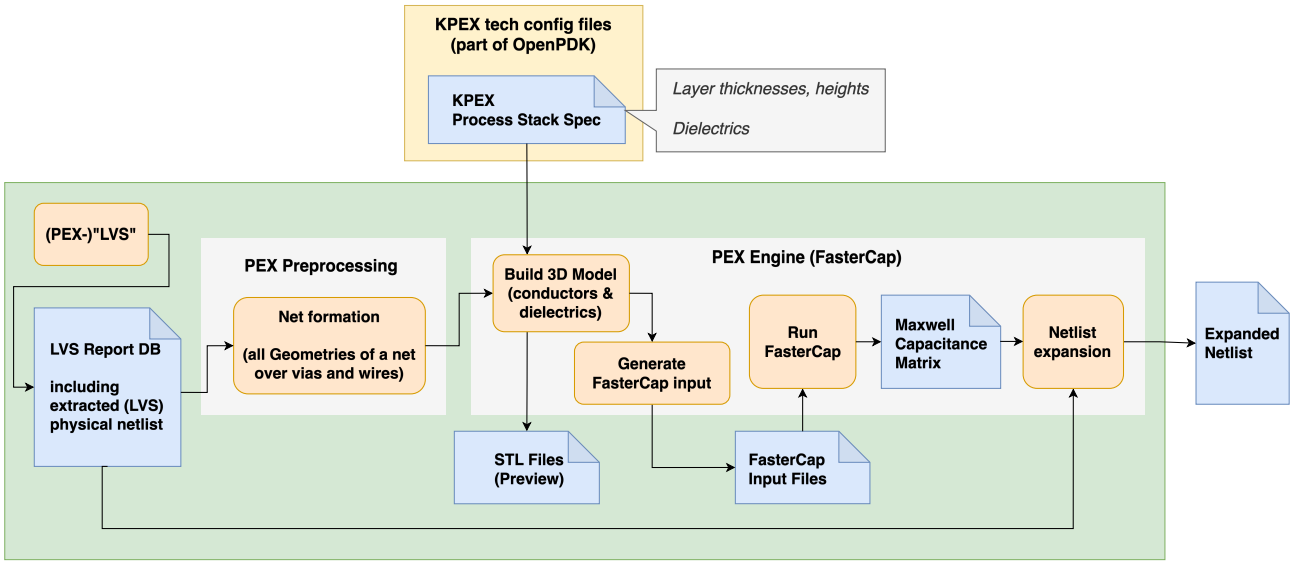


Figure 6: KPEX/FasterCap Engine

4.1 3D Input Geometries

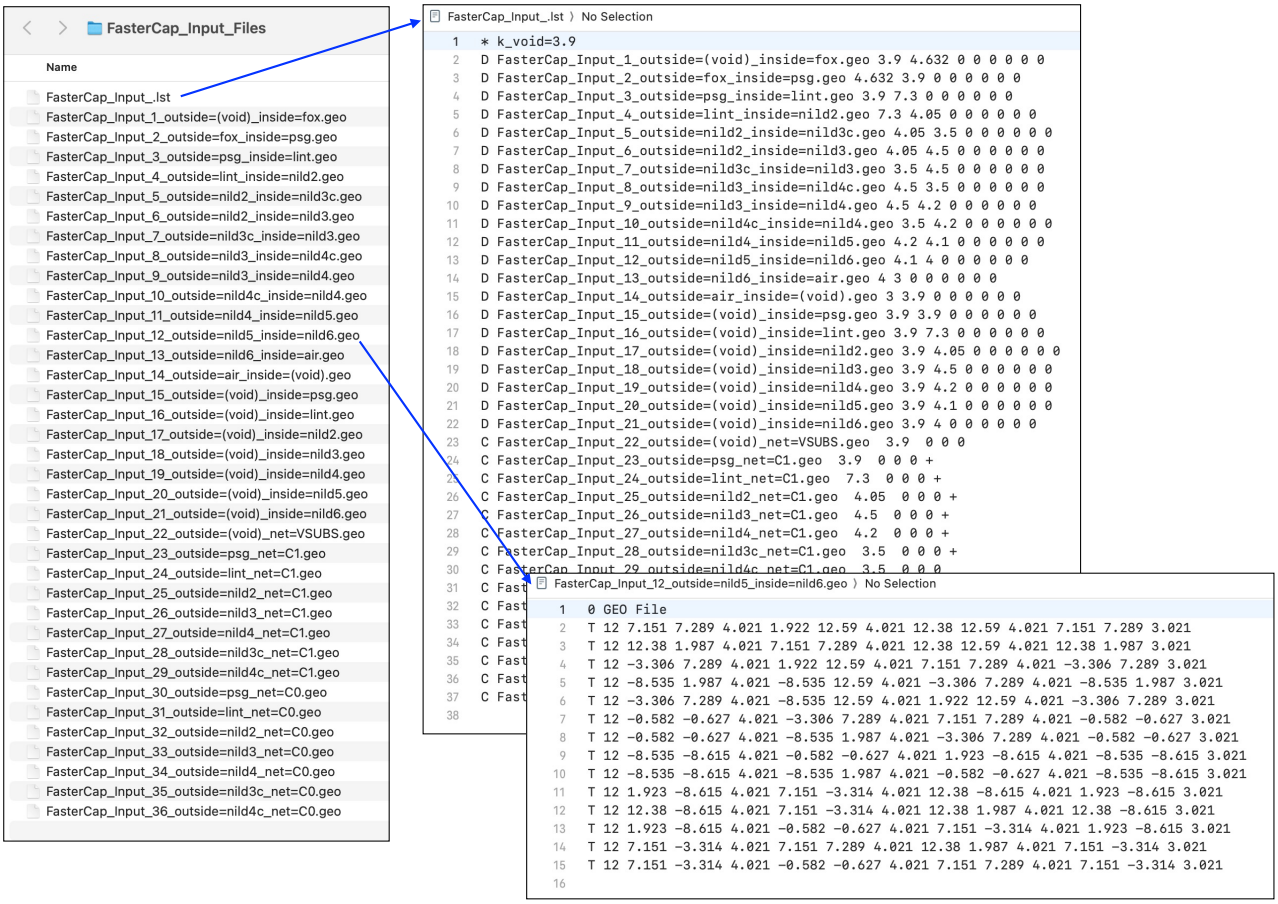


Figure 7: `FasterCap` 3D Input: File System Overview

The `FasterCap` input files and their format is documented in (Di Lorenzo 2019), a PDF version of the Windows-specific `*.chm` file is available at https://github.com/martinjankoehler/FasterCap/tree/master/doc/pdf.

`KPEX` generates 3D input geometries:

- `*.lst` file: Main input file
  - defines dielectric instances
  - defines conductor instances
  - each instance refers to a `*.geo` file
- `*.geo` files: Defines single geometry
  - defines shapes (e.g. triangles)
  - Each shape has a reference point to define inside/outsides

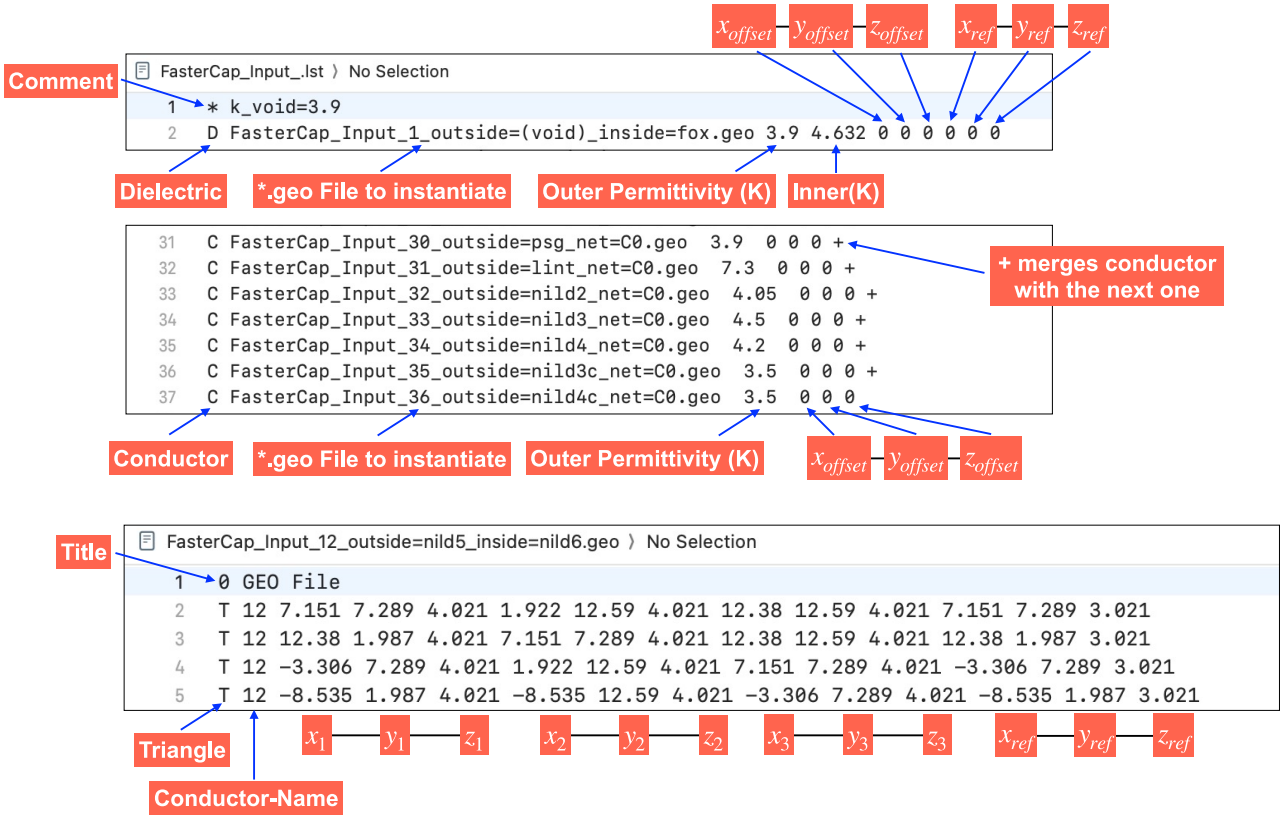


Figure 8: `FasterCap` 3D Input: File Format

4.2 Example: MOM Capacitor

Figure 9 depicts the MOM capacitor example of a from Section 2.3).

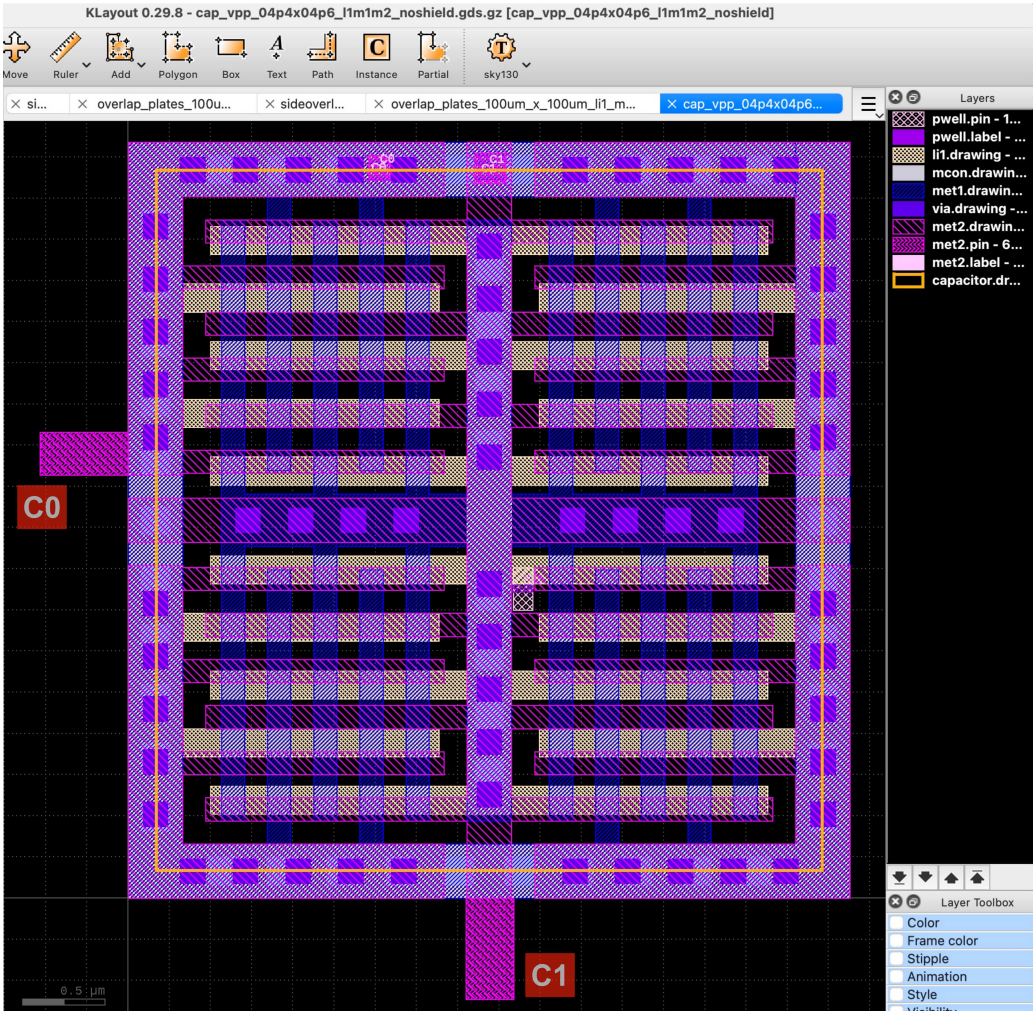
The corresponding schematic representation of Figure 10 contains 3 conductors (N_1, N_2 and N_3), and coupling capacitances:

- Capacitances between conductors: C_{ij} where $i \neq j$
  - C_{23} is the capacitance "intended" by the MOM designer
- Capacitances between conductors and ground: C_{ii}

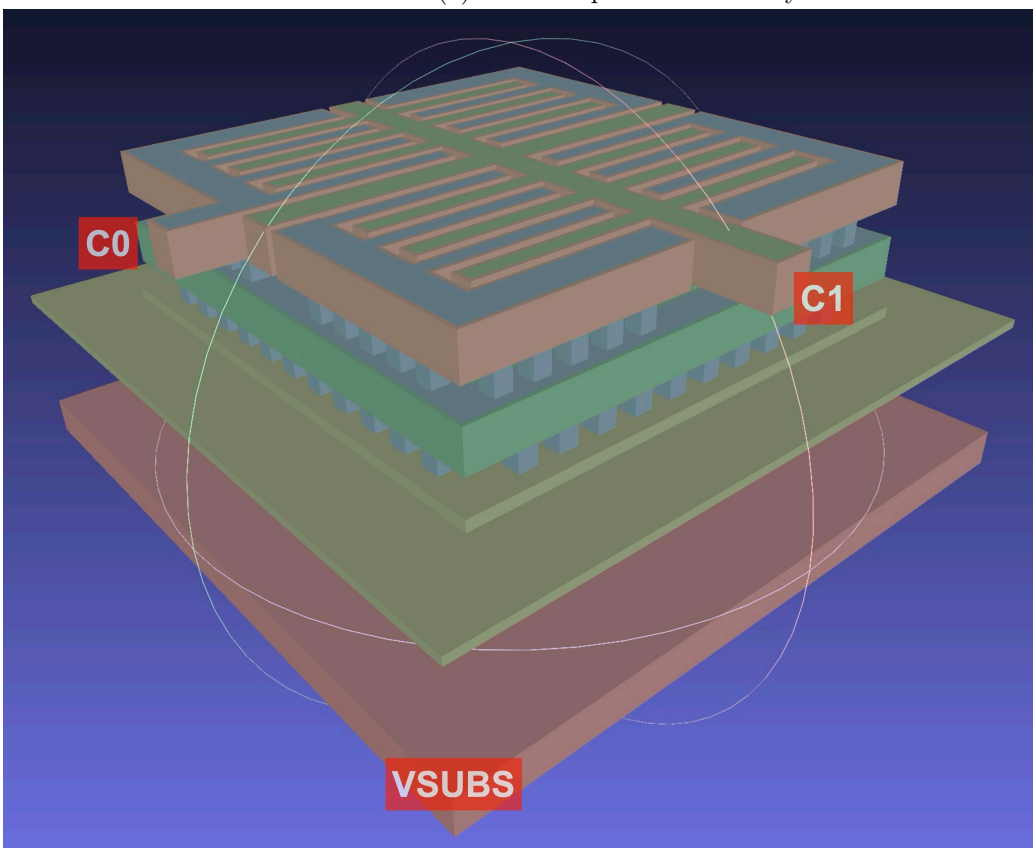
4.3 Output Maxwell Capacitance Matrix

A Maxwell capacitance matrix (Maxwell 1873) provides the relation between voltages on a set of conductors and the charges on these conductors, as described by the `FasterCap` author in the white paper (Di Lorenzo 2023).

`FasterCap` log output prints the Maxwell capacitance matrix (one for each iteration/refinement).



(a) MOM Capacitor: GDS Layout



(b) MOM Capacitor: `MeshLab` 3D Preview

Figure 9: MOM Capacitor

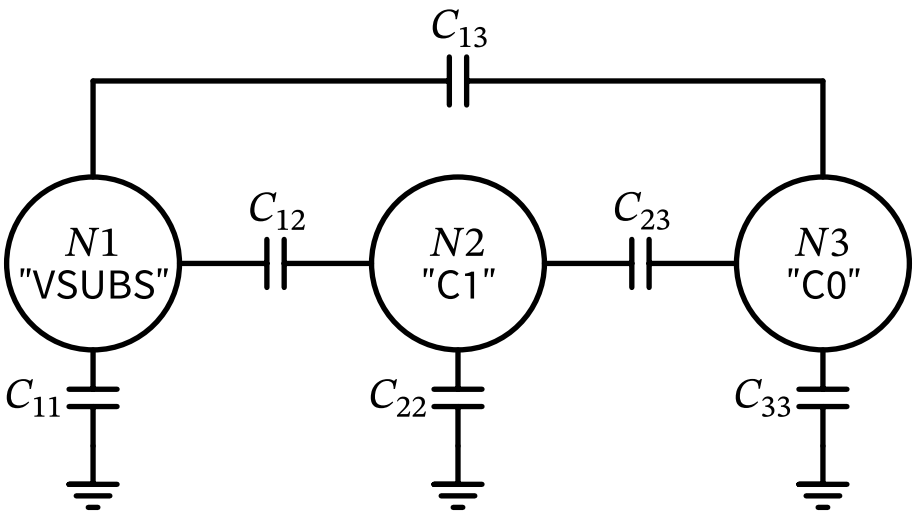


Figure 10: Schematic representation of the MOM capacitor.

```
Capacitance matrix is:
Dimension 3 x 3
g1_VSUBS  2.08888e-09 -1.46568e-10 -7.37007e-10
g2_C1  -1.64457e-10 1.47687e-08 -1.44368e-08
g3_C0  -7.68811e-10 -1.4528e-08 1.54806e-08

Weighted Frobenius norm of the difference between capacitance (auto option): 0.005031

Solve statistics:
Number of input panels: 18819 of which 5856 conductors and 12963 dielectric
Number of input panels to solver engine: 18819
Number of panels after refinement: 47371
Number of potential estimates: 6249961
Number of links: 19880193 (uncompressed 2244011641, compression ratio is 99.1%)
Max recursion level: 35
Max Mesh relative refinement value: 0.00126534
Time for reading input file: 0.014353s
Time for building super hierarchy: 0.001892s
Time for discretization: 0.562416s
Time for building potential matrix: 0.673007s
Time for precond calculation: 0.251937s
Time for gmres solving: 20.033398s
```

Figure 11: `FasterCap` Log Output: Maxwell Capacitance Matrix

- Matrix Properties:
  - Scaling: units have to be divided by 10^{-6}
  - rows and columns are the same (list of net names)
  - Row Cells:
    - off diagonals cells contains the coupling between row/col nets (times -1)
    - diagonal cells contains the sum of the absolute values of all other cells in the row
  - Matrix Symmetry:
    - in theory (ideal world), the matrix would be symmetric
    - in practice it's not

    - therefore `FastCap2` did average the off-diagonals
    - `FasterCap` does not average, so it's done as part of `KPEX`

$$C_{3\times3} = \begin{matrix} \\ \text{VSUBS} \\ \text{C0} \\ \text{C1} \end{matrix} \begin{matrix} \begin{matrix} \text{VSUBS} & \text{C0} & \text{C1} \end{matrix} \\ \begin{bmatrix} c_{11}+c_{12}+c_{13} & -c_{12} & -c_{13} \\ -c_{21} & c_{22}+c_{22}+c_{23} & -c_{23} \\ -c_{31} & -c_{32} & c_{31}+c_{32}+c_{33} \end{bmatrix} \end{matrix}$$

$$C_{avg} = \begin{bmatrix} c_{11} - |\text{avg}(c_{12}, c_{21})| - |\text{avg}(c_{13}, c_{31})| & |\text{avg}(c_{12}, c_{21})| & |\text{avg}(c_{13}, c_{31})| \\ |\text{avg}(c_{12}, c_{21})| & c_{22} - |\text{avg}(c_{12}, c_{21})| - |\text{avg}(c_{23}, c_{32})| & |\text{avg}(c_{23}, c_{32})| \\ |\text{avg}(c_{13}, c_{31})| & |\text{avg}(c_{23}, c_{32})| & c_{33} - |\text{avg}(c_{13}, c_{31})| - |\text{avg}(c_{23}, c_{32})| \end{bmatrix}$$

$$C_{femtofarad} = \begin{bmatrix} 2.09 - \text{avg}(0.147, 0.164) - \text{avg}(0.74, 0.77) & \text{avg}(0.147, 0.164) & \text{avg}(0.74, 0.77) \\ \text{avg}(0.147, 0.164) & 14.77 - \text{avg}(0.147, 0.164) - \text{avg}(14.44, 14.52) & \text{avg}(14.44, 14.52) \\ \text{avg}(0.74, 0.77) & \text{avg}(14.44\ 14.52) & 15.48 - \text{avg}(0.74, 0.77) - \text{avg}(14.44\ 14.52) \end{bmatrix}$$

$$= \begin{bmatrix} 1.18 & 0.16 & 0.75 \\ 0.16 & 0.13 & 14.48 \\ 0.75 & 14.48 & 0.25 \end{bmatrix}$$

Result CSV

Device	Net1	Net2	Capacitance [F]	Capacitance [fF]
Cext_0_1	VSUBS	C1	1.555125e-16	0.16f
Cext_0_2	VSUBS	C0	7.52909e-16	0.75f
Cext_1_2	C1	C0	1.44824e-14	14.48f
Cext_1_1	C1	VSUBS	1.307875e-16	0.13f
Cext_2_2	C0	VSUBS	2.45291e-16	0.25f

Figure 12: `FasterCap` Maxwell Capacitance Matrix: Interpretation

5 KPEX/MAGIC Engine

This engine is merely just a wrapper around `magic`, which prepares a TCL script that opens the layout file and starts MAGIC's PEX flow. See Section 2.4 to get started with a first extraction example.

The following chapter will illustrate concepts of the parasitic extraction done in `MAGIC`, also motivated by the fact that the engine in Section 6 will be based on those. Major parts of this illustration, the figures and concepts are based on work done by `MAGIC` maintainer Tim Edwards, especially his talk from FSiC conference 2022 (Edwards 2022) and a (conceptual follow-up) ChipsAlliance meeting on April 4, 2023, see (Edwards 2023b) and (Edwards 2023a). In addition, code review and debugging of the MAGIC codebase was performed.

5.1 MAGIC database units

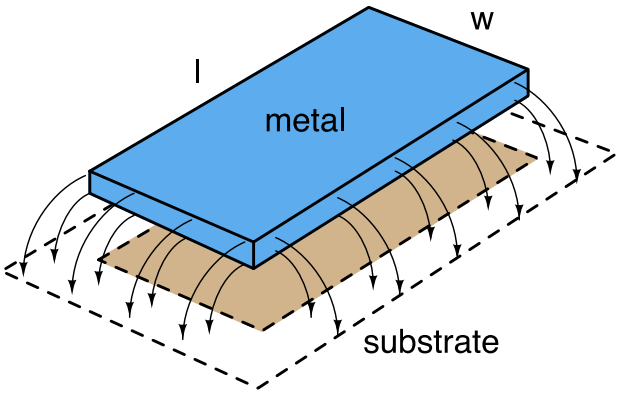
- To convert between MAGIC database units and μm, a scaling factor α is used, so that $L_{\mu m} = \frac{L_{dbu}}{\alpha}$
- E.g. for `sky130A`, $\alpha = 200.0$

5.2 Types of Parasitic Capacitances

MAGIC models multiple types of capacitances:

- *Substrate Overlap:* Overlap area of a metal with the substrate
- *Substrate Fringing:* Sidewall of a metal fringes out to substrate
- *Sidewall Capacitance:* Coupling between adjacent sidewalls on the same layer
- *Overlap Capacitance:* Overlap on different metal layers
- *Fringe Capacitance ("Side Overlap"):* Sidewall of a metal fringes out other metal layers

5.3 Substrate Capacitance



- Overlapping area:

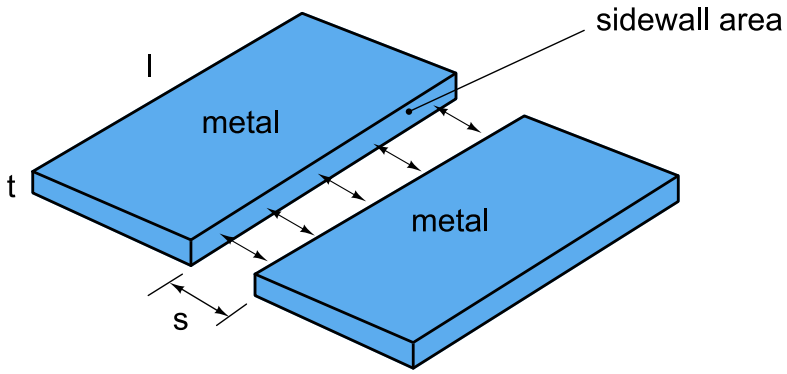
$$C_{area} = \frac{\epsilon_{si} * K}{d} * \text{area} \quad \left[\frac{F}{\mu m^2} * \mu m^2\right]$$

- Fringe ("Perimeter"):

$$C_{fringe\,to\,substrate} = \text{perimeter} * C_{perim} = (2l + 2w) * C_{perim}$$

- Coefficients like C_{perim} are part of the tech files (Parasitic Tables)

5.4 Sidewall Capacitance



$$C_{sidewall} = \frac{\epsilon_{si} * K}{s} * \text{sidewall area} \quad \left[\frac{F}{\mu m^2} * \mu m^2\right]$$
$$= \frac{\epsilon_{si} * K}{s} * t * l \quad \left[\frac{F}{\mu m^2} * \mu m * \mu m\right]$$
$$C_{sidewall} = \frac{C_{sidewall\,coeff}}{s} * l \quad \left[\frac{F}{\mu m} * \mu m\right]$$

The coefficient describes the coupling between the two facing sidewalls, but `MAGIC` walks every edge, so each of the two edges contributes half of it:

$$C_{sidewall\,per\,edge} = \frac{1}{2} * \frac{C_{sidewall\,coeff}}{s} * l$$

- Coefficients are part of the tech files (Parasitic Tables)
- Layer thickness t is normally multiplied into the coefficient
- Foundry tables give constant coefficient referenced to $s = 1$

> ⚠ `defaultsidewall` counts once per pair, since MAGIC 8.3.679
>
> Up to `MAGIC 8.3.678`, each edge was evaluated against the *full* `defaultsidewall` value of the tech file, so the sidewall capacitance came out twice as large as intended. Since those values had been distributed with the open PDKs for years, `MAGIC 8.3.679` redefined the meaning of the tech file value instead of changing every PDK — the PDKs stay as they are, and `MAGIC` applies half of the value per edge.
> `KPEX/2.5D` follows that definition, so `MAGIC 8.3.679` or newer is required for the two to agree.
> See https://github.com/martinjankoehler/magic/issues/6#issuecomment-5371056429.

5.5 Overlap Capacitance

- Overlapping area:

$$C_{area} = \frac{\epsilon_{si} * K}{d} * \text{area} \quad \left[\frac{F}{\mu m^2} * \mu m^2\right]$$

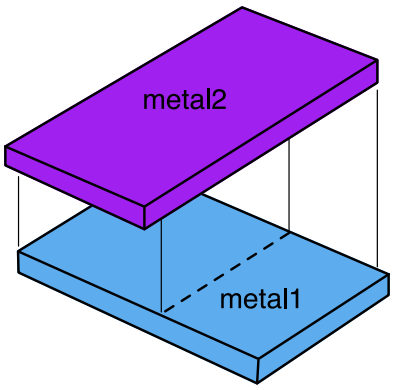


Figure 13: Overlap Capacitance

5.6 Fringe Capacitance

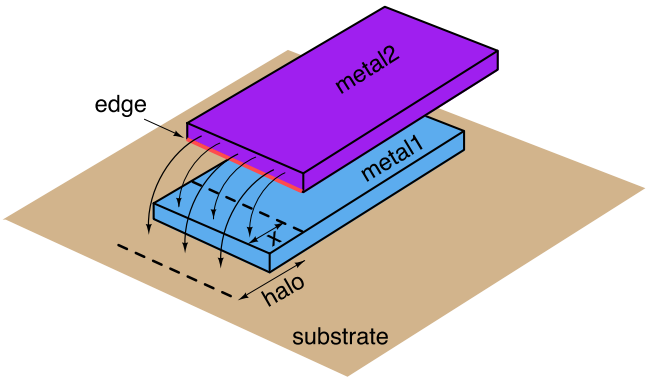


Figure 14: Fringe Capacitance: Overlapping (1)

- Causing sidewall (its bottom edge depicted red)
- Assume: Field is bounded by fringe halo (e.g. 8 μm away from edge)
- Fractions of fringe goes to metal1 or substrate

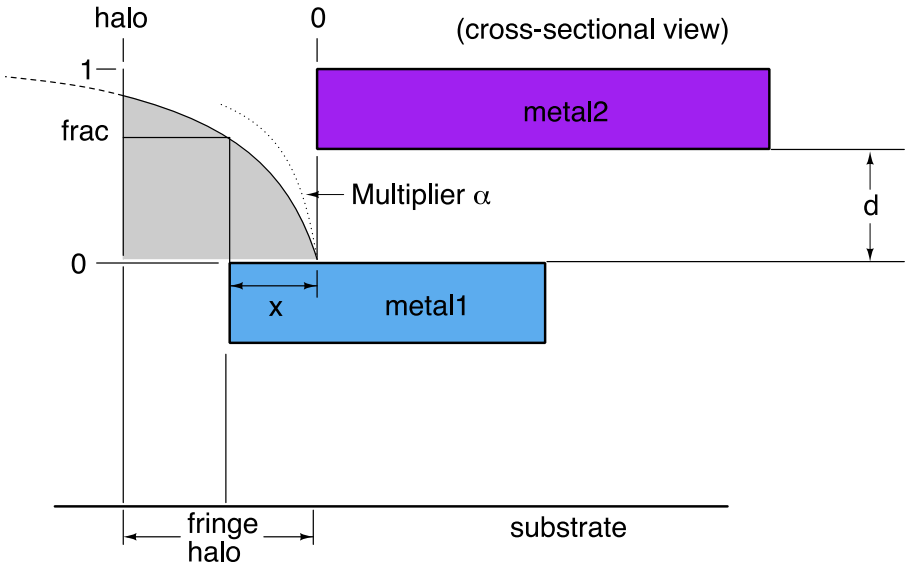


Figure 15: Fringe Capacitance: Overlapping (2)

- Multiplier α (comes from tech files: overlap table)
  - determines how quickly fringe capacitance drops with increasing distance

  - α is related to distance d between layers
  - α is proportional to $C_{overlap_{metal1 \leftrightarrow metal2}} = \frac{\epsilon_{si} * K}{d} * \text{area}$ (for fixed value of area $1\,\mu m^2$)

- Fringe Fractions:
  - $\text{frac}_{metal1} = \frac{2}{\pi} * \text{atan}(\alpha_{metal2 \rightarrow metal1} * x)$
  - $\text{frac}_{sub} = \frac{2}{\pi} * \text{atan}(\alpha_{metal2 \rightarrow sub} * (\text{halo} - x))$
  - $\frac{2}{\pi}$ is multiplied because of scaling to interval [0.0, 1.0], as $\text{atan}(\infty) = \frac{\pi}{2}$

- Overlap capacitance:
  - $C_{overlap} = \frac{\epsilon_{si} * K}{d} * \text{area} \quad (\text{with area} = 1 \mu m^2)$

- Coupling capacitance $metal1 \leftrightarrow metal2$:
  - $\alpha_{metal1 \leftrightarrow metal2} = \alpha_{scalefac} * C_{overlap_{metal1 \leftrightarrow metal2}}$
  - $\text{frac}_{metal1} = \frac{2}{\pi} * \text{atan}(\alpha_{metal1 \leftrightarrow metal2} * x)$
  - effective length = edge length $*$ frac_{metal1}
  - $C_{fringe_{metal2 \rightarrow metal1}}$ = effective length $* C_{sideoverlap_{metal2 \rightarrow metal1}}$

- Coupling capacitance $metal1 \leftrightarrow sub$:
  - $\alpha_{metal1 \leftrightarrow sub} = \alpha_{scalefac} * C_{overlap_{metal1 \leftrightarrow sub}}$
  - $\text{frac}_{sub} = \frac{2}{\pi} * \text{atan}(\alpha_{metal1 \leftrightarrow sub} * (\text{halo} - x))$
  - effective length = edge length $*$ frac_{sub}
  - $C_{fringe_{metal2 \rightarrow sub}}$ = effective length $* C_{sideoverlap_{metal2 \rightarrow sub}}$

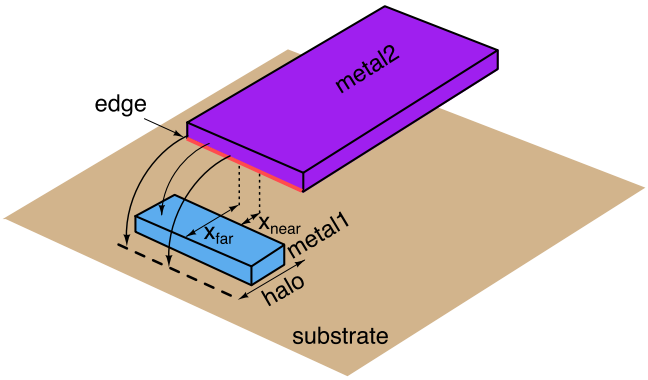


Figure 16: Fringe Capacitance: Non Overlapping (1)

- Partial side overlap
  - In case there is only a partial side overlap, the non-existing near fraction is subtracted from the far fraction
  - $metal1$ wire is offset, starts at x_{near}
  - $metal1$ ends at x_{far}

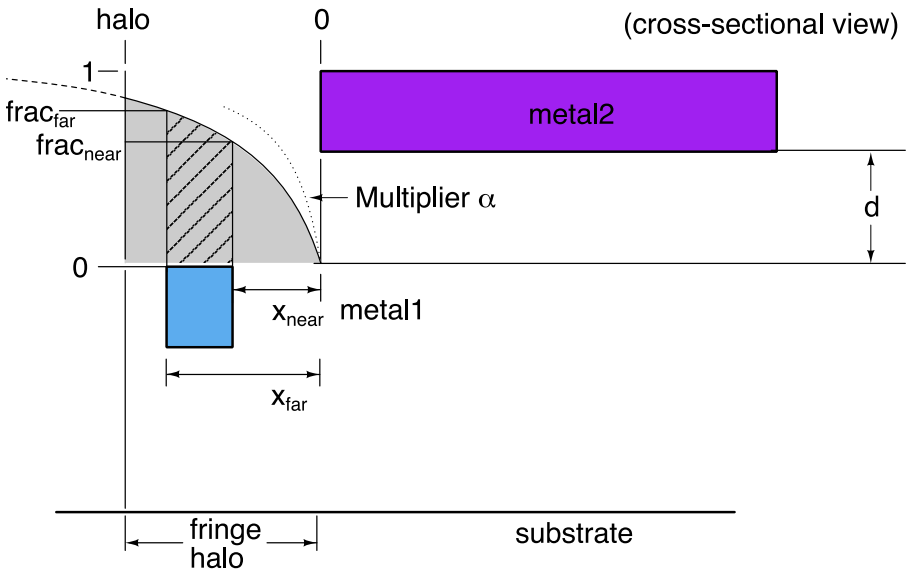


Figure 17: Fringe Capacitance: Non Overlapping (2)

- $\text{frac}_{near} = \frac{2}{\pi} * \text{atan}(\alpha * x_{near})$
- $\text{frac}_{far} = \frac{2}{\pi} * \text{atan}(\alpha * x_{far})$
- $\text{frac} = \text{frac}_{far} - \text{frac}_{near}$

5.7 Shielding Effects

Table 1: Shielding effects

Type	**Shielding To Substrate**	**Between layers**	**On same layer**
Overlap shielding	✓	✓	✗
Sidewall shielding	✗	✗	✓
Lateral fringe shielding	✗	✓	✓
Vertical fringe shielding	✓	✓	✗

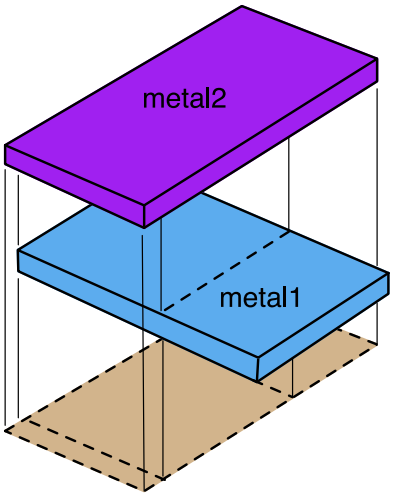
Note, given an analyzed sidewall (edge in 2D), lateral fringe shielding

- is caused by opposing shapes on the same layer
  - even by the same polygon
  - by other polygons (same net)
  - by other polygons (different net) — this is what we also look at when analyzing sidewall coupling
- will shield the fringing to the lower layers, i.e. in Figure 19 the coupling between the two shapes is the same in the above and below cases

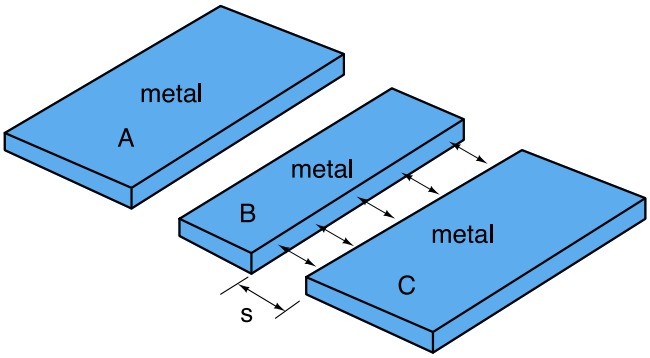
5.8 Parasitic Resistance

Magic constructs a graph of resistors between nodes

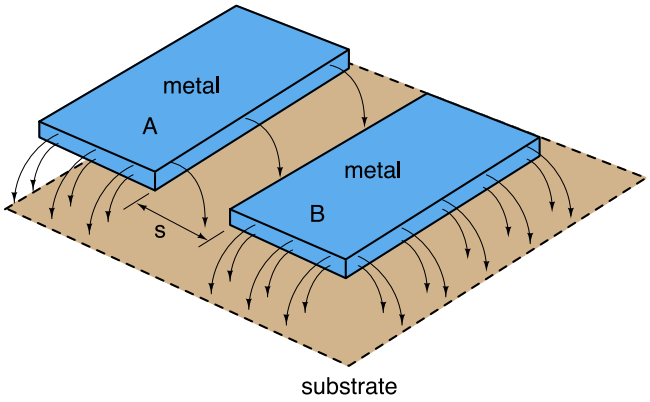
- device terminals
- pins
- junctions (vias or branching on the same metal layer)



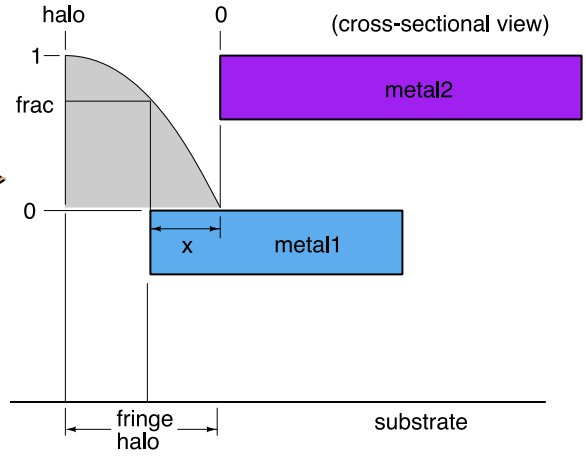
(a) Overlap shielding



(b) Sidewall shielding



(c) Lateral fringe shielding (1/2)



(d) Vertical fringe shielding

Figure 18: Shielding Effects

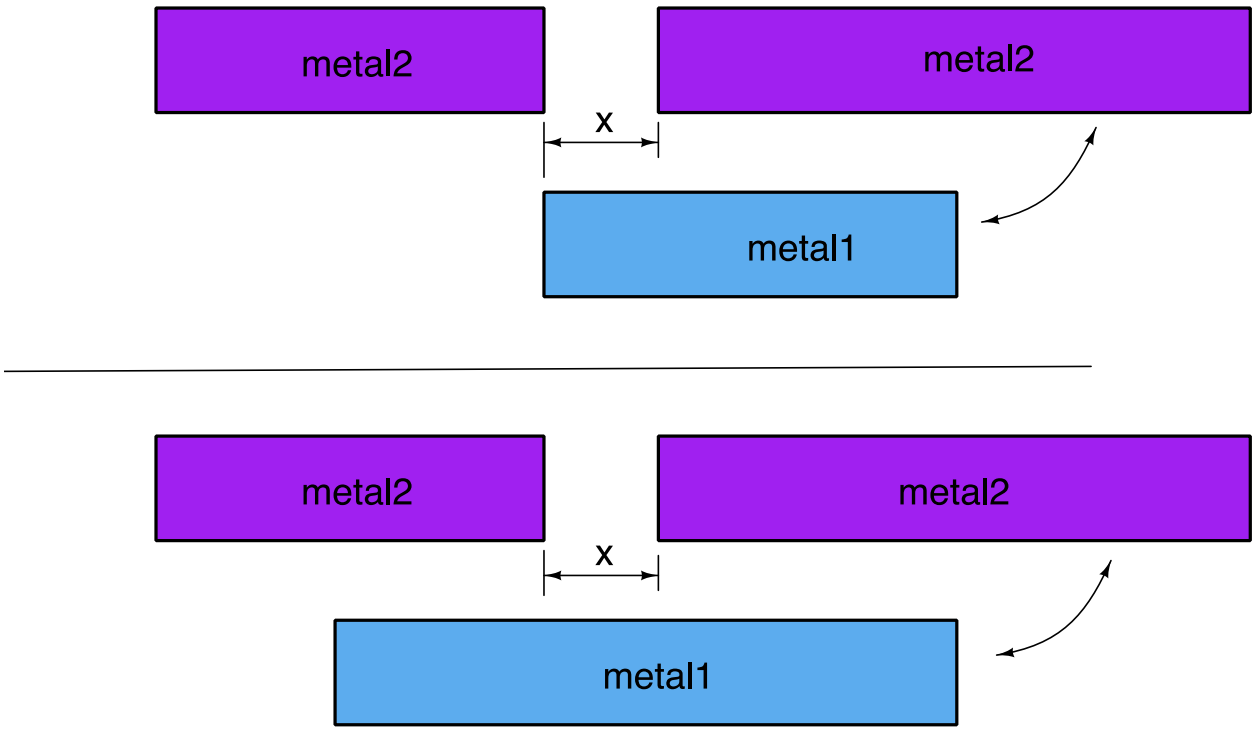
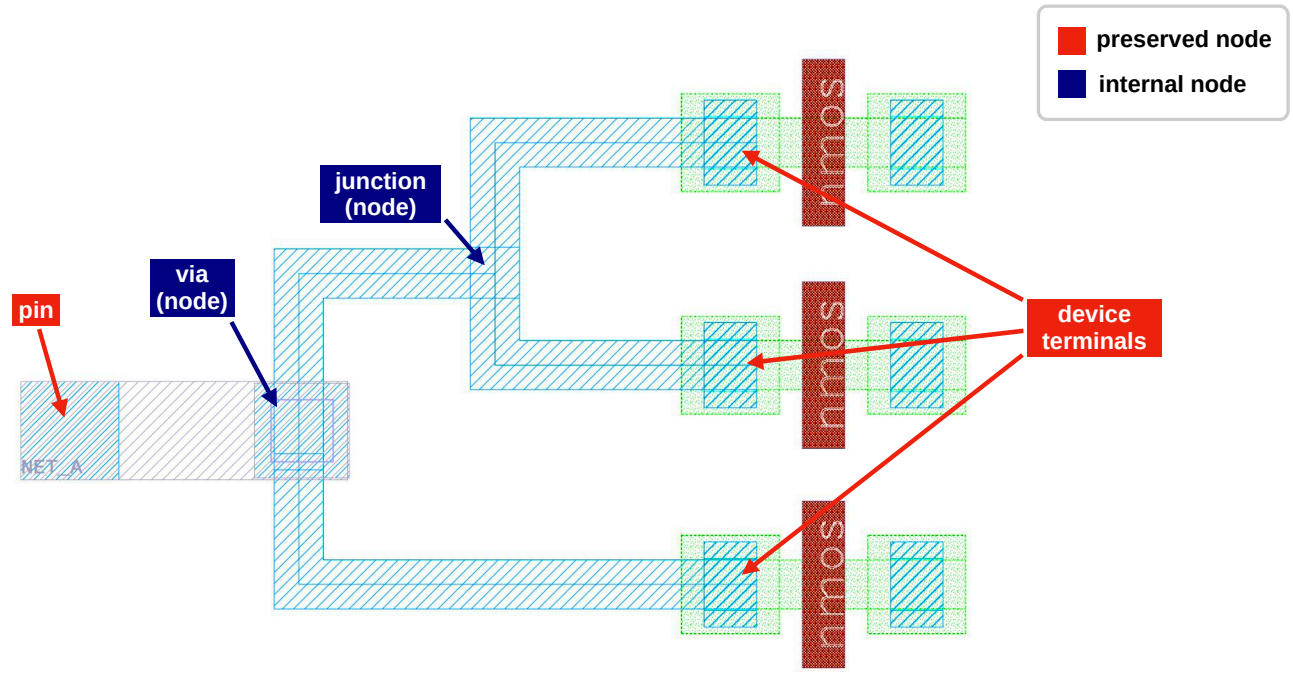
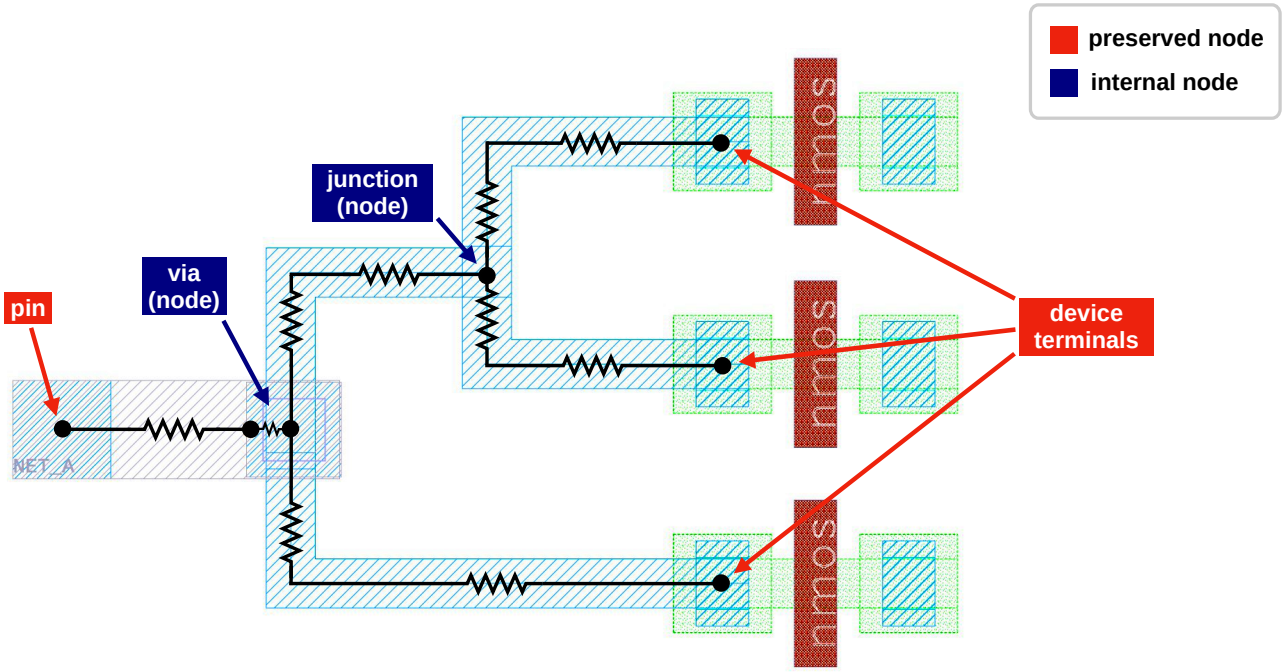


Figure 19: Lateral fringe shielding (2/2)





5.8.1 Wire resistance

- Given a wire with length l and height h, the basic formula is

$$R_{wire} = \frac{l}{h} * R_{coeff} \qquad \left[\frac{\mu m}{\mu m} * m\Omega\right]$$

- Coefficient R_{coeff} is part of the tech files (Parasitic Tables)
  - defined for every metal layer
  - in $m\Omega$ for $1\,\mu m^2$
  - Coefficient already includes the thickness aspect of the layer, so the formula works in 2D

5.8.2 Via resistance

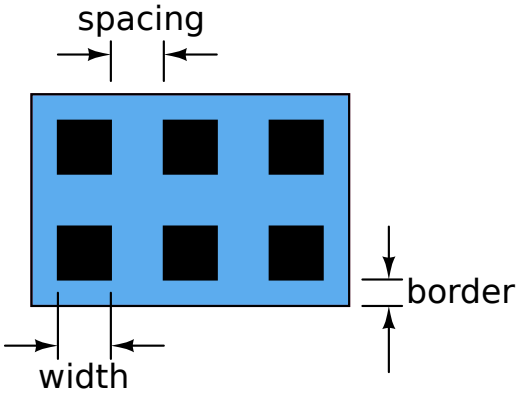


Figure 20: Via dimensions)

- Given
  - drawn via in the layout, with width w and height h
  - design rules defined for each via layer:

    - via width *viawidth*
    - *spacing* between vias (in case of a via array)
    - *border* on each side of the via

- MAGIC interpretation of the via drawing
  - determine number of vias in x and y direction, i.e. n_x and n_y
  - if x and y dimensions are below the minimum size, MAGIC counts 1 via in each direction
  - if the dimensions are larger, we calculate how much vias fit

$$R_{via} = \frac{R_{coeff}}{n_x * n_y} \quad \left[\frac{m\Omega}{\text{via count}}\right]$$

$$n_x = 1 + \left\lfloor \frac{w - (viawidth + 2 * border)}{viawidth + spacing} \right\rfloor$$

$$n_y = 1 + \left\lfloor \frac{h - (viawidth + 2 * border)}{viawidth + spacing} \right\rfloor$$

- Coefficient R_{coeff} is part of the tech files (Parasitic Tables)
  - defined for every via layer
  - in $m\Omega$ per via
  - Coefficient already includes the thickness aspect of the layer, so the formula works in 2D

6 KPEX/2.5D Engine

This analytical extraction engine is separate from the **PEX25D interchange format** described in Section 9.1. `kpex extract --2.5D` selects the engine; `kpex pex25d` generates a geometry and material description without running an engine.

Field solvers are precise, yet slow, they are useful to obtain a golden reference.

For most use cases, a faster engine is desirable. `KPEX/MAGIC` is such an engine, but the `MAGIC` code is tightly coupled with the database, layer/via design choices, and user interface of `MAGIC`. For example, it runs single-threaded.

Therefore, the `KPEX` 2.5D Engine intends to implement the concepts and formulas of `MAGIC` (see Section 5.2), but in a way that is best suited to the `KLayout` API, and is modular and open for parallel execution.

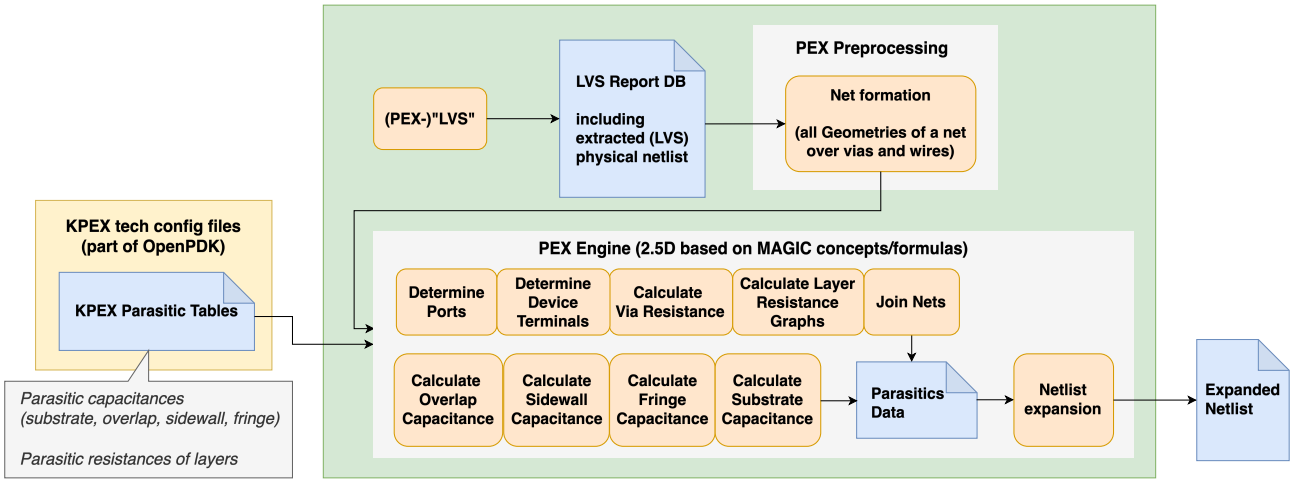


Figure 21: `KPEX` 2.5D Engine

In Figure 21, we see that as in the `KPEX/FasterCap` engine (see Section 4), a "LVS" script is used to extract the connectivity information and create an LVS report database.

As mentioned in Section 1.4, `KPEX` uses the KLayout API. During the coarse of this project, in cooperation with Matthias Köfferlein, the `KLayout` API was extended, in order to simplify parasitic extraction.

The engine uses those new API classes to extract parasitic capacitances:

- `klayout.db.PolygonNeighborhoodVisitor` (since `0.30.1`): used during the extraction of overlap capacitances (see Section 5.5)
- `klayout.db.EdgeNeighborhoodVisitor` (since `0.30.1`): used during the extraction of sidewall capacitances (see Section 5.4) and fringe capacitances (see Section 5.6)
- `klayout.db.PolygonWithProperties` / `klayout.db.EdgeWithProperties` / `klayout.db.EdgePairWithPr` / `klayout.db.BoxWithProperties` (since `0.30.1`): used to store the net information directly within geometry objects

The engine uses those new API classes to extract parasitic resistances:

- `klayout.pex` module (since KLayout `0.30.2`)
  - `klayout.pex.RNetExtractor`: used to extract resistance networks
  - `klayout.pex.RExtractorTech`: minimal process stack description for conductor and via layers

7 Comparison KPEX/2.5D with MAGIC

The initial goal of KPEX/2.5D is to basically come up with the same results as KPEX/MAGIC.

> **Note**
>
> Notes about this comparison:
>
> - Version `Magic 8.3 rev 486` is used, but augmented with debug logging[7], which will be shown and discussed for each example.
>   - NOTE: that revision still counted each sidewall edge against the full `defaultsidewall` value of the tech file, which `MAGIC 8.3.679` corrected (see Section 5.4). Sidewall capacitances are given below under the corrected definition, which is half of what the `CapDebug` output of rev 486 reports.
> - KPEX/MAGIC halo is set to `--magic_halo=100000`
>   - NOTE: the screenshots however were created with `--magic_halo=8` to give better illustrations
> - KPEX/2.5D halo is set to `--halo=100000`
> - The test patterns `overlap_shield_same_net_*` were compared with `MAGIC 8.3.681` without debug logging, so their MAGIC capacitances are given per net pair.

7.1 Test Pattern `single_plate_100um_x_100um_li1_over_substrate`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/single_plate_100um_x_100um_li1_over_substrate.gds.gz

Table 2: Extracted Parasitic Overlap Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Overlap	li1	li1	substrate	VSUBS	369.9	369.9	4

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Fringe (top)	li1	li1	substrate	VSUBS	4.07	4.07	5
Fringe (left)	li1	li1	substrate	VSUBS	4.07	4.07	6
Fringe (right)	li1	li1	substrate	VSUBS	4.07	4.07	7
Fringe (bottom)	li1	li1	substrate	VSUBS	4.07	4.07	8

```
Magic 8.3 revision 486 - Compiled on `date`.
----------------------------------------------------

CapDebug (extNodeAreaFunc/Area) layer li(90), net li_0_0#, area=400000000 (10000 µm^2) nreg_ca
CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer li(90), net li_0_0#, length=20000 (100 µm),
```

[7] A fork of MAGIC which includes debug logging about the different parasitic contributions is hosted here: https://github.com/martinjankoehler/magic

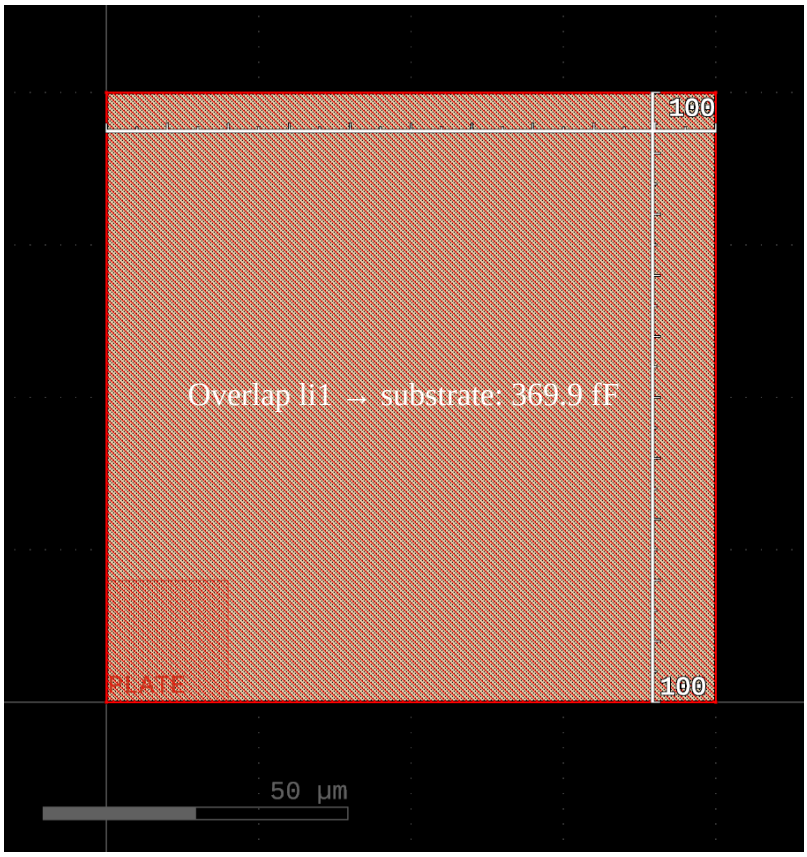


Figure 22: Overlap capacitance li1 to substrate.

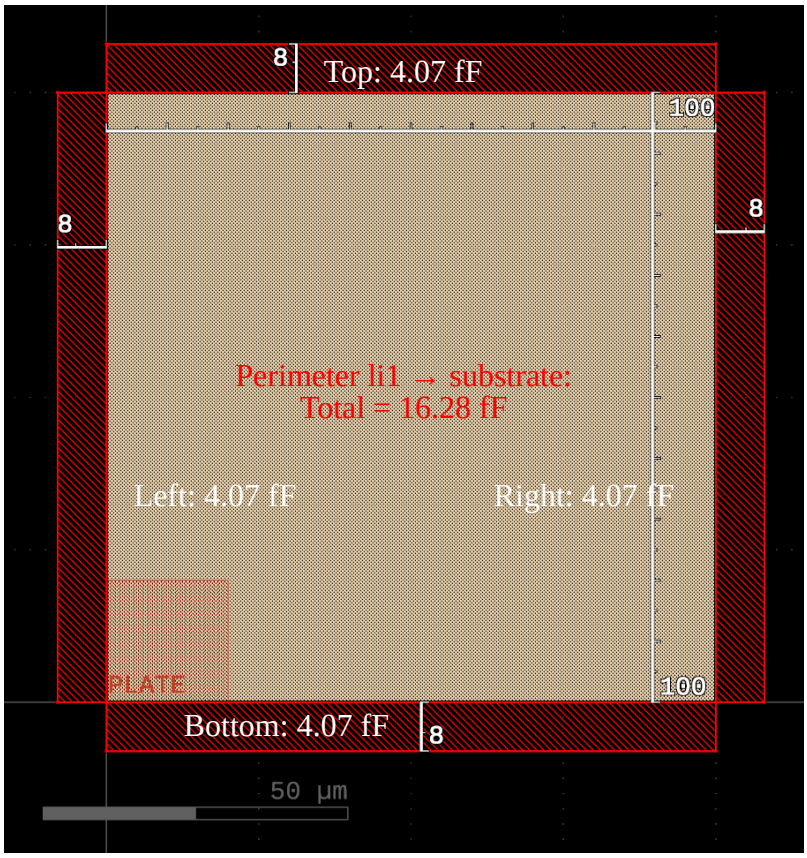


Figure 23: Perimeter (fringe) capacitance li1 to substrate.

```
CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer li(90), net li_0_0#, length=20000 (100 µm)
CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer li(90), net li_0_0#, length=20000 (100 µ
CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer li(90), net li_0_0#, length=20000 (100 µm
CapDebug (extSetResist): li_0_0# area=400000000 (10000 µm^2) perim=80000 (400 µm)
CapDebug ---
```

7.2 Test Pattern `sidewall_20um_length_distance_200nm_li1`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/sidewall_20um_length_distance_200nm_li1.gds.gz

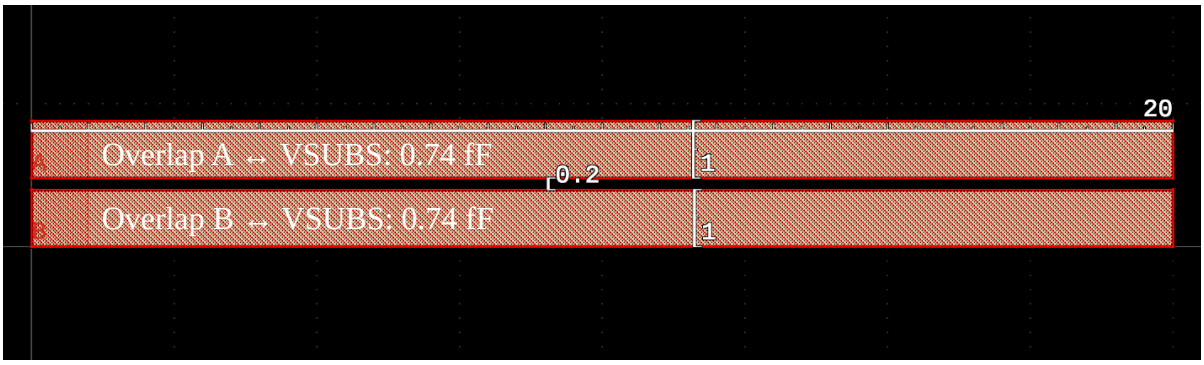


Figure 24: Overlap capacitance of nets on li1.

Table 3: Extracted Parasitic Overlap Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Overlap	li1	A	substrate	VSUBS	0.7398	0.74	4
Overlap	li1	B	substrate	VSUBS	0.7398	0.74	11

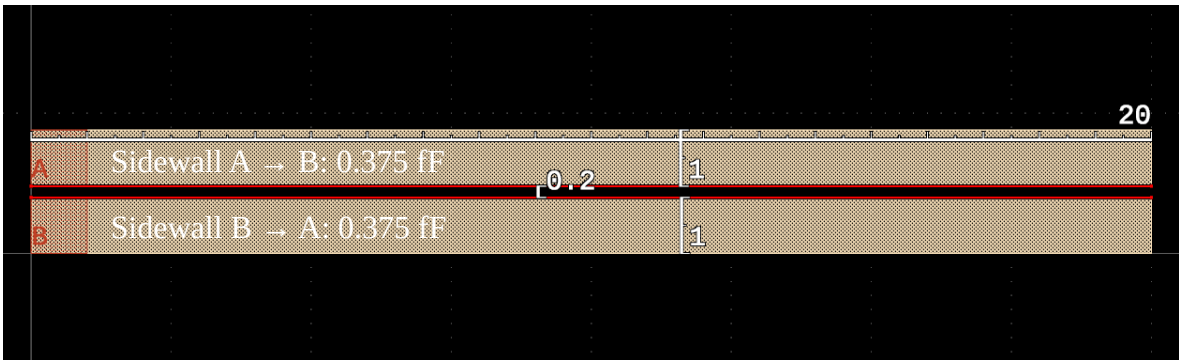


Figure 25: Sidewall capacitance between nets A and B on li1.

Table 4: Extracted Parasitic Sidewall Capacitances

Description	Layer	Net1	Net2	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Sidewall	li1	A	B	0.375	0.375	18
Sidewall	li1	B	A	0.375	0.375	21

Both edges together therefore couple `A` and `B` with 0.75 fF. Lines 18 and 21 of the `CapDebug` output below still report 0.75 fF per edge, adding up to 1.5 fF, because they were recorded with rev 486 — the `MAGIC` column above states that output under the definition of `MAGIC 8.3.679`.

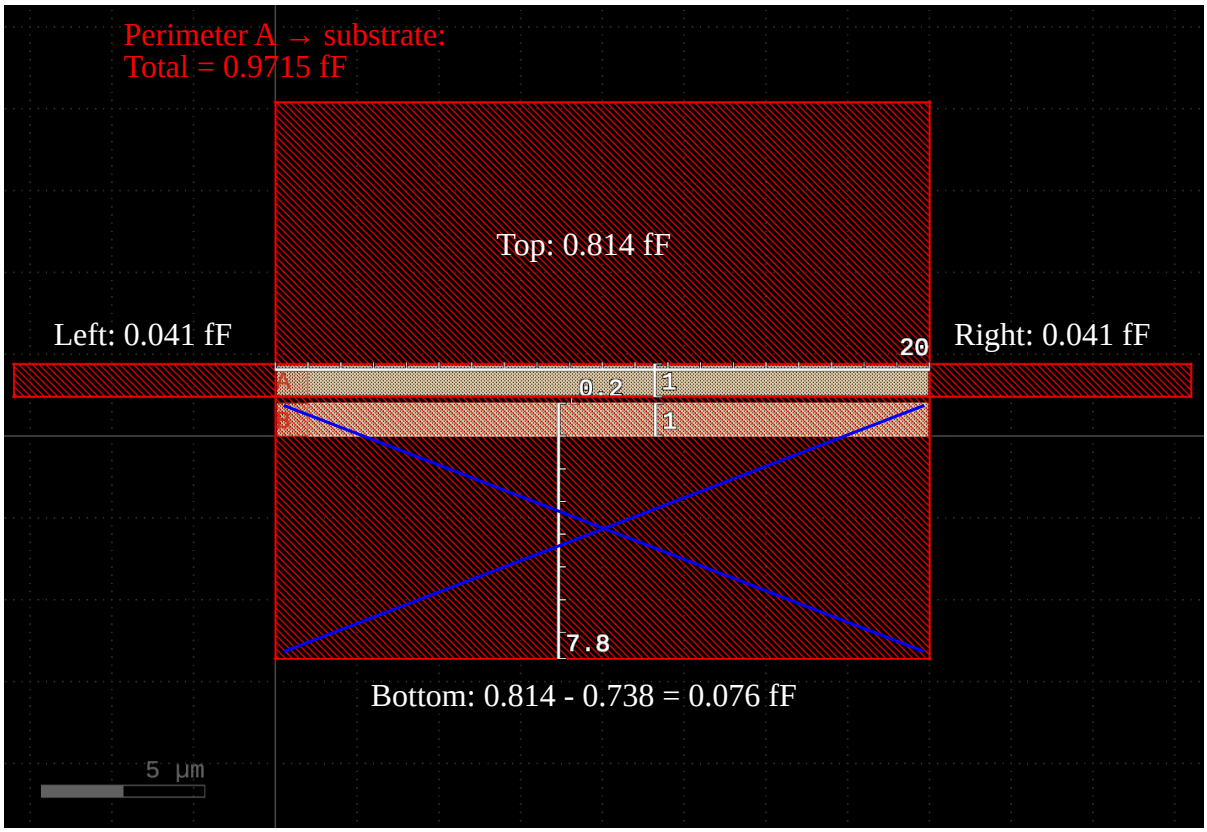


Figure 26: Fringe capacitance between net A and substrate.

Table 5: Extracted Parasitic Fringe Capacitances between nets A, B on li1 and substrate

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Fringe (top)	li1	A	substrate	VSUBS	0.814	0.814	5
Fringe (left)	li1	A	substrate	VSUBS	0.0407	0.041	6
Fringe (bottom)	li1	A	substrate	VSUBS	0.814	0.076	7
Fringe (bottom)	li1	A	substrate	VSUBS	-0.7379	—	19
Fringe (right)	li1	A	substrate	VSUBS	0.0407	0.041	8
Fringe (top)	li1	B	substrate	VSUBS	0.814	0.076	12
Fringe (top)	li1	B	substrate	VSUBS	-0.7379	—	22
Fringe (left)	li1	B	substrate	VSUBS	0.0407	0.041	13
Fringe (bottom)	li1	B	substrate	VSUBS	0.814	0.814	14
Fringe (right)	li1	B	substrate	VSUBS	0.0407	0.041	15

```
1 Magic 8.3 revision 486 - Compiled on `date`.
2 ------------------------------------------------
3
4 CapDebug (extNodeAreaFunc/Area) layer li(90), net li_0_240#, area=800000 (20 µm^2) nreg_cap +=
5 CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer li(90), net li_0_240#, length=4000 (20 µm),
```

```
CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer li(90), net li_0_240#, length=200 (1 µm),
CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer li(90), net li_0_240#, length=4000 (20 µ
CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer li(90), net li_0_240#, length=200 (1 µm),
CapDebug (extSetResist): li_0_240# area=800000 (20 µm^2) perim=8400 (42 µm)
CapDebug ---
CapDebug (extNodeAreaFunc/Area) layer li(90), net li_0_0#, area=800000 (20 µm^2) nreg_cap += 0
CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer li(90), net li_0_0#, length=4000 (20 µm), n
CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer li(90), net li_0_0#, length=200 (1 µm), nr
CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer li(90), net li_0_0#, length=4000 (20 µm)
CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer li(90), net li_0_0#, length=200 (1 µm), n
CapDebug (extSetResist): li_0_0# area=800000 (20 µm^2) perim=8400 (42 µm)
CapDebug ---
CapDebug (sidewall): A-B (layer li), overlap=4000 (20 µm), sep=40 (0.2 µm), e->ec_cap=12.75 (0
CapDebug (obsolete_fringe (blocked)): A -= 0.737881 fF …    now A == 1.711319 fF
    overlapMult=0.003699 (3.699 aF/µm^2) dnear=40 (0.2 µm), snear=0.0935129 (0.000467564 µm),
CapDebug (sidewall): A-B (layer li), overlap=4000 (20 µm), sep=40 (0.2 µm), e->ec_cap=12.75 (0
CapDebug (obsolete_fringe (blocked)): B -= 0.737881 fF …    now B == 1.711319 fF
    overlapMult=0.003699 (3.699 aF/µm^2) dnear=40 (0.2 µm), snear=0.0935129 (0.000467564 µm),
exttospice finished.
```

7.3 Test Pattern `sideoverlap_simple_plates_li1_m1`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/sideoverlap_simple_plates_li1_m1.gds.gz



Figure 27: Overlap capacitances to substrate.

Table 6: Extracted Parasitic Overlap Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Overlap	li1	li1	substrate	VSUBS	3.699	3.7	7
Overlap	met1	met1	substrate	VSUBS	232.02	232.02	15

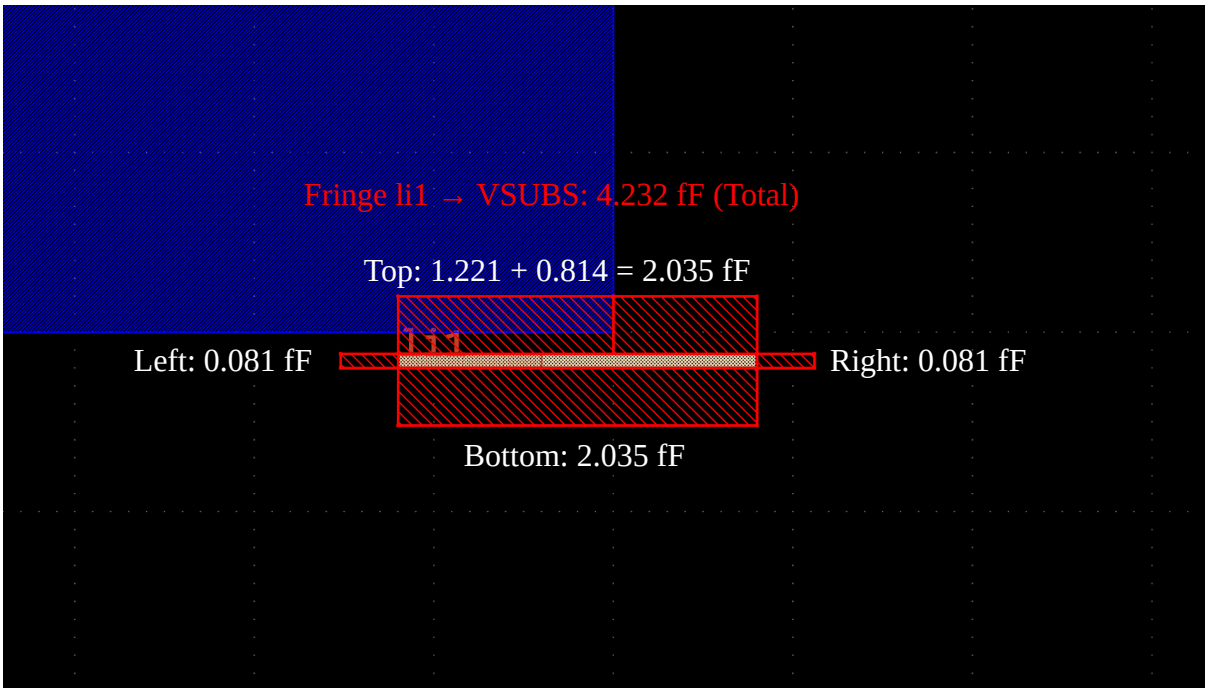


Figure 28: Fringe capacitances li1 to substrate.

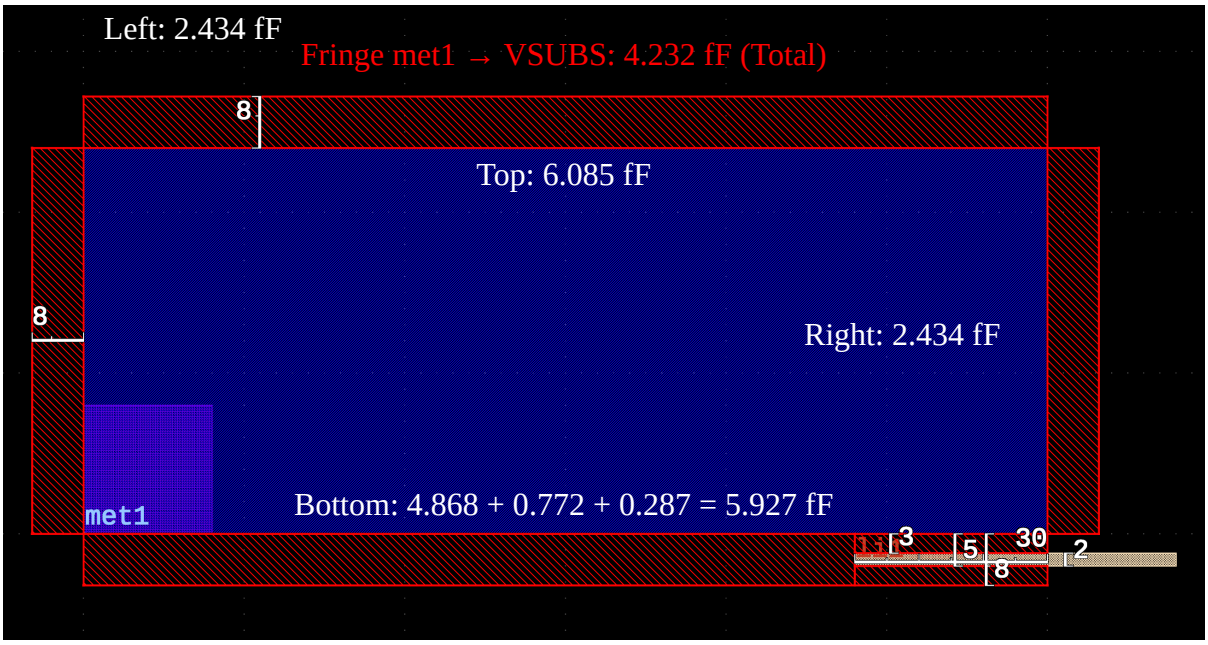


Figure 29: Fringe capacitances met1 to substrate.

Table 7: Extracted Parasitic Fringe Capacitances between metals and substrate

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Fringe (top)	li1	li1	substrate	VSUBS	2.035	2.035	9
Fringe (bottom)	li1	li1	substrate	VSUBS	2.035	2.035	11
Fringe (left)	li1	li1	substrate	VSUBS	0.081	0.081	10
Fringe (right)	li1	li1	substrate	VSUBS	0.081	0.081	12
Fringe (top)	met1	met1	substrate	VSUBS	6.0855	6.085	17
Fringe (bottom)	met1	met1	substrate	VSUBS	6.0855	5.927	19
Fringe (bottom)	met1	met1	substrate	VSUBS	-0.1579	—	19
Fringe (left)	met1	met1	substrate	VSUBS	2.4342	2.434	18
Fringe (right)	met1	met1	substrate	VSUBS	2.4342	2.434	20

Note

`MAGIC` handles shielding differently than `KPEX/2.5D`, it first assumes no shield exists, and later subtracts portions to arrive at the shielded value. That's why the bold cells differ. Line 15 - 20 accumulate contributions, whereas line 34 subtracts a contribution:

- Line 15: `CapDebug (extNodeAreaFunc/Area) layer m1(97), net m1_10000_10000#, area=360000000 (9000 µm^2) nreg_cap += 232.02 fF`
- Line 17: `CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer m1(97), net m1_10000_10000#, length=30000 (150 µm), nreg_cap += 6.0855 fF (now nreg_cap = 238.105 fF)`
- Line 18: `CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer m1(97), net m1_10000_10000#, length=12000 (60 µm), nreg_cap += 2.4342 fF (now nreg_cap = 240.54 fF)`
- Line 19: `CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer m1(97), net m1_10000_10000#, length=30000 (150 µm), nreg_cap += 6.0855 fF (now nreg_cap = 246.625 fF)`
- Line 20: `CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer m1(97), net m1_10000_10000#, length=12000 (60 µm), nreg_cap += 2.4342 fF (now nreg_cap = 249.059 fF)`
- Line 34: `nreg_cap -= 0.157966 fF (now nreg_cap = 248.901 fF)`

Comparison:

- `KPEX/2.5D`: $4.86834 + 0.77242 + 0.2867 = 5.9275$
- `MAGIC`: $6.0855 - 0.157966 = 5.9275$

Table 8: Extracted Parasitic Fringe Capacitances between metals

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	MAGIC [fF]	KPEX/2.5D [fF]	MAGIC Lines
Fringe (top)	li1	li1	met1	met1	0.598	0.06	26
Fringe (bottom)	met1	met1	li1	li1	0.0654	0.065	36

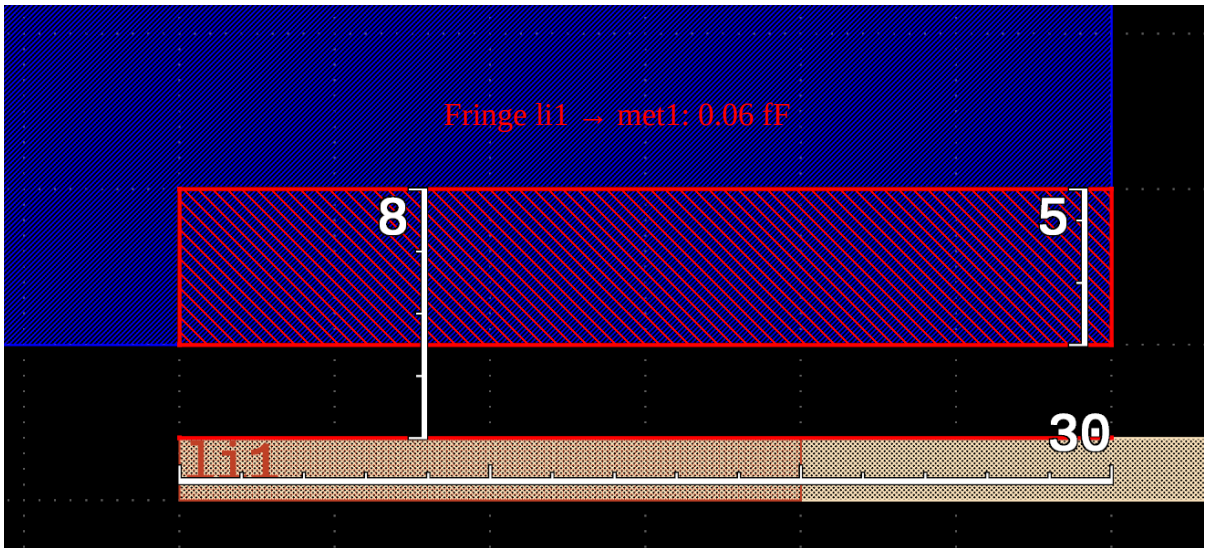


Figure 30: Fringe capacitances li1 to m1.

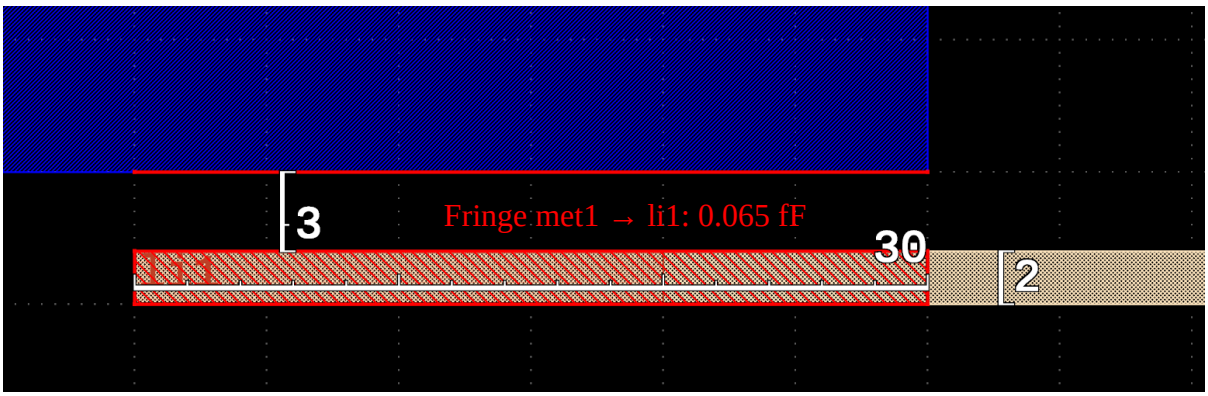


Figure 31: Fringe capacitances m1 to li1.

```
Magic 8.3 revision 486 - Compiled on `date`.

----------------------------------------------------

Extracting sideoverlap_simple_plates_li1_m1 into /Users/martin/Source/klayout-pex/output_sky13

CapDebug (extNodeAreaFunc/Area) layer li(90), net li_34000_9000, area=4000000 (100 µm^2) nreg_

CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer li(90), net li_34000_9000, length=10000 (50
CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer li(90), net li_34000_9000, length=400 (2 µ
CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer li(90), net li_34000_9000, length=10000
CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer li(90), net li_34000_9000, length=400 (2
CapDebug (extSetResist): li_34000_9000 area=4000000 (100 µm^2) perim=20800 (104 µm)
CapDebug ---
CapDebug (extNodeAreaFunc/Area) layer m1(97), net m1_10000_10000#, area=360000000 (9000 µm^2)

CapDebug (extNodeAreaFunc/Perimeter/TopSide) layer m1(97), net m1_10000_10000#, length=30000 (
CapDebug (extNodeAreaFunc/Perimeter/LeftSide) layer m1(97), net m1_10000_10000#, length=12000
CapDebug (extNodeAreaFunc/Perimeter/BottomSide) layer m1(97), net m1_10000_10000#, length=3000
CapDebug (extNodeAreaFunc/Perimeter/RightSide) layer m1(97), net m1_10000_10000#, length=12000
CapDebug (extSetResist): m1_10000_10000# area=360000000 (9000 µm^2) perim=84000 (420 µm)
CapDebug ---
CapDebug (extSideOverlapHalo): (li-m1) length=30.000000 µm, mult=0.011420, dnear=3.000000 µm (
CapDebug (extSideOverlapHalo) (li-substrate): mult=0.003699, snear=0.730477, sfar=0.893413 (sf
CapDebug (extSideOverlapHalo): efflength=1.72351 µm, cap+=0.299029 aF, cap=59.8059 aF, e->ec_c
CapDebug (sideoverlaphalo): met1-li1 += 0.059806 fF …   now met1-li1 == 0.059806 fF
CapDebug ---
CapDebug (extSideOverlapHalo): (m1-li) length=30.000000 µm, mult=0.011420, dnear=3.000000 µm (
CapDebug (extSideOverlapHalo) (m1-substrate): mult=0.002578, snear=0.634619, sfar=0.764408 (sf
CapDebug (extSideOverlapHalo): efflength=1.0996 µm, cap+=0.327131 aF, cap=65.4263 aF, e->ec_ca
CapDebug (extSideOverlapHalo/obsolete_perimcap) layer m1(97), net met1,
    efflength=3.89366 µm (778.733) = (sfrac(0.129789) - subfrac(0)) * length(30 µm)
    exts_perimCap[m1][0] = 0.20285
    nreg_cap -= 0.157966 fF (now nreg_cap = 248.901 fF)
CapDebug (obsolete_perimcap): met1 -= 0.157966 fF … now met1 == 248.901423 fF
CapDebug (sideoverlaphalo): met1-li1 += 0.065426 fF …   now met1-li1 == 0.125232 fF
```

7.4 Test Pattern `overlap_shield_same_net_100um_x_100um_li1_m1_m2`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/overlap_shield_same_net_100um_x_100um_li1_m1_m2.gds.gz

Plates of $100\,m \times 100\,m$: `li1` and `met1` of net `A`, joined by an `mcon` in a corner, under `met2` of net `B`. `met1` shields `li1` from `met2`, although both are of net `A`.

Table 9: Extracted Parasitic Overlap Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	KPEX/2.5D [fF]
Overlap	met2	B	met1	A	1338.6
Overlap	met2	B	li1	A	—
Overlap	li1	A	substrate	VSUBS	369.9

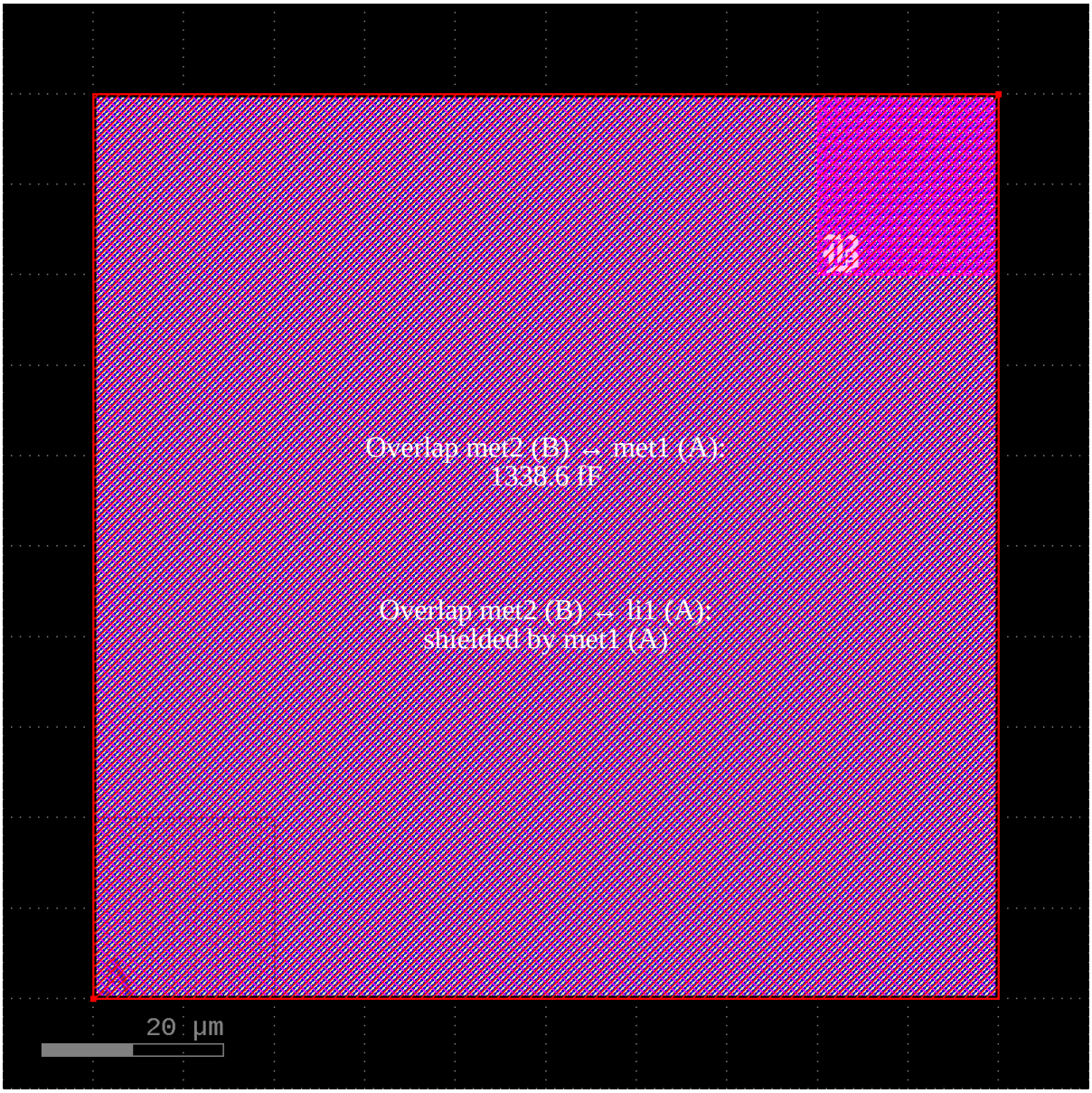


Figure 32: Overlap capacitances between the nets A and B, met1 (A) covers li1 (A).

Table 10: Extracted Parasitic Fringe Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	KPEX/2.5D [fF]
Fringe (all sides)	li1	A	substrate	VSUBS	16.28
Fringe (all sides)	met1	A	substrate	VSUBS	16.228
Fringe (all sides)	met2	B	substrate	VSUBS	15.104

Table 11: Extracted Parasitic Capacitances per Net Pair

Net 1	Net 2	MAGIC [fF]	KPEX/2.5D [fF]
A	B	1338.6	1338.6
A	VSUBS	402.41	402.408
B	VSUBS	15.104	15.104

> **Note**
>
> Up to #299, only shapes of other nets shielded in `KPEX/2.5D`, so `A B` also got the overlap of `met2` and `li1` below `met1`: $1338.6 + 375.6 = 1714.2\,fF$.

7.5 Test Pattern `overlap_shield_same_net_half_100um_x_100um_li1_m1_m2`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/overlap_shield_same_net_half_100um_x_100um_li1_m1_m2.gds.gz

The same plates, but `met1` covers only the left half of `li1`: `met2` overlaps `met1` on the left half and `li1` on the right half.

Table 12: Extracted Parasitic Overlap Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	KPEX/2.5D [fF]
Overlap	met2	B	met1	A	669.3
Overlap	met2	B	li1	A	187.8
Overlap	li1	A	substrate	VSUBS	369.9

Table 13: Extracted Parasitic Fringe Capacitances

Description	Layer Top	Net Top	Layer Bottom	Net Bottom	KPEX/2.5D [fF]
Fringe (all sides)	li1	A	substrate	VSUBS	16.28
Fringe (all sides)	met1	A	substrate	VSUBS	8.214
Fringe (right)	met1	A	met2	B	4.796
Fringe (all sides)	met2	B	substrate	VSUBS	15.104

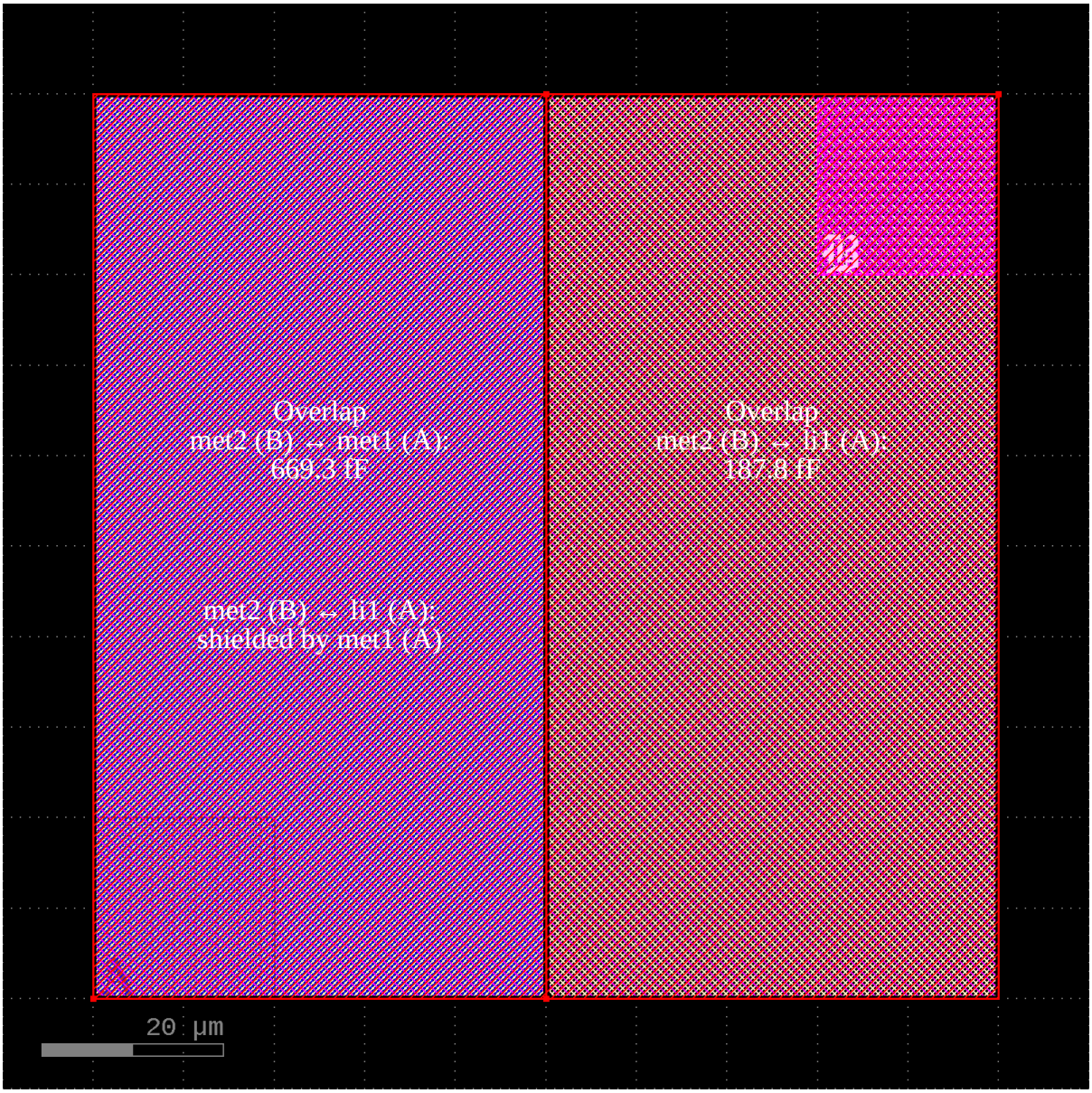


Figure 33: Overlap capacitances between the nets A and B, met1 (A) covers the left half of li1 (A).

Table 14: Extracted Parasitic Capacitances per Net Pair

Net 1	Net 2	MAGIC [fF]	KPEX/2.5D [fF]
A	B	861.9	861.896
A	VSUBS	394.39	394.394
B	VSUBS	15.104	15.104

> **Note**
>
> Up to #299, `KPEX/2.5D` also counted the overlap of `met2` and `li1` below `met1` (the left half), so `A B` was $861.896 + 187.8 = 1049.696\,fF$.

7.6 Test Pattern `r_single_wire_li1`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/r_single_wire_li1.gds.gz

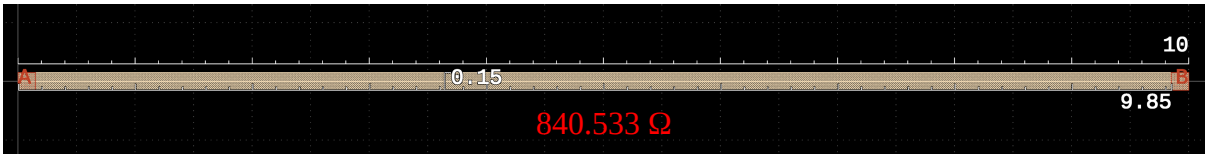


Figure 34: Resistance of a single wire on li1.

- Wire length $L = 9.85\,\mu m$
- Wire height $H = 0.15\,\mu m$
- Layer is `li1`
- Parasitic Table Coefficient $R_{coeff}(li1) = 12800\,m\Omega$

If we apply the formula illustrated in Section 5.8.1, we get

$$R_{wire} = \frac{L}{H} * R_{coeff}$$

$$R_{wire} = \frac{9.85\,\mu m}{0.15\,\mu m} * 12800\,m\Omega = 84053\dot{3}\,m\Omega = 840.53\dot{3}\,\Omega$$

```
Magic 8.3 revision 486 - Compiled on `date`.
------------------------------------------------------
...
Warning:  Ports "A" and "B" are electrically shorted.

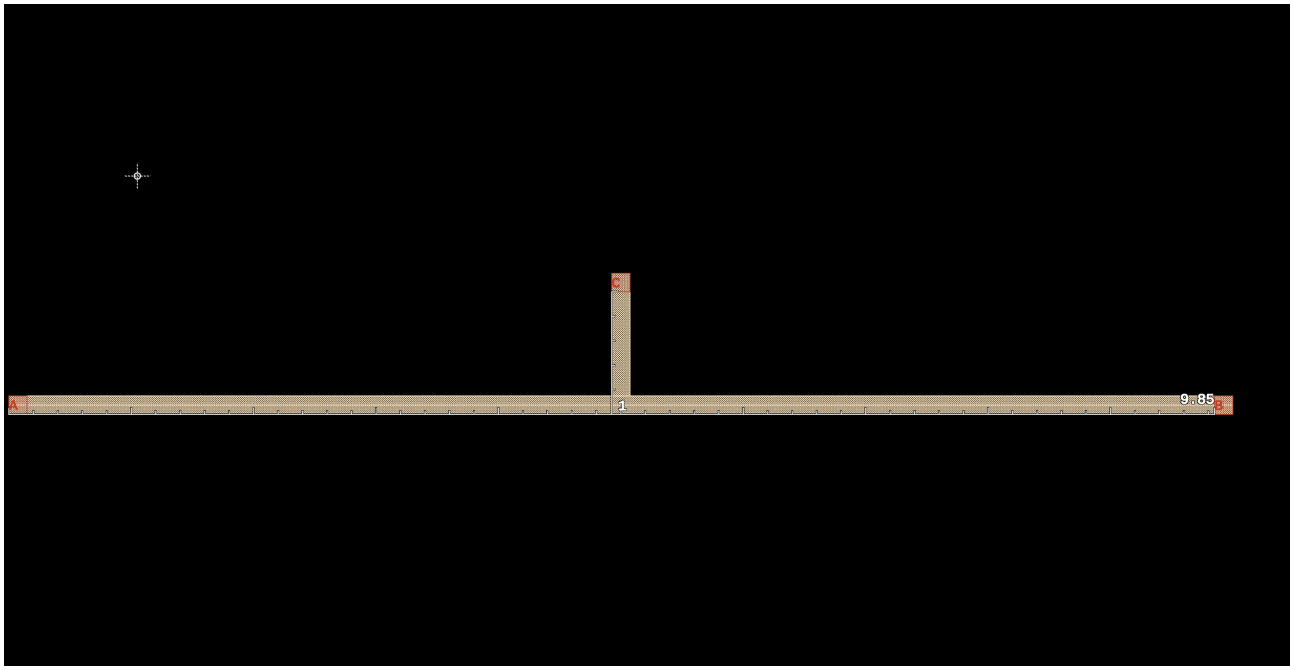
Location is (1970, -15); drivepoint (1970, -15)
Location is (0, -15); drivepoint (0, -15)

ResCalcEastWest: A (0, -0.075) <-> B (9.85, -0.075) @ li
    exts_sheetResist[li]=12800, length=9.85, height=0.15, resistor->rr_value = 840533 mΩ
ResCalcEastWest: A (0, -0.075) <-> B (9.85, -0.075) @ li
    exts_sheetResist[li]=12800, length=9.85, height=0.15, resistor->rr_value = 840533 mΩ
Total Nets: 2
```

> **Note**
>
> MAGIC recognizes that ports A and B are shorted. The purpose of these test patterns is to look at minimized problems in isolation, like the resistance of the wire here. They are not realistic examples, so it appears to be a bug in that corner case, that the resistance is reported twice.

7.7 Test Pattern r_wire_voltage_divider_li1

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/r_wire_voltage_divider_li1.gds.gz



With MAGIC, we get (again, R3-R8 are redundant):

```
.subckt r_wire_voltage_divider_li1 B A C
R0 A C.n0 426.668
R1 C.n0 B 413.868
R2 C.n0 C 72.5338
.ends
```

Table 15: Extracted Parasitic Resistances between pins

Description	Layer	Node1	Node2	MAGIC [Ω]	KPEX/2.5D [Ω]
Wire	li1	A	internal N1	426.668	426.667
Wire	li1	B	internal N1	413.868	413.867
Wire	li1	C	internal N1	72.5338	72.533

7.8 Test Pattern r_contact_1x1_minsize_mcon

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/r_contact_1x1_minsize_mcon.gds.gz

- Layers:

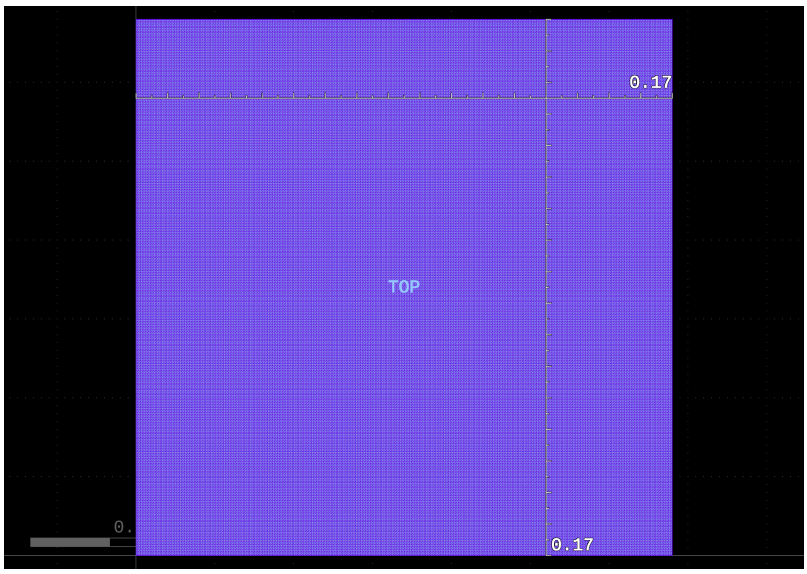


Figure 35: Resistance of a minimum sized via on mcon.

  - Top Layer: `met1`
  - Via: `mcon`
  - Bottom Layer: `li1`

- Design rules:
  - Via width: $viawidth = 0.17\,\mu m$
  - Via spacing: $spacing = 0.19\,\mu m$
  - Via border: $border = 0.0\,\mu m$
- Parasitic Table Coefficient $R_{coeff}(mcon) = 9300\,m\Omega$
- Drawn via:
  - $w = 0.17\,\mu m$
  - $h = 0.17\,\mu m$

If we apply the formula illustrated in Section 5.8.2, we get

$$n_x = 1 + \left\lfloor \frac{w - (viawidth + 2 * border)}{viawidth + spacing} \right\rfloor$$

$$n_x = 1 + \left\lfloor \frac{0.17\,\mu m - (0.17 + 2 * 0.0)\,\mu m}{0.17\,\mu m + 0.19\,\mu m} \right\rfloor = 1$$

$$n_y = 1 + \left\lfloor \frac{h - (viawidth + 2 * border)}{viawidth + spacing} \right\rfloor$$

$$n_y = 1 + \left\lfloor \frac{0.17\,\mu m - (0.17 + 2 * 0.0)\,\mu m}{0.17\,\mu m + 0.19\,\mu m} \right\rfloor = 1$$

$$R_{via} = \frac{R_{coeff}}{n_x * n_y} \quad \left[\frac{m\Omega}{\text{via count}}\right]$$

$$R_{via} = \frac{9300\,m\Omega}{1 * 1} = 9300\,m\Omega = 9.3\,\Omega$$

Table 16: Extracted Parasitic Resistances

Device	Description	Layer	Node1	Node2	MAGIC [Ω]	KPEX/2.5D [Ω]
R2	Wire	mcon	internal `$0.16`	pin `BOT`		0.0
R1	Via	li1	internal `$0.16`	internal `$1.23`	9.3	9.3
R3	Wire	met1	internal `$1.23`	pin `TOP`		0.0

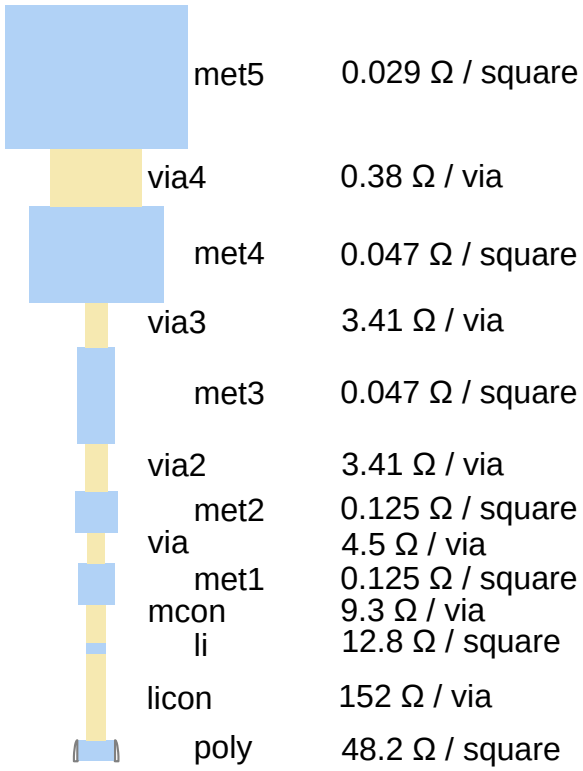
MAGIC extraction results are shown below.

```
Magic 8.3 revision 486 - Compiled on `date`.
----------------------------------------------------
...
Warning:  Ports "TOP" and "BOT" are electrically shorted.
Location is (0, 0); drivepoint (0, 0)
Location is (0, 0); drivepoint (0, 0)

ResDoContacts: (null) (0.085, 0.085) <-> (null) (0.085, 0.085) @ v0
    W = 0.17 µm, H = 0.17 µm
    exts_viaResist[v0]=9300, viawidth=0.17 µm, spacing=0.19 µm, border=0 µm
    squaresx=1, squaresy=1, resistor->rr_value = 9300 mΩ
```

7.9 Test Pattern `r_via_stack_1x1_minsize_poly_to_met5`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/r_via_stack_1x1_minsize_poly_to_met5.gds.gz



On each metal layer, a pin is placed directly in middle of the via.

Table 17: Extracted Parasitic Resistances between pins

Device	Description	Layer	Node1	Node2	MAGIC [Ω]	KPEX/2.5D [Ω]
R2	Wire	poly	internal $0.17	pin poly		0.0
R1	Via	licon	internal $0.17	internal $1.18		152.0
R4	Wire	li1	internal $1.18	pin li1		0.0
R3	Wire	li1	internal $1.18	internal $2.18		0.0
R8	Via	mcon	internal $2.18	internal $3.25		9.3
R10	Wire	met1	internal $3.25	pin met1		0.0
R9	Wire	met1	internal $3.25	internal $4.25		0.0
R11	Via	via	internal $4.25	internal $5.26		4.5
R13	Wire	met2	internal $5.26	pin met2		0.0
R12	Wire	met2	internal $5.26	internal $6.26		0.0
R14	Via	via2	internal $6.26	internal $7.43		3.41
R16	Wire	met3	internal $7.43	pin met3		0.0
R15	Wire	met3	internal $7.43	internal $8.43		0.0
R17	Via	via3	internal $8.43	internal $9.40		3.41
R18	Wire	met4	internal $9.40	pin met4		0.0
R6	Wire	met4	internal $10.40	internal $9.40		0.0
R5	Via	via4	internal $10.40	internal $11.27		0.38
R7	Wire	met5	internal $11.27	pin met5		0.0

KPEX/2.5D extraction results are shown below.

```
Device;Net1;Net2;Capacitance [fF];Resistance [Ω]
R2;$0.17;poly;;0.0
R1;$0.17;$1.18;;152.0
R4;$1.18;li1;;0.0
R3;$1.18;$2.18;;0.0
R8;$2.18;$3.25;;9.3
R10;$3.25;met1;;0.0
R9;$3.25;$4.25;;0.0
R11;$4.25;$5.26;;4.5
R13;$5.26;met2;;0.0
R12;$5.26;$6.26;;0.0
R14;$6.26;$7.43;;3.41
R16;$7.43;met3;;0.0
R15;$7.43;$8.43;;0.0
R17;$8.43;$9.40;;3.41
R18;$9.40;met4;;0.0
R6;$10.40;$9.40;;0.0
R5;$10.40;$11.27;;0.38
R7;$11.27;met5;;0.0
```

MAGIC extraction results are shown below.

```
.subckt r_via_stack_1x1_minsize_poly_to_met5 poly li1 met1 met2 met3 met4 met5
R0 li1.n3 poly 175.37
R1 li1.n4 li1.n3 9.3005
R2 li1.n3 li1 6.4005
R3 li1.n4 li1.n2 4.5005
```

```
R4 li1.n1 li1.n0 3.4105
R5 li1.n2 li1.n1 3.4105
R6 li1.n0 met5 0.3834
R7 met1 li1.n4 0.063
R8 li1.n2 met2 0.0545541
R9 li1.n1 met3 0.0232879
R10 li1.n0 met4 0.00687288
; removed the redundant devices (5.3.540)
.ends
```

7.10 Test Pattern `nfet_li1_redux`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/nfet_li1_redux.gds.gz

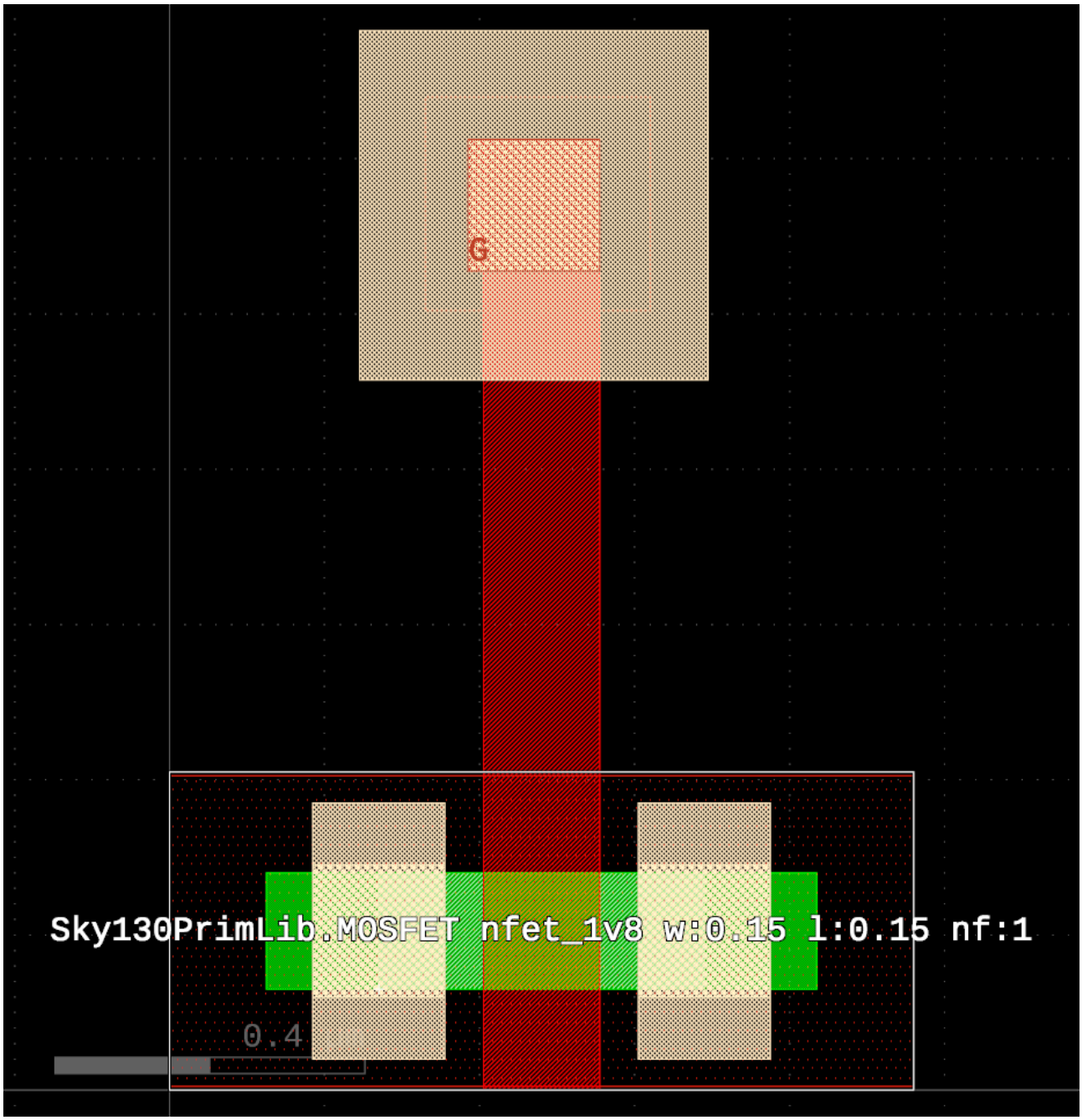


Table 18: Extracted Parasitic Resistances

Device	Description	Layer	Node1	Node2	KPEX/2.5D [Ω]
R1	Wire	li1	internal `G.$1.17`	pin `G`	2.418
R2	Via	licon	internal `G.$0.16` (poly)	internal `G.$1.17` (li1)	152.0
R3	Wire	poly	`G.P0.16` (`G` device terminal)	internal `G.$0.16`	316.321

KPEX/2.5D extraction results are shown below.

```
Device;Net1;Net2;Capacitance [fF];Resistance [Ω]
R1;G;G.$1.17;;2.418
R2;G.$0.16;G.$1.17;;152.0
R3;G.$0.16;G.P0.16;;316.321
```

MAGIC extraction results are shown below.

```
.subckt nfet_li1 VNB G S D
X0 a_111_26# G a_25_26# VSUBS sky130_fd_pr__nfet_01v8 ad=0.042 pd=0.86 as=0.042 ps=0.86 w=0.15
R0 G G.n0 451.651
.ends
```

> **Note**
>
> Notes about this comparison:
>
> - We can see that MAGIC did not extract the interconnection resistances
> - Only exception is `G` ↔ `G.n0`, but node `G.n0` is dangling and not connected

7.11 Test Pattern `nfet_li1`

GDS: https://github.com/iic-jku/klayout-pex/blob/main/testdata/designs/sky130A/test_patterns/nfet_li1.gds.gz

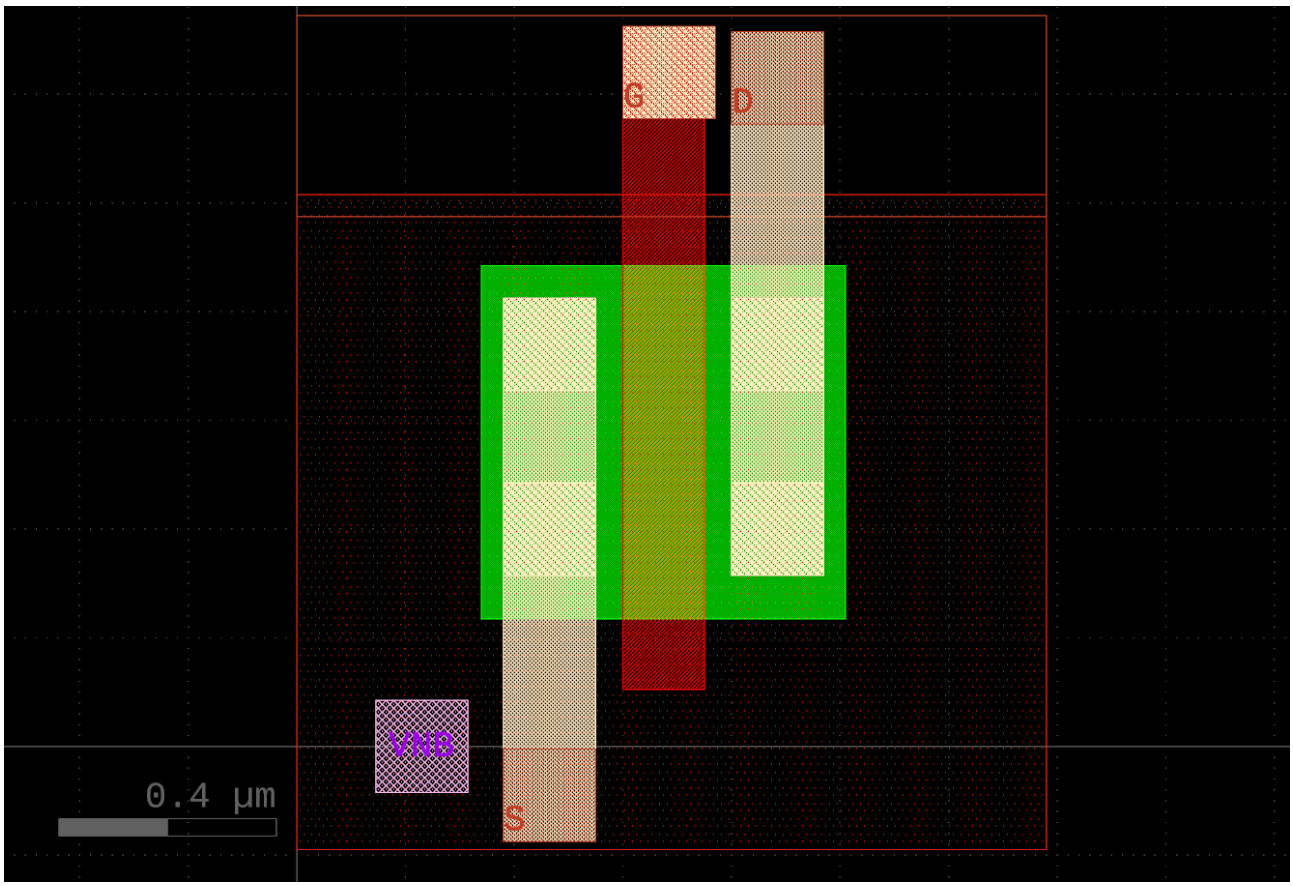


Figure 36: nfet example (2D layout view)

Figure 37 gives an 3D overview over the 3 nets.

KPEX/2.5D extraction results are shown below.

Table 19: Extracted Parasitic Resistances

Device	Description	Layer	Net	Node1	Node2	MAGIC [Ω]	KPEX/2.5D [Ω]
R1	Wire	li1	D	pin D	internal D.$3.18		30.494
R4	Wire	li1	D	internal D.$1.18	internal D.$3.18		25.6
R2	Via	licon	D	internal D.$0.13 (nsdm)	internal D.$1.18 (li1)		185.0
R5	Via	licon	D	internal D.$2.13 (nsdm)	internal D.$3.18 (li1)		185.0
R3	Wire	nsdm	D	internal D.$0.13	D.P0.13 (D device terminal)		0.0
R6	Wire	nsdm	D	internal D.$2.13	D.P0.13 (D device terminal)		0.0
R7	Wire	nsdm	G	pin G	internal G.$1.18		6.4
R8	Via	licon	G	internal G.$0.17 (poly)	internal G.$1.18 (li1)		152.0
R9	Wire	poly	G	internal G.$0.17	G.P0.17 (G device terminal)		234.381
R10	Wire	li1	S	pin S	internal S.$1.18		43.294

[8]The 3D model of this figure was generated by KPEX/FasterCap and saved as STL files. The screenshot was created by opening the STL files in MeshLab.

Device	Description	Layer	Net	Node1	Node2	MAGIC [Ω]	KPEX/2.5D [Ω]
R13	Wire	nsdm	S	internal S.$1.18	S.$3.18 (S device terminal)		0.0
R11	Via	licon	S	internal S.$0.13 (nsdm)	internal S.$1.18 (li1)		185.0
R14	Via	licon	S	internal S.$2.13 (nsdm)	internal S.$3.18 (li1)		185.0
R12	Wire	nsdm	S	internal S.$0.13	S.P0.13 (S device terminal)		0.0
R15	Wire	nsdm	S	internal S.$2.13	S.P0.13 (S device terminal)		0.0

Warning

To improve readability, the illustrations in Figure 38 and Figure 39 do not depict all nodes correctly in regard of the the z-axis orientation

- nodes on metal layers are depicted residing on the top side of the layers and not inside or below the wires / vias
- currently, we treat the metal layer z-axis as an equipotential
- so in reality, R1, R4, R7, R9 would be the whole wire section

Figure 38 illustrates the parasitic resistance contributions for net D (all nets can be seen in Figure 37). Net G is illustrated in Figure 39. Net S is not worth showing, as it is similar to net D.

MAGIC extraction results are shown below.

```
.subckt nfet_li1 VNB G S D
X0 D G S VNB sky130_fd_pr__nfet_01v8 ad=0.169 pd=1.82 as=0.169 ps=1.82 w=0.65 l=0.15
R0 G G.n0 373.693
.ends
```

Note

Notes about this comparison:

- We can see that MAGIC did not extract the interconnection resistances
- Only exception is G ↔ G.n0, but node G.n0 is dangling and not connected

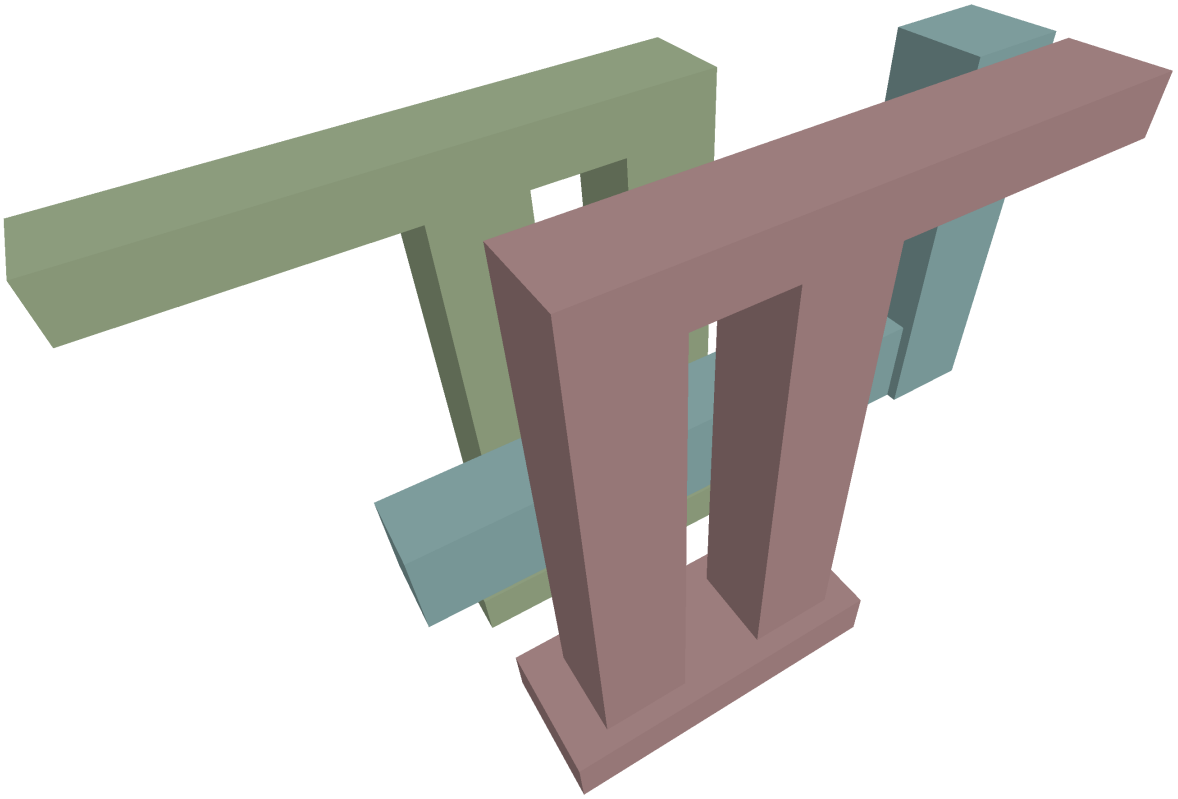


Figure 37: nfet example (3D view[8])

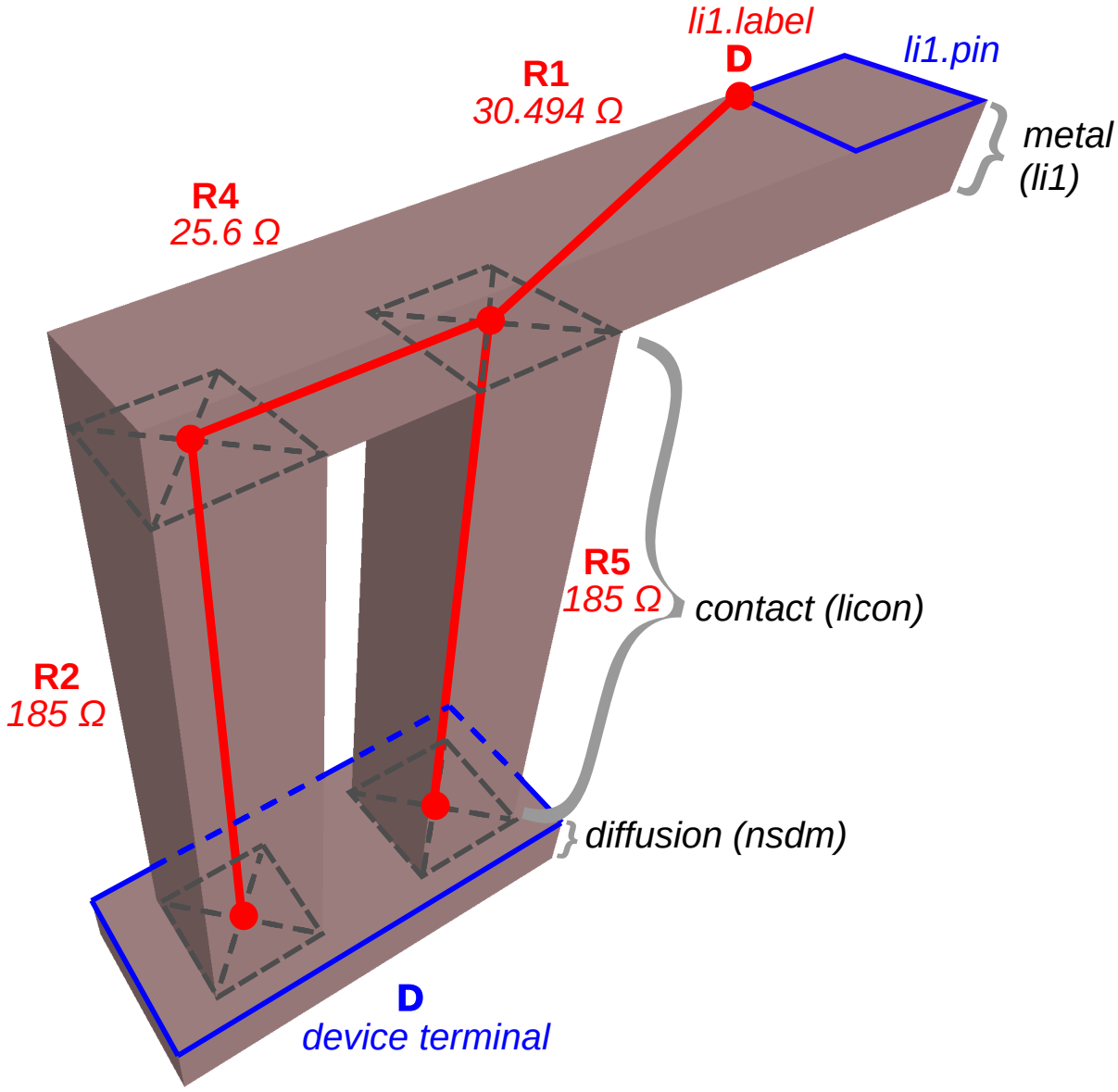


Figure 38: Resistance extraction of nfet example (only net D)"

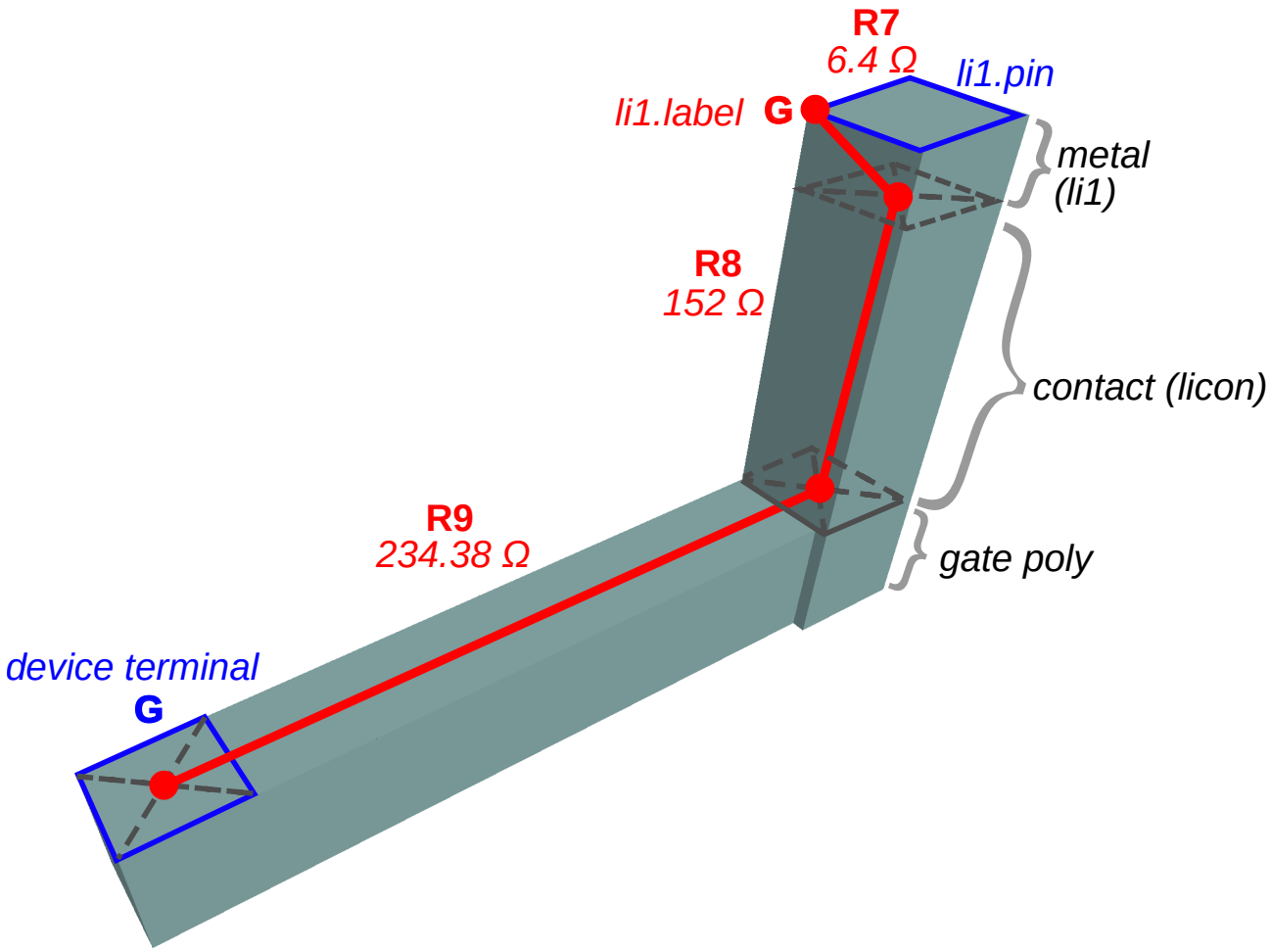


Figure 39: Resistance extraction of nfet example (only net G)"

8 Netlist reduction: Feasibility study

8.1 TICER algorithm: TIme-Constant Equilibration Reduction

This section reviews the original TICER paper (Sheehan 1999), and illustrates the main ideas.

Given N-terminal star network:

- node N is the center
- terminals labeled from 0 to $N-1$ (in Figure 40, m is used instead of $N-1$)
- each branch, going from the terminal T_i to N, contains a resistor and a capacitor in parallel
- resistor values are given as conductance (easier parallel summation): g_{ij} [℧]
- capacitances values: c_{ij} [F]

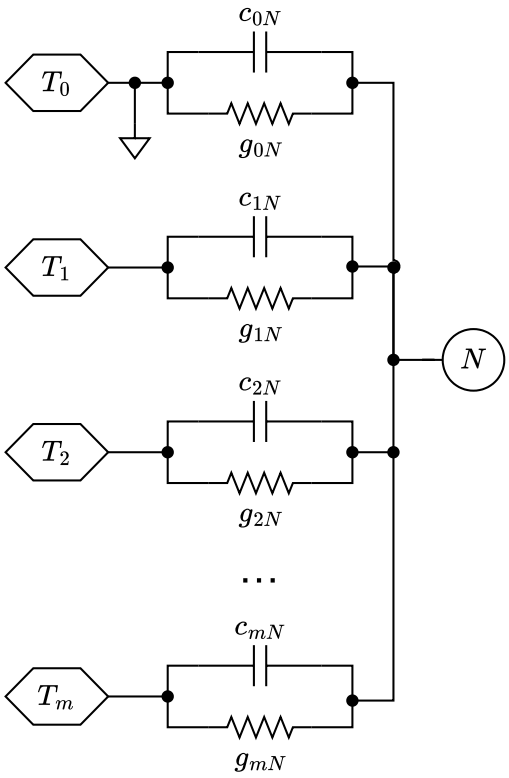


Figure 40: N-terminal star network

Sum of resistances γ_N, and capacitances χ_N:

$$\gamma_N = \sum_{k=0}^{N-1} g_{kN}, \quad \chi_N = \sum_{k=0}^{N-1} c_{kN} \tag{1.2}$$

The time constant τ_N of a given node N then is:

$$\tau_N = \frac{\chi_N}{\gamma_N} = \frac{\sum_{k=0}^{N-1} c_{kN}}{\sum_{k=0}^{N-1} g_{kN}} \tag{1.3}$$

Basic node elimination idea: Given a desired frequency window, using the time-constants, we can classify nodes into 3 groups: slow / normal / quick. Then we can eliminate slow and quick nodes, as the normal nodes will suffice to preserve the behavior for the frequency window:

RC circuit in Laplace domain:

$$(s\mathbf{C} + \mathbf{G})\mathbf{v} = \mathbf{Y}\mathbf{v} = \mathbf{J} \tag{2.1}$$

- $s \in \mathbb{C}$: Complex Frequency
- $\mathbf{C} \in \mathbb{R}^{N-1\times N-1}$: nodal capacitances
- $\mathbf{G} \in \mathbb{R}^{N-1\times N-1}$: nodal conductances
- $\mathbf{v} \in \mathbb{R}^{N-1}$: nodal voltages
- $\mathbf{J} \in \mathbb{R}^{N-1}$: current sources at the nodes

This can be written as a block system:

$$\begin{bmatrix} \tilde{\mathbf{Y}} & \mathbf{y} \\ \mathbf{y}^T & s\chi_N + \gamma_N \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{v}} \\ v_N \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{J}} \\ j_N \end{bmatrix} \tag{2.2}$$

> **Note**
>
> In equation (2.2), Sheehan uses the tilde accent to name the remainder of the matrix $\mathbf{Y}$ as $\tilde{\mathbf{Y}}$, when looking only at the last row and column (same principle applies to vector $\mathbf{v}$ and $\tilde{\mathbf{v}}$):
>
> $$\mathbf{Y} = \left[\begin{array}{cccc:c} y_{11} & y_{12} & \cdots & y_{1,N-1} & y_{1N} \\ y_{21} & y_{22} & \cdots & y_{2,N-1} & y_{2N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \hdashline y_{N1} & y_{N} & \cdots & y_{N,N-1} & y_{NN} \end{array}\right] = \left[\begin{array}{ccccc} & & & & y_{1N} \\ & \tilde{\mathbf{Y}} & & & y_{2N} \\ & & & & \vdots \\ \hdashline y_{N1} & y_{N2} & \cdots & y_{N,N-1} & y_{NN} \end{array}\right] = \left[\begin{array}{c:c} \tilde{\mathbf{Y}} & \mathbf{y} \\ \hdashline \mathbf{y}^T & y_{NN} \end{array}\right] \tag{1}$$

8.1.1 Comments on page 2, equation (2.3)

To arrive at equations (2.3), (2.4), and (2.5) from equation (2.2), the process involves solving for v_N, the voltage at the node to be eliminated (node N), and substituting it back into the system of equations.

Starting with equation (2.2):

$$\begin{bmatrix} \tilde{\mathbf{Y}} & \mathbf{y} \\ \mathbf{y}^T & s\chi_N + \gamma_N \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{v}} \\ v_N \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{J}} \\ j_N \end{bmatrix} \tag{2.2}$$

we have two block equations:

$$\tilde{\mathbf{Y}}\tilde{\mathbf{v}} + \mathbf{y}v_N = \tilde{\mathbf{J}} \qquad \text{(First block equation of 2.2)}$$

$$\mathbf{y}^T\tilde{\mathbf{v}} + (s\chi_N + \gamma_N)v_N = j_N \qquad \text{(Second block equation of 2.2)}$$

Solve the second block equation for v_N:

$$v_N = \frac{j_N - \mathbf{y}^T\tilde{\mathbf{v}}}{s\chi_N + \gamma_N}$$

Substitute the expression for v_N in the first block equation:

$$\tilde{\mathbf{Y}}\tilde{\mathbf{v}} + \mathbf{y}\left(\frac{j_N - \mathbf{y}^T\tilde{\mathbf{v}}}{s\chi_N + \gamma_N}\right) = \tilde{\mathbf{J}}$$

Simplifying, we get:

$$\left(\tilde{\mathbf{Y}} - \frac{\mathbf{y}\mathbf{y}^T}{s\chi_N + \gamma_N}\right)\tilde{\mathbf{v}} = \tilde{\mathbf{J}} - \frac{\mathbf{y}j_N}{s\chi_N + \gamma_N}$$

This leads to the modified system:

$$(\tilde{Y} - \mathbf{E})\tilde{\mathbf{v}} = \tilde{\mathbf{J}} - \mathbf{F} \tag{2.3}$$

Where:

$$\mathbf{E} = \frac{\mathbf{y}\mathbf{y}^T}{s\chi_N + \gamma_N}$$

$$\mathbf{F} = \frac{\mathbf{y}j_N}{s\chi_N + \gamma_N}$$

> **Note**
>
> There's a typo in the original paper in (2.3), the printed variable $\mathbf{v_N}$ should instead be $\tilde{\mathbf{v}}$.

So we arrive at:

$$\mathbf{E}_{ij} = \frac{(g_{iN} + sc_{iN})(g_{jN} + sc_{jN})}{s\chi_N + \gamma_N} \tag{2.4}$$

$$\mathbf{F}_i = \frac{g_{iN} + sc_{iN}}{s\chi_N + \gamma_N} j_N \tag{2.5}$$

8.1.2 Quick Nodes

Suppose:

- we eliminate a quick node N
- $s\chi_N \ll \gamma_N$
- equivalently, $|s\tau_N| \ll 1$
- we approximate $\mathbf{E}_{ij}$ from (2.4)
- we eliminate higher-order terms, containing factors like s^n for $n \geq 2$

$$\begin{aligned}\mathbf{E}_{ij} &= \frac{(g_{iN} + sc_{iN})(g_{jN} + sc_{jN})}{s\chi_N + \gamma_N} \\ &= \frac{g_{iN}g_{jN}}{s\chi_N + \gamma_N} + \frac{s\left(g_{jN}c_{iN} + g_{iN}c_{jN}\right)}{s\chi_N + \gamma_N} + \frac{s^2 c_{iN}c_{jN}}{s\chi_N + \gamma_N}\end{aligned}$$

$$\mathbf{E}_{ij} \approx \frac{g_{iN}g_{jN}}{\cancel{s\chi_N} + \gamma_N} + s\frac{g_{jN}c_{iN} + g_{iN}c_{jN}}{\cancel{s\chi_N} + \gamma_N} + \frac{\cancel{s^2 c_{iN}c_{jN}}}{\cancel{s\chi_N} + \gamma_N}$$

$$\mathbf{E}_{ij} \approx \frac{g_{iN}g_{jN}}{\gamma_N} + s\frac{g_{jN}c_{iN} + g_{iN}c_{jN}}{\gamma_N} \tag{3.2}$$

Eliminating a quick node:

Recipe for translating (3.2) into a modified circuit:

- remove all resistors and capacitors connecting other nodes to N
- insert new resistors and capacitors between former neighbors i, j of N according to these rules:

– If nodes i and j had been connected to N through conductances g_{iN} and g_{jN}, insert conductance $\frac{g_{iN}g_{jN}}{\gamma_N}$ between i and j
– If node i had a capacitor c_{iN} to N, and node j had a conductance g_{jN} to N, insert a capacitor $\frac{c_{iN}g_{jN}}{\gamma_N}$ between i and j

8.1.3 Slow Nodes

Suppose:

- we eliminate a slow node N
- $s\chi_N \gg \gamma_N$
- equivalently, $|s\tau_N| \gg 1$
- we approximate $\mathbf{E}_{ij}$ from (2.4)
- we retain terms containing s
- to preserve DC characteristics, we use $\frac{g_{iN}g_{jN}}{\gamma_N}$ in place of whatever constant terms come from the equation
- higher-order terms in γ_N and s are neglected to simplify the expression.

Starting with (2.4):

$$\mathbf{E}_{ij} = \frac{(g_{iN} + sc_{iN})(g_{jN} + sc_{jN})}{s\chi_N + \gamma_N} \tag{2.4}$$

Given the approximation:

$$\frac{1}{s\chi_N + \gamma_N} \approx \frac{1}{s\chi_N}\left(1 - \frac{\gamma_N}{s\chi_N}\right) \tag{4.2}$$

Substitute (4.2) into (2.4):

$$\begin{aligned}\mathbf{E}_{ij} &\approx \frac{(g_{iN} + sc_{iN})(g_{jN} + sc_{jN})}{s\chi_N}\left(1 - \frac{\gamma_N}{s\chi_N}\right) \\ &\approx \left(\frac{g_{iN}g_{jN}}{s\chi_N} + \frac{\cancel{s}\left(g_{jN}c_{iN} + g_{iN}c_{jN}\right)}{\cancel{s}\chi_N} + \frac{\cancel{s^2}c_{iN}c_{jN}}{\cancel{s}\chi_N}\right)\left(1 - \frac{\gamma_N}{s\chi_N}\right)\end{aligned}$$

Expand:

$$\begin{aligned}\mathbf{E}_{ij} = &\frac{g_{iN}g_{jN}}{s\chi_N} - \frac{g_{iN}g_{jN}\gamma_N}{s^2\chi_N^2} \\ &+ \frac{g_{jN}c_{iN} + g_{iN}c_{jN}}{\chi_N} - \frac{\left(g_{jN}c_{iN} + g_{iN}c_{jN}\right)\gamma_N}{s\chi_N^2} \\ &+ \frac{sc_{iN}c_{jN}}{\chi_N} - \frac{\cancel{s}c_{iN}c_{jN}\gamma_N}{\cancel{s}\chi_N^2}\end{aligned}$$

After we remove constant terms:

$$\mathbf{E}_{ij} \approx \frac{g_{iN}g_{jN}}{s\chi_N} - \frac{g_{iN}g_{jN}\gamma_N}{s^2\chi_N^2} - \frac{\left(g_{jN}c_{iN} + g_{iN}c_{jN}\right)\gamma_N}{s\chi_N^2} + s\frac{c_{iN}c_{jN}}{\chi_N}$$

Remove higher-order terms in γ_N and s:

$$\mathbf{E}_{ij} \approx \frac{g_{iN} g_{jN}}{s\chi_N} + s\frac{c_{iN} c_{jN}}{\chi_N}$$

Sheehan arrives at (4.1):

$$\mathbf{E}_{ij} \approx \frac{g_{iN} g_{jN}}{\gamma_N} + s\frac{c_{iN} c_{jN}}{\chi_N} \tag{4.1}$$

- Rules:
  - If nodes i and j had been connected to N through conductances g_{iN} and g_{jN}, insert conductance $\frac{g_{iN} g_{jN}}{\gamma_N}$ from i to j
  - If node i had a capacitor c_{iN} to N, and node j had a capacitor c_{jN} to N, insert a capacitor $\frac{c_{iN} c_{jN}}{\gamma_N}$ from i to j

8.2 TICER 2007 paper: Pseudocode

This section reviews some aspects of the 2007 TICER paper (Sheehan 2007).

User-supplied algorithm parameters:

- Set of **fixed** nodes (designated to be left alone, i.e., not to be eliminated)
- **MaxDeg** $\in \mathbb{N} \setminus \{0, 1\}$: number of passes (user-supplied parameter)
- $f^{\max}$: maximum operating frequency of interest
- $\epsilon \in \mathbb{R}_{[0,1]}$: small number enforcing the requirement that $|s\tau_N| \ll 1$

In the paper, figure 5 and 6 demonstrate the algorithm for eliminating a quick node (here is a python conversion of those).

```
def eliminate_quick_node(N: Node):
   neighbors = nodes_incident_to(node=N)
   for i in neighbors:
      g_iN = conductance[i, N]
      if g_iN == 0:
         continue
      for j in neighbors:
         if i <= j:
            continue
         g_jN = conductance[j, N]
         c_jN = capacitance[j, N]
         gamma_N = incident_conductance_sum(node=N)
         if g_jN > 0:
            add_conductance(i, j, g_iN * g_jN / gamma_N)
         if c_jN > 0:
            add_capacitance(i, j, g_iN * c_jN / gamma_N)
   remove_all_incident_conductances_and_capacitances(i, N)
```

```
def ticer(freq_max: float, epsilon: float, max_deg: int):
   for deg in range(2, max_deg):
      Q = queue.Queue(nodes_not_fixed())
      while not Q.empty():
         N = Q.get()  # pop
         if number_of_incident_resistors(N) > deg:
            continue
```

```
            tau_N = time_constant(N)
            if 2 * math.pi * freq_max * tau_N <= epsilon:
               # ensure the neighbors are in the queue,
               # because the neighbors' incident resistors and time constants
               # might have been changed, so they should be reconsidered for elimination
               missing_neighbors = [n in nodes_incident_to(node=N) if n not in Q]
               q.push(missing_neighbors)

               eliminate_quick_node(N)

   # NOTE: "leaf node" N means, number_of_incident_resistors(N) == 1
   for N in leaf_nodes():
      if N in fixed_nodes():  # N is protected
         continue

      tau_N = time_constant(N)
      if 2 * math.pi * freq_max * tau_N <= epsilon:
         eliminate_quick_node(N)
```

8.2.1 Pseudocode for slow node elimination

Sheehan mentions that elimination of slow nodes is of reduced importances, and therefore does not explicitly provide pseudocode for this case, instead refers to the original TICER paper (Sheehan 1999).

So if we fill in the gaps we arrive at:

```
def eliminate_slow_node(N: Node):
   neighbors = nodes_incident_to(node=N)
   for i in neighbors:
      g_iN = conductance[i, N]
      c_iN = capacitance[i, N]
      for j in neighbors:
         if i <= j:
            continue
         g_jN = conductance[j, N]
         c_jN = capacitance[j, N]
         gamma_N = incident_conductance_sum(node=N)
         if g_iN > 0 and g_jN > 0:
            add_conductance(i, j, g_iN * g_jN / gamma_N)
         if c_iN > 0 and c_jN > 0:
            add_capacitance(i, j, c_iN * c_jN / gamma_N)
   remove_all_incident_conductances_and_capacitances(i, N)
```

To integrate this into the algorithm, we additionally need a minimum frequency $f^{\min}$ parameter supplied by the user.

```
def ticer(freq_min: float, freq_max: float, epsilon: float, max_deg: int):
   for deg in range(2, max_deg):
      Q = queue.Queue(nodes_not_fixed())
      while not Q.empty():
         N = Q.get()  # pop
```

```
        if number_of_incident_resistors(N) > deg:
            continue
        tau_N = time_constant(N)
        if 2 * math.pi * freq_max * tau_N <= epsilon:
            # NOTE: ensure the neighbors are in the queue,
            # because the neighbors' incident resistors and time constants
            # might have been changed, so they should be reconsidered for elimination
            missing_neighbors = [n in nodes_incident_to(node=N) if n not in Q]
            q.push(missing_neighbors)

            eliminate_quick_node(N)

        if 2 * math.pi * freq_min * tau_N <= epsilon:
            # NOTE: ensure the neighbors are in the queue,
            # because the neighbors' incident resistors and time constants
            # might have been changed, so they should be reconsidered for elimination
            missing_neighbors = [n in nodes_incident_to(node=N) if n not in Q]
            q.push(missing_neighbors)

            eliminate_slow_node(N)

# NOTE: "leaf node" N means, number_of_incident_resistors(N) == 1
for N in leaf_nodes():
    if N in fixed_nodes():  # N is protected
        continue

    tau_N = time_constant(N)
    if 2 * math.pi * freq_max * tau_N <= epsilon:
        eliminate_quick_node(N)
```

8.3 Example from TICER 2007 paper

Let's reduce the example presented by Sheehan in the from the 2007 paper (Sheehan 2007), as shown in Figure 41. The goal is to eliminate the quick node v_3.

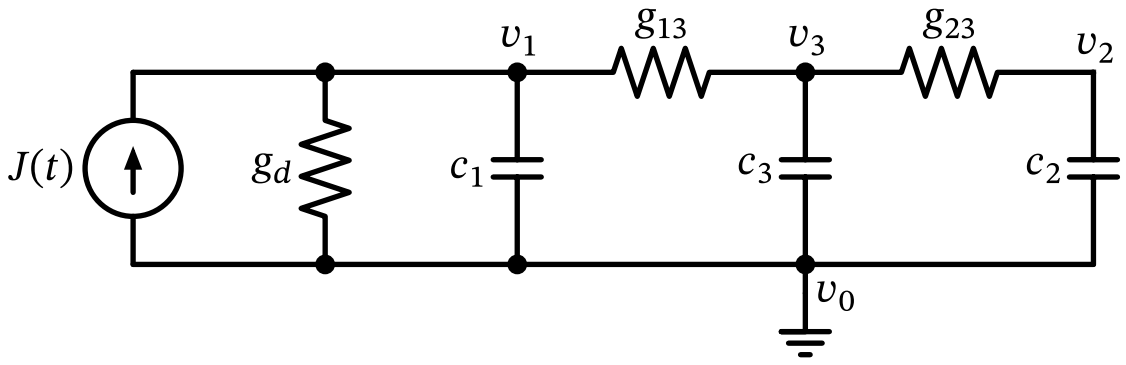


Figure 41: Passive RC circuit shown in Sheehan (2007).

- Time constant $\tau_{v_3} = \frac{c_3}{g_{13}+g_{23}}$
- Quick node condition: is_quick$(N) := 2\pi f^{\max}\tau_N < \epsilon$

Elimination of node N_2:

- Incident devices R_{13}, R_{23}, C_3 are removed
- New devices for neighbors
  - Between nodes (v_0, v_1):
    - add $c_{10} = \frac{g_{13} * c_3}{g_{13}+g_{23}}$
  - Between nodes (v_0, v_2):
    - add $c_{20} = \frac{g_{23} * c_3}{g_{13}+g_{23}}$
  - Between nodes (v_1, v_2):
    - add $g_{12} = \frac{g_{13} * g_{23}}{g_{13}+g_{23}}$

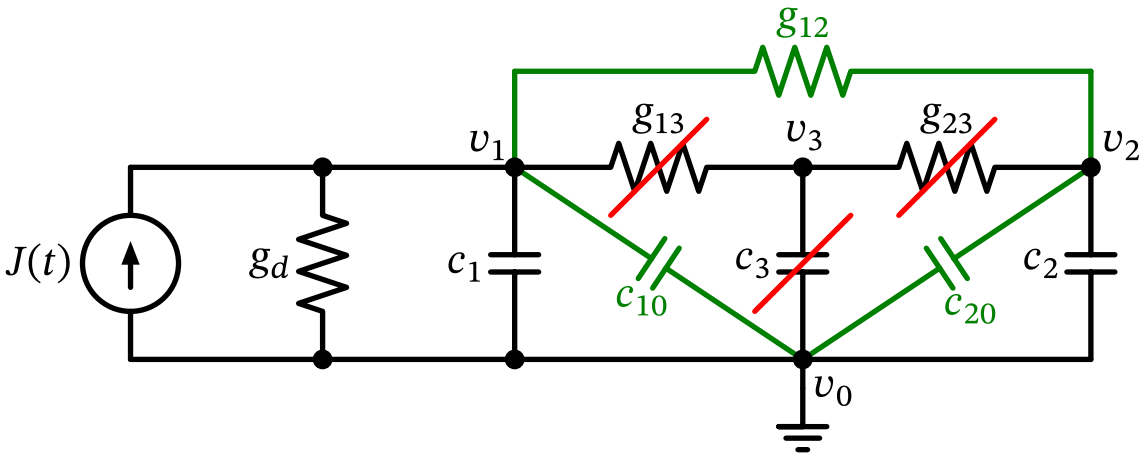


Figure 42: Passive RC circuit after the elimination of v_3.

8.3.1 Modified Nodal Analysis in the Laplace domain

RC circuit in Laplace domain:

$$(s\mathbf{C} + \mathbf{G})\mathbf{v} = \mathbf{Y}\mathbf{v} = \mathbf{J} \tag{2.1}$$

Solve for voltages:

$$\mathbf{v} = (s\mathbf{C} + \mathbf{G})^{-1}\mathbf{J} = Y^{-1}\mathbf{J}$$

Transfer function:

$$H(s) = \frac{\mathbf{v}(s)}{\mathbf{J}(s)} = Y^{-1}(s) = (s\mathbf{G} + \mathbf{C})^{-1}$$

E.g. the transfer function (for input $J_1(s)$, output $V_2(s)$) is:

$$H(s) = \frac{\text{adj}(s\mathbf{C} + \mathbf{G})_{21}}{det(s\mathbf{C} + \mathbf{G})}$$

8.3.2 Analysis of the original circuit

So for the original circuit of Figure 41, we have 3 nodes, in addition to GND (v_0).

So we have 3 nodal equations:

$$\begin{aligned} c_1\dot{v_1} + g_d v_1 + g_{13}(v_1 - v_3) &= j_1(t) \\ c_2\dot{v_2} + g_{23}(v_2 - v_3) &= 0 \\ c_3\dot{v_3} + g_{13}(v_3 - v_1) + g_{23} * (v_3 - v_2) &= 0 \end{aligned}$$

Laplace domain (MNA[9] form):

$$\left(s \begin{bmatrix} c_1 & 0 & 0 \\ 0 & c_2 & 0 \\ 0 & 0 & c_3 \end{bmatrix} + \begin{bmatrix} g_d + g_{13} & 0 & -g_{13} \\ 0 & g_{23} & -g_{23} \\ -g_{13} & -g_{23} & g_{13} + g_{23} \end{bmatrix} \right) \begin{bmatrix} V_1(s) \\ V_2(s) \\ V_3(s) \end{bmatrix} = \begin{bmatrix} J_1(s) \\ 0 \\ 0 \end{bmatrix}$$

So given the values from Sheehan's numerical example:

- $g_d = g_{13} = g_{23} = 1$
- $c_1 = c_2 = 1$
- $c_3 = 0.01$

Node v_3 is a quick node:

- Time constant: $\tau_{v_3} = \frac{c_3}{g_{13}+g_{23}} = \frac{0.001}{1+1} = 0.0005$
- Frequency: $f = \frac{1}{2\pi\tau_{v_3}} \approx 318.31\,\text{Hz}$

We have a system with

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.01 \end{bmatrix}, \mathbf{G} = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Transfer function:

$$H(s) = \frac{1}{0.01s^3 + 2.03s^2 + 4.02s + 1}$$

8.3.3 Analysis of the reduced circuit

The reduced circuit in Figure 42 only has 2 remaining nodes (besides GND).

So we have 2 nodal equations:

$$\begin{aligned} (c_1 + c_{10})\dot{v_1} + g_d v_1 + g_{12}(v_1 - v_2) &= j_1(t) \\ (c_2 + c_{20})\dot{v_2} + g_{12}(v_2 - v_1) &= 0 \end{aligned}$$

Laplace domain (MNA[10] form):

$$\left(s \begin{bmatrix} c_1 + c_{10} & 0 \\ 0 & c_2 + c_{20} \end{bmatrix} + \begin{bmatrix} g_d + g_{12} & -g_{12} \\ -g_{12} & g_{12} \end{bmatrix} \right) \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = \begin{bmatrix} J_1(s) \\ 0 \end{bmatrix}$$

New devices:

[9] Modified Nodal Analysis
[10] Modified Nodal Analysis

- $c_{10} = \frac{g_{13} * c_3}{g_{13}+g_{23}} = \frac{1*0.01}{1+1} = 0.005$
- $c_{20} = \frac{g_{23} * c_3}{g_{13}+g_{23}} = \frac{1*0.01}{1+1} = 0.005$
- $g_{12} = \frac{g_{13} * g_{23}}{g_{13}+g_{23}} = \frac{1*1}{1+1} = 0.5$

We have a system with:

$$\mathbf{C} = \begin{bmatrix} 1.005 & 0 \\ 0 & 1.005 \end{bmatrix}, \mathbf{G} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Transfer function:

$$H(s) = \frac{0.5}{1.010025s^2 + 2.01s + 0.5}$$

8.3.4 Comparison of original and reduced circuits

In the Bode plots, **?@fig-bode-plot-ft-vs-mag** and **?@fig-bode-plot-ft-vs-phase**, we see that magnitude and phase responses stays the same up to a certain frequency.

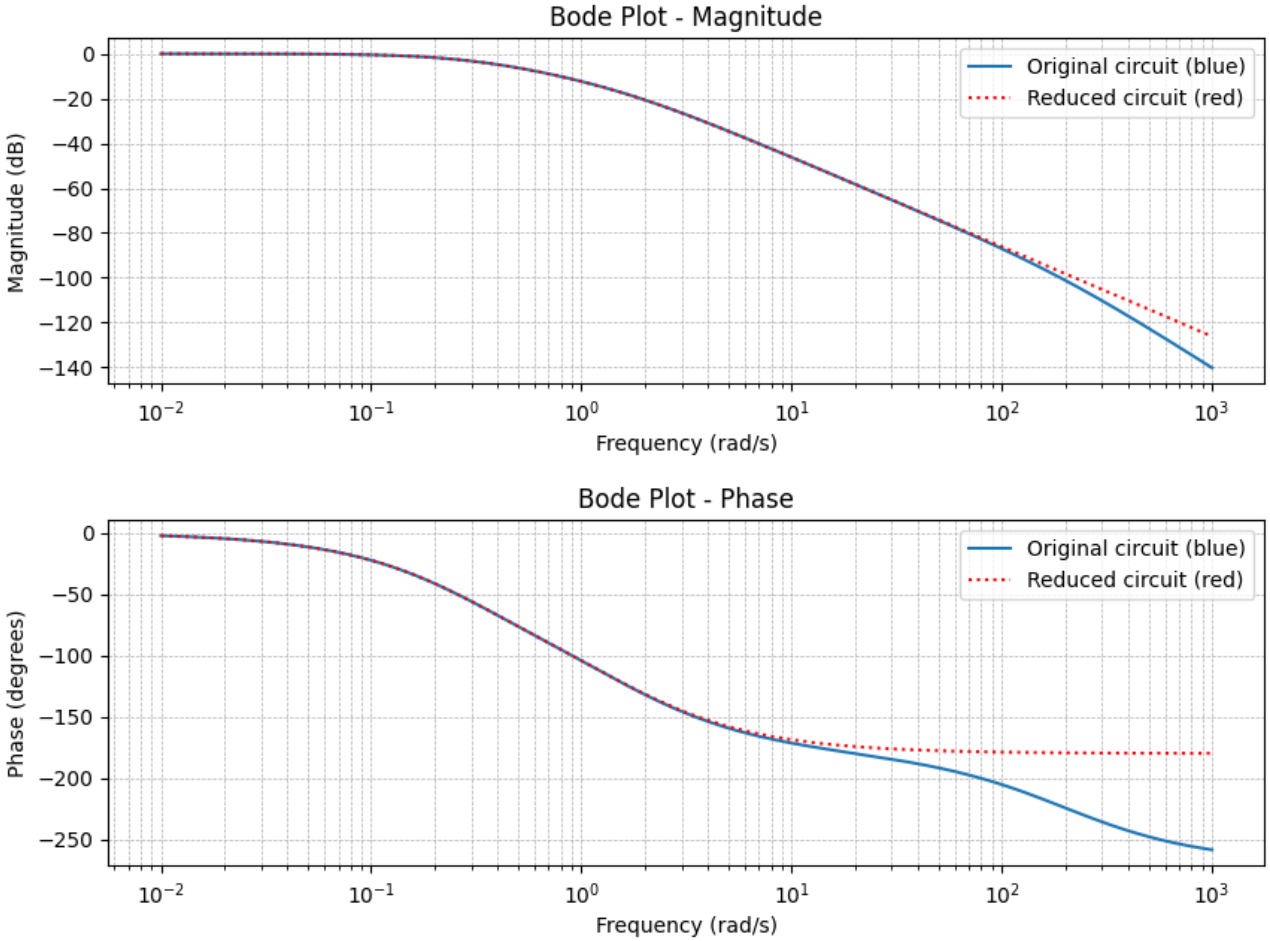
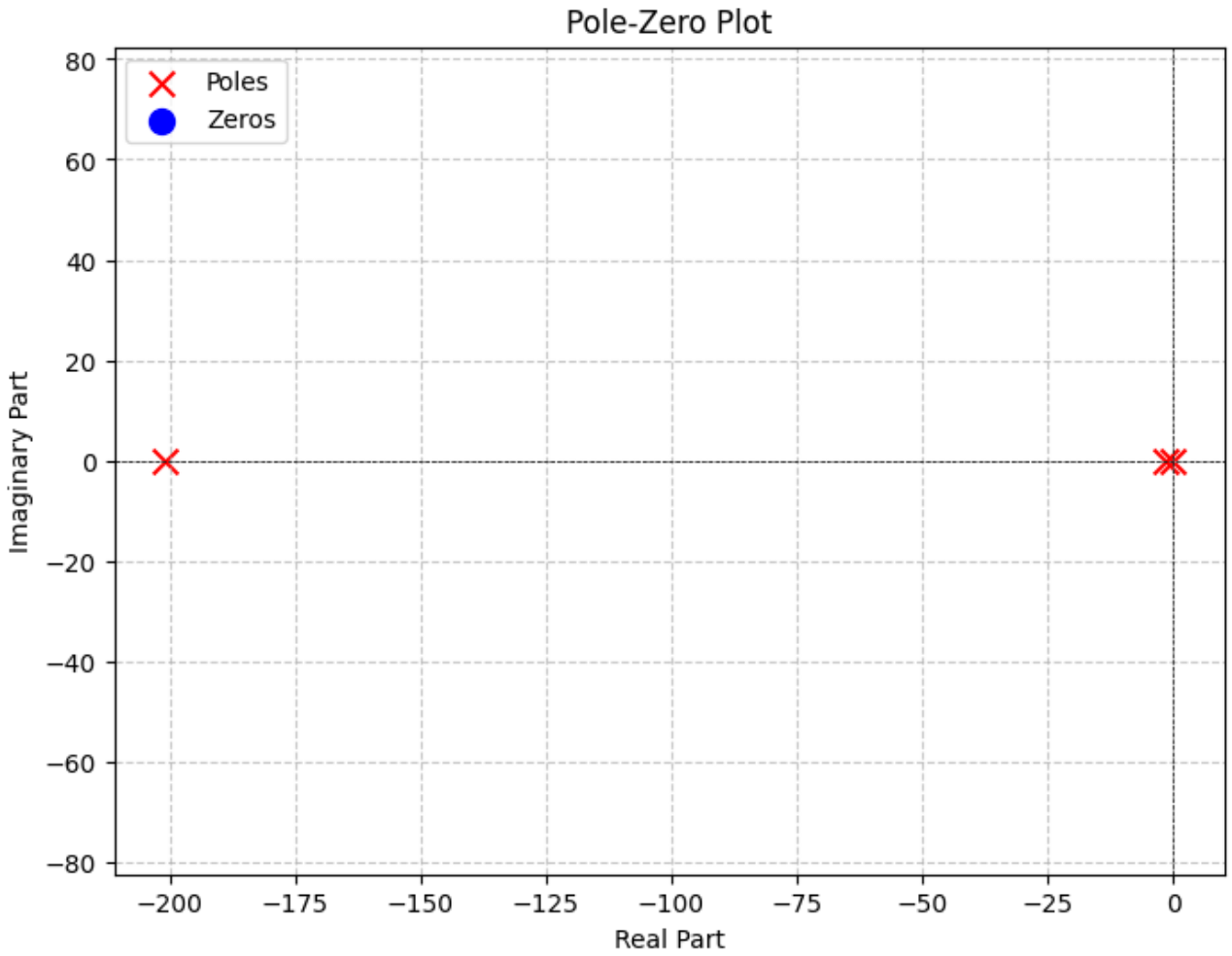
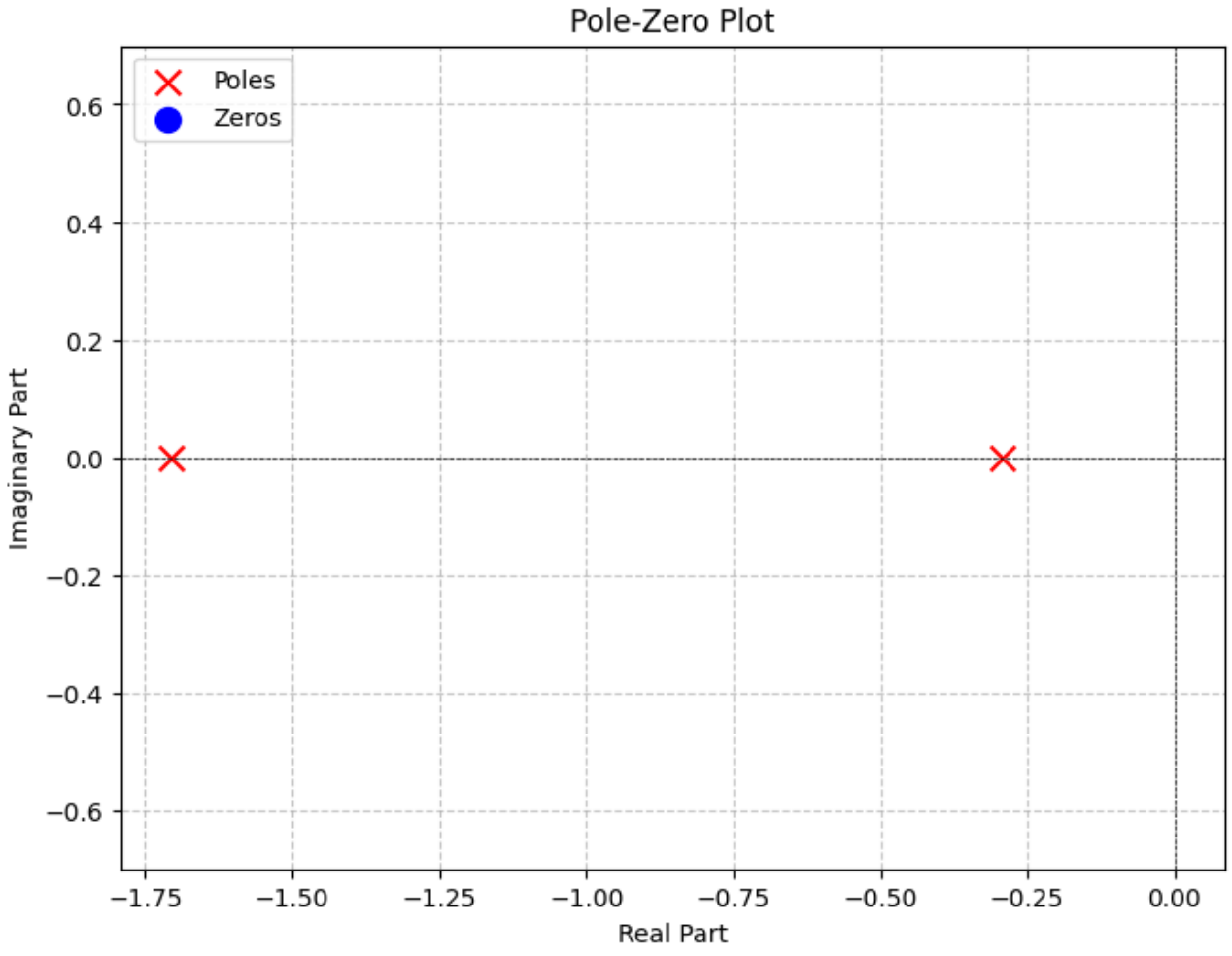


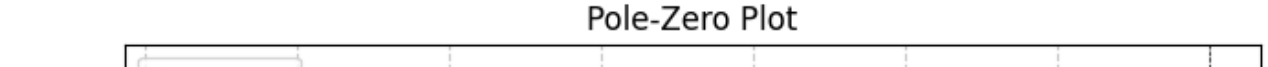
Figure 43: Bode Plots

In the original circuit we have an additional pole at $s = 201$, which is expendable (**?@fig-pole-zero-plot-original**).

In **?@fig-pole-zero-plot-original-without-201** we remove this pole, for better comparison with the reduced circuit (**?@fig-pole-zero-plot-reduced**), where that pole is no longer present, but the others still are.


Pole-Zero Plot
Poles
Zeros
Imaginary Part
Real Part
80
60
40
20
0
−20
−40
−60
−80
−200
−175
−150
−125
−100
−75
−50
−25
0


Pole-Zero Plot
Poles
Zeros
Imaginary Part
Real Part
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−1.75
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00


Pole-Zero Plot

9 Appendix

9.1 PEX25D File Format: Represent Interconnect Geometry and Materials

The **PEX25D format** (Parasitic Extraction 2.5D) describes interconnect geometry and materials for exchange between tools. It combines two-dimensional (XY) conductor shapes with a vertical process stack, so a solver adapter can reconstruct the three-dimensional structure without the original layout or PDK files. The same structure can describe an electrostatic capacitance problem and, with resistance parameters and terminals, a DC resistance problem.

This appendix specifies the PEX25D format and explains its protobuf API and command-line tools. The reference implementation writes `PEX25D 1.0-rc1`. The format version is independent of the KLayout-PEX package version.

> **i** The PEX25D format and the KPEX/2.5D engine
>
> The **PEX25D format** defines an interchange description. `kpex pex25d` generates files in this format; `pex25d` inspects and processes them.
> **KPEX/2.5D** is the analytical extraction engine described in Section 6, selected with `kpex extract --2.5D`. It uses calibrated technology coefficients and shielding rules. Generating or resolving PEX25D files does not run that engine, and the format does not contain its calibrated capacitance tables.

9.1.1 Why use PEX25D?

- **Exchange a physical problem between tools.** A layout producer and a solver adapter can agree on conductor identities, geometry, materials, and terminals through one documented interface. The adapter does not need to repeat LVS or understand the producer's PDK representation.
- **Compare different solvers on the same structure.** Export one PEX25D description through different solver adapters to compare extracted results using the same conductor geometry, material properties, and net identities. This helps distinguish solver differences from differences in input preparation. For a meaningful comparison, also account for each adapter's meshing, boundary treatment, and supported format features.
- **Inspect and reproduce solver input.** Save a readable text file for review, a compact protobuf scene for repeated runs, or a small hand-written example for a bug report. Validation can run before meshing or starting an expensive field solve.
- **Change process corners consistently.** Only the ground plane and metal profiles declare absolute heights. Via endpoints, dielectric bands, and film growth refer to those profiles, so a change to a metal height propagates through the stack. `INCLUDE` can separate a technology stack from reusable geometry.
- **Share one resolver across consumers.** The protobuf API separates what the author wrote from the resolved scene. Adapters consume the latter and share the interpretation of `CONNECTS`, `BETWEEN`, `WRAPS`, and terminal intersections.
- **Pass prepared input directly to C++ engines.** Python can prepare and resolve the structure, then forward a binary `PEX25DScene` to a C++ consumer using generated protobuf bindings. The C++ engine needs neither a PEX25D text parser nor a second implementation of stack resolution. This supports keeping preparation in Python while moving computation-intensive work to C++.
- **Preserve exact dimensions.** Coordinates are integer grid units internally. Coincident vertical faces remain equal after relative heights are computed, without accumulating floating-point discrepancies.

The PEX25D format describes the physical interconnect structure. An analytical consumer still needs its own technology data, potentially matched through `META technology` and the layer names; this does not imply that the current KPEX/2.5D engine reads PEX25D files.

The following are outside the scope of version 1.x:

- Extracted results and fitted parasitic models, including calibrated capacitance tables.
- Inductance.
- Frequency-dependent effects, such as skin effect or dispersive permittivity.
- Self-heating.
- Electromigration.
- Substrate resistance; the ground plane is an ideal 0 V reference.

9.1.2 From layout to solver input

The workflow has five distinct operations:

Operation	Tool or API	Result
Generate from layout connectivity and PDK stack	`kpex pex25d`	`PEX25DFile`, optionally also `PEX25DScene`
Translate another tool's scene description	Importer; `pex25d import`	`PEX25DFile`
Read or re-encode a description	Text reader; `pex25d convert`	The same artifact kind in another encoding
Resolve relative geometry	Resolver; `pex25d resolve`	`PEX25DScene`
Mesh and export for a solver	Exporter; `pex25d export`	Native solver input files

Importers and exporters are plugins: `klayout-pex` ships some of them, further ones are installed as separate Python packages.

The intended project architecture keeps layout and technology preparation in **Python** and moves computation-intensive portions into **C++** engines. The resolved binary scene is the interface between those stages: Python produces it, and C++ reads the same protobuf message through generated bindings. This is the direction for moving computation to C++; it does not mean all current engines have already been ported.

`PEX25DFile` is the literal, unresolved model: it retains profile references and relative descriptions. The text reader flattens `INCLUDE` files and groups records by type. Comments, whitespace, and original record order are not preserved, so conversion is not a byte-for-byte text round trip.

`PEX25DScene` carries absolute Z extents, dielectric roots and wrap depths, conductor geometry grouped by layer, and terminal intersections. When a domain was requested, it contains the resulting box. Without a domain record, the resolver retains finite geometry bounds and leaves boundary selection to the adapter. The scene still describes 2.5D geometry; it is not a triangulated solver mesh.

Resolution is one-way. A scene has no representation in the PEX25D text syntax, and `convert` cannot reconstruct the original file from it. Keep the file when you want to edit relative stack definitions later.

9.1.3 Artifact filenames and encodings

The filename identifies both the artifact kind (unresolved file or resolved scene) and its encoding. The tools infer these from the following suffixes:

Conventional filename	Protobuf message	Encoding
`name.pex25d`	–	PEX25D textual format
`name.pex25d.pb`	`PEX25DFile`	Binary protobuf
`name.pex25d.textpb`	`PEX25DFile`	Protobuf text
`name.pex25d.scene.pb`	`PEX25DScene`	Binary protobuf
`name.pex25d.scene.textpb`	`PEX25DScene`	Protobuf text

Reading the PEX25D textual format produces a `PEX25DFile` message in memory.

A trailing `.gz` enables gzip for any of these filenames. Protobuf text (`textpb`) uses protobuf field names and braces; it is a different syntax from the PEX25D textual format.

9.1.4 A complete example

Download two_nets.pex25d to try the commands below without a layout or an LVS run. This small, illustrative stack is not calibrated PDK data. Conductor `A` crosses from `met1` to `met2` through one explicit via cut and has two terminals. Conductor `B` is a nearby metal polygon.

```
# Illustrative stack; these values do not represent a calibrated PDK.
PEX25D 1.0-rc1
UNITS LENGTH um GRID 0.0001
META technology example
META source_cell two_nets
META source_dbu 0.001

GROUND_PLANE subs Z_OFFSETS -0.4 -0.1
METAL met1 Z_OFFSETS 1.0 1.4
METAL met2 Z_OFFSETS 2.0 2.4
VIA via1 CONNECTS met1 met2

DIELECTRIC_BACKGROUND air PERMITTIVITY 1.0
DIELECTRIC_SIMPLE fox WRAPS subs PERMITTIVITY 3.9 \
    BETWEEN subs met1
DIELECTRIC_CONFORMAL lint WRAPS met1 PERMITTIVITY 7.3 \
    THICKNESS_OVER_WRAPPED 0.1 \
    THICKNESS_BESIDE_WRAPPED 0.08 \
    THICKNESS_ON_FIELD 0.0
DIELECTRIC_SIMPLE ild WRAPS lint PERMITTIVITY 4.1 \
    BETWEEN met1 met2

RESISTANCE TEMPERATURE 25.0
RESISTANCE METAL met1 SHEET 0.125
RESISTANCE METAL met2 SHEET 0.125
RESISTANCE VIA via1 PER_CUT 4.5

CONDUCTOR A neta
CONDUCTOR B netb
BOX CONDUCTOR A LAYER met1 LL 0.0 0.0 UR 1.0 0.5
BOX CONDUCTOR A LAYER via1 LL 0.2 0.1 UR 0.4 0.3
BOX CONDUCTOR A LAYER met2 LL 0.0 0.0 UR 1.0 0.5
POLYGON CONDUCTOR B LAYER met1 \
    OUTER 2.0 0.0 3.0 0.0 3.0 1.0 2.0 1.0
TERMINAL input CONDUCTOR A LAYER met1 KIND PIN \
    LL 0.0 0.0 UR 0.1 0.5
TERMINAL output CONDUCTOR A LAYER met2 KIND PIN \
    LL 0.9 0.0 UR 1.0 0.5

DOMAIN_MARGIN X 4.0 Y 4.0 Z 2.0

```

The via spans **1.4–2.0 µm**: the top of `met1` to the bottom of `met2`. The `lint` film reaches 0.1 µm above `met1` and 0.08 µm beside it. `ild` fills the band from the **bottom** of `met1` at 1.0 µm to the bottom of `met2` at 2.0 µm. Where both dielectrics are present, `lint` wins because it has wrap depth 1 and `ild` has depth 2.

For capacitance, `A` is equipotential despite its two terminals. For resistance, those terminals define the two nodes between which its distributed interconnect is reduced. `B` has no terminals, so it participates in capacitance but contributes no resistance network.

9.1.5 PEX25D textual format reference

In the syntax templates below, angle brackets identify values to supply; square brackets mark optional clauses. Neither is written literally into a file.

9.1.5.1 Records, versions, and grid units

Files are UTF-8 and line-oriented. Keywords are case-sensitive. Whitespace separates tokens, `#` starts a comment outside a quoted value, and a trailing backslash continues a record on the next line. Quote metadata values or paths containing whitespace. Clauses **within** a record must follow the documented order.

```
PEX25D <major>.<minor>[-<suffix>]
UNITS LENGTH <um|nm|m> GRID <grid>
```

`PEX25D` must be the first record of the top-level file. `UNITS` must precede numeric records, including numbers brought in by `INCLUDE`. After these, records can appear in any order and may reference profiles declared later.

A reader accepts minor versions with the same supported major version. It skips an unknown top-level record with a warning; it rejects an unsupported major version. An unknown clause in a known record is an error. A clause keyword such as `WRAPS` or `THICKNESS_ON_FIELD` at the beginning of a record is also an error, since it may indicate a missing continuation backslash. An optional version suffix consists of letters, digits, dots, or hyphens and does not affect compatibility; writers should preserve it.

Coordinates, Z offsets, thicknesses, and margins use the declared length unit and must be multiples of `GRID`. Decimal and scientific notation are accepted, including `.5`, `5.`, and `1e-3`. There are no per-value unit overrides or thousands separators. To check a length, divide by `GRID`, round to the nearest integer, and require the quotient to be within 10^{-6} of that integer. A value outside that tolerance is an error, not permission to snap the input geometry.

For `UNITS LENGTH um GRID 0.0001`, a coordinate of `0.3262` is stored as **3262 grid units**. Protobuf represents the grid itself as an exact rational. It stores coordinates in signed 64-bit integers, not floating-point micrometres. Derived vertical positions therefore remain exact. Lateral offsets of non-Manhattan edges can fall off-grid; the specification requires rounding those *derived offsets* to the nearest grid point.

Permittivity and resistance parameters are material values and are **not** grid-quantized. The PEX25D grid and the source layout database unit (DBU) are distinct: `GRID 0.0001` and a layout DBU of `0.001` µm differ by a factor of ten.

9.1.5.2 Metadata and includes

```
META <key> <value>
INCLUDE <path>
```

Metadata records describe provenance and do not change geometry or material interpretation. A key may occur only once across the entire include tree. Unknown metadata keys are silently accepted. Defined keys are `technology`, `process_corner`, `source_cell`, `source_layout_hash` (for example, `sha256:<digest>`), `source_dbu`, and `generator`. Prefix private keys with a vendor tag such as `x-kpex-`. If `source_dbu` is supplied, the grid must divide it exactly.

An included file contains records without its own `PEX25D` or `UNITS` header. Relative paths are resolved against the including file's directory; nested includes are allowed, cycles are errors. For input from stdin, relative includes resolve against the current working directory.

For example, keep the header in `design.pex25d`, put profiles and material values in `stack-typical.inc`, and put conductor shapes in `geometry.inc`:

```
PEX25D 1.0-rc1
UNITS LENGTH um GRID 0.0001
META technology example
META process_corner typical
INCLUDE stack-typical.inc
INCLUDE geometry.inc
```

Both included files use the top-level length unit and grid. Changing the stack include selects another physical corner; it does not supply a new calibration table to an analytical engine.

9.1.5.3 Ground plane, metals, and vias

```
GROUND_PLANE <name> Z_OFFSETS <zlow> <zhigh>
METAL <name> Z_OFFSETS <zlow> <zhigh>
VIA <name> CONNECTS <below> <above>
```

There must be exactly one ground plane. It occupies an infinite XY slab at 0 V. Ground-plane and metal Z ranges are absolute, with `zlow < zhigh`. Metals become solids only where conductor shapes are drawn.

A via fills the vertical gap between its ordered endpoints:

$$z_{\text{low,via}} = z_{\text{high,below}}, \qquad z_{\text{high,via}} = z_{\text{low,above}}.$$

Endpoints may name profiles with Z extents, including the ground plane for substrate or well taps. Contacts landing on different layers need distinct via profiles; contacts with equal heights but different resistances may also need distinct profiles.

Ground-plane, metal, via, dielectric, and background names share a unique **profile namespace**. Conductor shortnames have a separate namespace, and terminal names another. Shape and terminal `LAYER` references must select a metal or via, never the ground plane or a dielectric.

A metal profile without shapes is legal. Such an **anchor** can locate the bottom of a contact at an excluded device layer without introducing that layer's solid. It has no resistance requirement and does not enlarge the computational domain or overlap the ground plane merely because its Z range lies inside it.

9.1.5.4 Dielectrics and material precedence

```
DIELECTRIC_SIMPLE <name> WRAPS <profile> PERMITTIVITY <k> \
    BETWEEN <below> <above>
DIELECTRIC_CONFORMAL <name> WRAPS <profile> PERMITTIVITY <k> \
    THICKNESS_OVER_WRAPPED <top> \
    THICKNESS_BESIDE_WRAPPED <side> \
    THICKNESS_ON_FIELD <field>
DIELECTRIC_BACKGROUND <name> PERMITTIVITY <k>
```

`PERMITTIVITY` is relative, isotropic, and frequency-independent; vacuum has a value of 1.0. Exactly one background is required. It fills whatever no conductor, ground plane, or dielectric occupies.

`DIELECTRIC_SIMPLE` occupies a band across the whole XY plane, from the **bottom face of `below` to the bottom face of `above`**. This differs from `VIA CONNECTS`, which starts at the lower profile's *top*. `BETWEEN` determines the band's geometry; `WRAPS` determines its precedence. A simple fill must wrap the outermost film anchored on its lower object, or the lower object itself if it has no film.

`DIELECTRIC_CONFORMAL` grows from the surface of the profile it wraps:

Thickness	Meaning
`THICKNESS_OVER_WRAPPED`	Growth above its top face; zero is flush with that face
`THICKNESS_BESIDE_WRAPPED`	Growth beside its side faces
`THICKNESS_ON_FIELD`	Film where the wrapped object is absent, measured upward from its bottom face; zero leaves no field film

Thicknesses are incremental along a `WRAPS` chain. If one film extends 0.006 µm beside a metal and the next extends 0.0431 µm beside that film, the outer extent is 0.0491 µm beyond the metal. A sidewall film is expressed by a conformal record with zero field thickness; there is no separate `DIELECTRIC_SIDEWALL` record.

Profiles describe grown solids containing the wrapped object, rather than pre-subtracted shells. Material occupancy determines the visible material:

1. Conductors and the ground plane take precedence over dielectrics.
2. Among dielectrics claiming a point, the smallest `WRAPS` depth wins. A film on a metal, via, or ground plane has depth 1; wrapping another dielectric adds one.
3. The background fills unclaimed space.

`WRAPS` references must be acyclic. Overlapping dielectrics at equal depth are ambiguous and invalid; changing record order cannot fix them. Re-anchor the fill or film to express the intended precedence. These rules make stack order in the file a matter of readability rather than material interpretation.

9.1.5.5 Conductors and XY shapes

```
CONDUCTOR <shortname> <net>
BOX CONDUCTOR <shortname> LAYER <profile> LL <x> <y> UR <x> <y>
POLYGON CONDUCTOR <shortname> LAYER <profile> \
    OUTER <x1> <y1> <x2> <y2> ... [HOLE <x1> <y1> <x2> <y2> ...]
```

Each shape is extruded over its layer's Z extent. A conductor's shapes are unioned and may be disconnected. Different conductor bodies must not overlap in volume. The meaning of a shared net depends on the analysis:

Analysis	Meaning of conductors sharing a net
Capacitance	All are constrained to one potential; results are combined per net
Resistance	Each conductor forms a separate network between its own terminals; external connections belong to the enclosing flow

The exact, case-sensitive net name `FLOATING` is reserved for unconnected shielding bodies such as dummy fill. Each floating conductor has its own unknown potential, is reported under its shortname, and must not have terminals. Floating bodies are not shorted together just because they all use that reserved net name.

Boxes use lower-left and upper-right corners. Polygon rings are implicitly closed, so do not repeat the first vertex. An outer ring may be concave; outer and hole rings must be simple. Holes must lie strictly inside their outer ring and may not touch or intersect another ring. Winding direction is immaterial. Use separate polygon records for disconnected regions or an island inside a hole.

Every via cut is an explicit shape. There is no via-array record, and readers do not infer cuts from spacing or enclosure rules. This preserves dielectric gaps between cuts and makes `PER_CUT` resistance unambiguous.

9.1.5.6 Resistance parameters

```
RESISTANCE TEMPERATURE <celsius>
RESISTANCE METAL <profile> SHEET <ohm_per_square> [TC1 <value>] [TC2 <value>]
RESISTANCE VIA <profile> PER_CUT <ohm> [TC1 <value>] [TC2 <value>]
```

Resistance records are optional. There may be at most one reference temperature, one sheet resistance per metal profile, and one per-cut resistance per via profile. `SHEET` is in ohms per square, and `PER_CUT` in ohms per cut, independently of the length unit. KPEX technology data stores resistance in milliohms; a producer converting that data divides by 1000.

Missing resistance means **unspecified**, not zero. A resistance consumer must reject a profile carrying geometry if it has no resistance value. Unused anchor profiles are exempt. A capacitance consumer can ignore resistance parameters.

Sheet resistance is a calibrated electrical property. Do not reconstruct it from the drawn metal thickness or invert it to infer bulk resistivity. Two geometrically equal contacts can likewise have different per-cut resistances, for example when they land on different diffusion types.

The optional temperature coefficients have units of K^{-1} and K^{-2}:

$$R(T) = R_{\mathrm{ref}} \left[1 + TC1\,(T - T_{\mathrm{ref}}) + TC2\,(T - T_{\mathrm{ref}})^2\right].$$

The reader and resolver carry these values through; the consumer applies them at its operating temperature. An absent reference temperature does not imply 25 °C. The enclosing flow must supply it if needed.

9.1.5.7 Terminals and compact-device boundaries

```
TERMINAL <name> CONDUCTOR <shortname> LAYER <profile> [KIND <kind>] \
    LL <x> <y> UR <x> <y>
```

A terminal is the intersection of its rectangular region with the selected conductor's geometry on one metal or via profile. It must select a non-empty region. Several disjoint pieces may form **one electrical node**, but a terminal may not partially select a via cut: each selected cut must be covered completely.

`KIND PIN` denotes a port or probe point. `KIND DEVICE_TERMINAL` marks the boundary to a compact device model. Omitting `KIND` leaves the role unspecified. Names are unique among terminals and become network node names.

A metal terminal makes its selected region equipotential over the layer's thickness; a 3D resistance adapter may choose a drive face. A via terminal is a node at the **lower face** of the selected cuts. It does not short through the via, so the contact resistance above that node remains part of the extracted network. A conductor with fewer than two terminals contributes no resistance elements.

> **!** Terminal size changes the resistance problem
>
> Everything selected by a terminal is shorted together. A marker covering an entire wire or gate can therefore remove the resistance you intend to extract. Size the region to the pin or device boundary it represents, rather than copying a large enclosing shape.

Geometry owned by a compact model is excluded from the interconnect description to avoid counting its parasitics twice. The specification places the device/interconnect boundary below the contact: diffusion and the device's internal spreading resistance belong to the device, while the contact and routing above it belong to interconnect extraction. An anchor profile can locate that contact without drawing diffusion geometry.

The lower face selected by `DEVICE_TERMINAL` is an **open boundary of the extracted region**, for both capacitance and resistance. This prevents a device contact coincident with the ground-plane surface from becoming a short to 0 V or adding device junction capacitance. Only the marked portion is opened. A substrate tap without a device terminal retains ordinary electrical contact with the ground plane. These are specification requirements for consumers; the current FasterCap adapter does not yet implement device-terminal open faces.

9.1.5.7.1 KLayout-generated terminals

Terminal generation is under development. The current development implementation derives `PIN` records from labels inside layout pin markers. Each pin region extends one layout DBU from the label in every direction, preserving wire resistance even when the pin marker itself covers a long wire.

Device terminals use LVS terminal geometry on existing conductive profiles. Several disjoint pieces can share one node when their bounding box selects exactly those pieces; an abutting boundary selects a strip one DBU wide inside the interconnect. Generation rejects a box that would select additional interconnect or a node that spans multiple profiles. Names use `pin:<label>` and `device:<instance>:<terminal>`, with percent encoding for syntax-sensitive characters and numeric suffixes for uniqueness.

This work also adds exact clipping of Manhattan polygons, including holes and disconnected intersections. Partial clipping of non-Manhattan polygons remains unsupported. Earlier resolver implementations support clipped boxes and whole polygons only. **Diffusion/contact terminal generation is still pending**; the presence of pin and gate terminals must not be taken as a complete device boundary model.

9.1.5.8 Computational domain

```
DOMAIN_MARGIN X <xmargin> Y <ymargin> Z <zmargin>
DOMAIN_BOX LL <x> <y> <z> UR <x> <y> <z>
```

These optional records help adapters select a finite computational region; they do not prescribe boundary conditions. `DOMAIN_MARGIN` expands the finite geometry bounds in both X and Y directions and only upward in Z. The ground-plane top forms the lower boundary. An adapter may increase the requested clearance but must not reduce it.

Finite geometry includes conductor shapes and laterally bounded conformal films, including their lateral growth. The ground plane, background, simple dielectric bands, field-covering films, and unused anchor profiles do not enlarge those bounds.

`DOMAIN_BOX` specifies the box directly and takes precedence over `DOMAIN_MARGIN` if both occur, with a warning. Finite non-ground-plane geometry must fit inside; touching a box face is allowed. The lower Z face normally coincides with the ground-plane top. Without either record, domain choice belongs to the adapter. The current FasterCap/FastCap2 exporter uses the resolved box's XY extent to bound laterally infinite materials; it does not turn the box into a prescribed six-face boundary condition.

9.1.6 Command-line tools

The `klayout-pex` package supplies `kpex` and the separate `pex25d` entry point. Check command availability with `--help`; older installed releases may not have these commands.

```
kpex pex25d --help
pex25d --help
pex25d validate --help
pex25d export --help
```

Reading, validating, converting, resolving, and summarizing an existing artifact do not require a layout, LVS, or a running solver, and their implementation does not import KLayout. Generating from a layout and exporting through the current meshing backend do require KLayout. This is a runtime distinction; the distributed `klayout-pex` package still declares KLayout as a dependency.

9.1.6.1 Generate from a layout: `kpex pex25d`

This subcommand runs layout connectivity extraction through LVS, combines it with the selected PDK's stack and resistance data, and writes artifacts in the PEX25D format before solver execution. The layout example in Section 2.2 shows both outputs. At least one output must be requested, and the two paths must differ.

Option	Use
`--output_file PATH`	Write an unresolved file; `.pex25d`, `.pex25d.pb`, or `.pex25d.textpb` selects its encoding
`--output_scene PATH`	Write a resolved scene; use `.pex25d.scene.pb` or `.pex25d.scene.textpb`
`--format FORMAT`	Select `auto`, `text`, `pb`, or `textpb`; one setting applies to both requested outputs
`--grid UM`	Grid in µm; default `0.0001`, must represent stack dimensions and divide the layout DBU
`--domain_margin UM`	Emit equal X, Y, and upper-Z margins; default is no domain record

Option	Use
`--validate BOOL`	Run reference validation; `y` (default) or `n`
`--comments BOOL`	Add syntax hints to text output; `y` or `n` (default)
`--with_source_refs BOOL`	Retain per-record provenance in protobuf output; `y` or `n` (default)

Layout, PDK, top-cell, blackboxing, and material selection options are listed by `kpex pex25d --help`. The artifact's kind is selected by the output option, so requesting `--format text` together with `--output_scene` is invalid. Filenames ending in `.gz` produce compressed output.

`--strict`, `--werror`, `--diagnostics`, and `--diagnostics_out` are also available, with the meanings described in Section 9.1.6.2.

Generation can leave artifacts on disk even when validation reports errors. Always inspect the exit status and diagnostics before using them as solver input. In the current Sky130 stack, the `nild6`/`capild` anchoring can trigger `PEX25D-E0260`. For an isolated terminal smoke test, `--diel all,-nild6` avoids that conflict, but removes a dielectric from the physical problem and is not a substitute for correcting the stack for capacitance extraction.

9.1.6.2 Validate and inspect

Run these commands in the directory containing the downloaded example:

```
pex25d validate two_nets.pex25d
pex25d validate two_nets.pex25d --strict
pex25d show two_nets.pex25d
pex25d show two_nets.pex25d --section layers --section terminals
```

Normal validation checks syntax, units, references, wrap anchoring and depth conflicts, resistance records, and terminal selection. `--strict` adds geometric checks for boxes and rings, hole containment, and conductor volume overlap. `show` summarizes the artifact as supplied; resolve it first when you want to inspect derived layer heights.

`--section` is repeatable and accepts `header`, `meta`, `layers`, `dielectrics`, `conductors`, `terminals`, `resistance`, `domain`, and `all`. Human-readable summaries and diagnostic wording are not stable APIs.

For automated checks of a custom PEX25D writer, save a structured report and use the exit status:

```
pex25d validate two_nets.pex25d --strict --werror \
  --diagnostics json --diagnostics_out validation.json
pex25d validate two_nets.pex25d --strict \
  --diagnostics pb --diagnostics_out validation.pb
```

JSON and binary protobuf reports carry stable `PEX25D-Ennnn` codes, severity, validation tier, and source positions where available. Use an explicit report path to keep the report separate from console logging. `--werror` makes warnings fail the run; without it, warnings alone do not fail validation.

Exit status	Meaning
0	Success
1	Validation or resolution errors, or warnings with `--werror`

Exit status	Meaning
`2`	Invalid command line, I/O failure, missing dependency, or export failure
`3`	An operation reports that it is not implemented
`4`	An importer or exporter plugin failed unexpectedly (traceback with `--log_level debug`)

The validator is a reference check, not a guarantee that every adapter implements every specification rule. For example, the current depth-tie check detects competing profiles on the same wrap root; it is not a complete geometric intersection check for films on different roots. Terminal-boundary support must also be checked at the adapter level.

9.1.6.3 Convert encodings and resolve a scene

```
pex25d convert two_nets.pex25d -o two_nets.pex25d.pb
pex25d convert two_nets.pex25d.pb -o two_nets.pex25d.textpb
pex25d convert two_nets.pex25d.pb -o two_nets-copy.pex25d --comments
pex25d convert two_nets.pex25d -o two_nets.pex25d.pb.gz

pex25d resolve two_nets.pex25d -o two_nets.pex25d.scene.pb --strict
pex25d convert two_nets.pex25d.scene.pb -o two_nets.pex25d.scene.textpb
pex25d validate two_nets.pex25d.scene.pb --strict
pex25d show two_nets.pex25d.scene.pb --section layers --section domain
```

`convert` changes encoding within one kind; it does not validate the complete physical problem. `resolve` computes the scene and accepts `--strict`, `--werror`, and the diagnostics options. Validate before passing a resolved scene to an exporter as well: the current export path resolves file input, but does not automatically run the reference validator on an already-resolved scene.

Use `--with_source_refs` on text conversion or resolution when you want the protobuf artifact to retain locations in an include tree. Diagnostics produced while reading text have source positions even without that flag. `--comments` adds syntax hints when writing PEX25D text; unlike the `kpex` option, it is a flag and takes no `y` argument.

9.1.6.4 Import from another tool

```
pex25d importers
pex25d import design.uf --from openrcx-uf -o design.pex25d
```

An importer translates another tool's scene description into an unresolved `PEX25DFile`. `pex25d import` validates the result and writes it even when it is invalid, so that it can be reviewed; the exit status tells whether it is valid. `--supporting NAME=PATH` passes further inputs the importer asks for, e.g. a process stack, and `--option NAME=VALUE` passes importer options. The built-in `openrcx-uf` importer (OpenRCX Universal Pattern Format) is a stub for now and exits with status `3`.

9.1.6.5 Export native solver input

```
pex25d exporters
pex25d export two_nets.pex25d.scene.pb \
  --to fastercap --out_dir solver/fastercap --prefix two_nets
pex25d export two_nets.pex25d \
  --to fastcap2 --out_dir solver/fastcap2 --prefix two_nets
pex25d export two_nets.pex25d --to stl --out_dir solver/stl
```

`--to` selects an exporter by its name, or by `package:name` when two installed packages provide the same name. `pex25d exporters` lists the installed ones. `klayout-pex` ships three exporters:

- `fastercap` and `fastcap2` write a `.lst` entry file and its referenced surface files; keep them together.
- `stl` writes the solids of the scene as STL files, one per dielectric and net, for viewing (e.g. in `MeshLab`).

File input is resolved automatically. Export generates meshes and input files, **without running the solver or producing a capacitance matrix**. The solver executable is needed when those files are subsequently solved. `kpex extract` is the separate integrated flow for running engines from a layout.

Exporter options are passed with `--option NAME=VALUE`, which can be repeated (since `klayout-pex` 0.6.3). VALUE is read as JSON where it parses (`10`, `1e-9`, `true`), otherwise as text. For FasterCap and FastCap2, `--stl`, `--field_margin`, `--delaunay_amax`, `--delaunay_b`, and `--geo_check` are shorthands for their options. The default lateral field margin is 8 µm when the scene has no explicit or margin-derived domain.

The built-in exporters write capacitance input. They do not extract a resistance network from `SHEET`, `PER_CUT`, or terminal records. Exporters for the FEM solvers Elmer and Palace are available as separate packages, klayout-pex-plugin-elmer and klayout-pex-plugin-palace.

9.1.6.6 Pipelines and explicit encodings

A path of `-` means stdin for input and stdout for output. Artifact output reserves stdout and redirects logging and diagnostics to stderr. Use explicit kinds and encodings in a binary pipeline:

```
pex25d resolve two_nets.pex25d -o - --out_kind scene --out_format pb \
  | pex25d show - --in_kind scene --in_format pb --section layers
```

For filenames without conventional suffixes, supply **both** `--in_kind file|scene` and `--in_format text|pb|textpb`, or the corresponding `--out_kind` and `--out_format`. These are options of the standalone tool; `kpex pex25d` uses its output options and `--format` instead. Stdin defaults to unresolved text, while `resolve` defaults to binary scene output on stdout. Gzip is inferred from filenames; use a shell gzip tool when compressing a stream.

9.1.7 Protobuf API: Python Preparation and C++ Consumers

The PEX25D schemas define the file and scene messages, exact units, geometry, dielectric and resistance records, terminals, domain, diagnostics, and source references. Generated bindings let another language read binary artifacts without implementing the text parser. An adapter should consume `PEX25DScene` and use the shared resolver for file input.

One import is enough: `from klayout_pex import pex25d` provides the verbs `read`, `write`, `resolve`, `validate`, `import_file`, `export` and `show`, the artifact and diagnostics types, every error the pipeline raises, and `pex25d.proto` for the generated messages (`pex25d.proto.PEX25DFile()`, or a module such as `pex25d.proto.dielectric` for its enums). Names listed in `pex25d.__all__` are the supported surface, while the submodules stay importable for the classes behind the verbs, e.g. `resolver.Resolver` and `validator.Validator`.

`read()` and `write()` take a path in any supported encoding and derive kind and encoding from the conventional suffixes (Section 9.1.3); `read_text()` and `write_text()` work on the bytes of the text format. `validate()` fills a diagnostics report and never raises for an invalid message; `resolve()` computes the scene and raises `ResolveError` when resolution cannot proceed. The package depends on `protobuf` alone, so it can be installed as a reference reader and validator without `klayout`.

This Python example loads and resolves the downloadable file, then requests native FasterCap input and STL solids through the exporter API:

```python
from klayout_pex import pex25d

report = pex25d.DiagnosticsReport(warnings_are_errors=True)
description = pex25d.read("two_nets.pex25d", report=report, with_source_refs=True)
scene = pex25d.resolve(description, report=report)
pex25d.validate(description, report=report, strict=True, scene=scene)
if report.exit_code:
    raise RuntimeError("PEX25D validation failed")

pex25d.write(scene, "two_nets.pex25d.scene.pb")
paths = pex25d.export(
    scene,
    target="fastercap",
    output_dir_path="solver/api",
    prefix="two_nets",
    options={"write_stl": True},
)
print(paths[0])
```

A C++ consumer can read the uncompressed binary scene written above using the bindings generated from `pex25d_scene.proto` and its imports. For example, this complete program loads a scene and reports its size; the extraction engine would then operate on that message:

```cpp
#include <fstream>
#include <iostream>
#include "kpex/pex25d/pex25d_scene.pb.h"

int main(int argc, char **argv) {
    if (argc != 2) {
        std::cerr << "Usage: read_scene FILE.pex25d.scene.pb\n";
        return 2;
    }
    std::ifstream input(argv[1], std::ios::binary);
    kpex::pex25d::PEX25DScene scene;
    if (!input || !scene.ParseFromIstream(&input)) {
        std::cerr << "Cannot read PEX25DScene\n";
        return 1;
    }
    std::cout << scene.layers_size() << " layers, "
              << scene.conductors_size() << " conductors\n";
    return 0;
}
```

Compile consumers with the generated `.pb.cc` files and the protobuf C++ runtime. A successful protobuf parse establishes that the binary message can be decoded; reference validation still belongs in the preparation flow, and an engine must check the format version and the features it supports. Keep the `.scene` kind explicit when exchanging files, since both unresolved files and resolved scenes can use binary protobuf encoding.

When consuming the scene directly, convert grid coordinates using `grid_numerator / grid_denominator` in the declared length unit. Check `scene.domain.origin` before using `scene.domain.box`: the domain message may exist only to carry geometry bounds. Preserve conductor net identities, floating-body independence, resistance units, and terminal boundary semantics when mapping the scene to a solver.

9.2 Plugin API

KPEX can be extended by plugins, which are separate Python packages. Once a plugin is installed (e.g. via pip3), KPEX finds it automatically, there is no need to change or re-install KPEX itself.

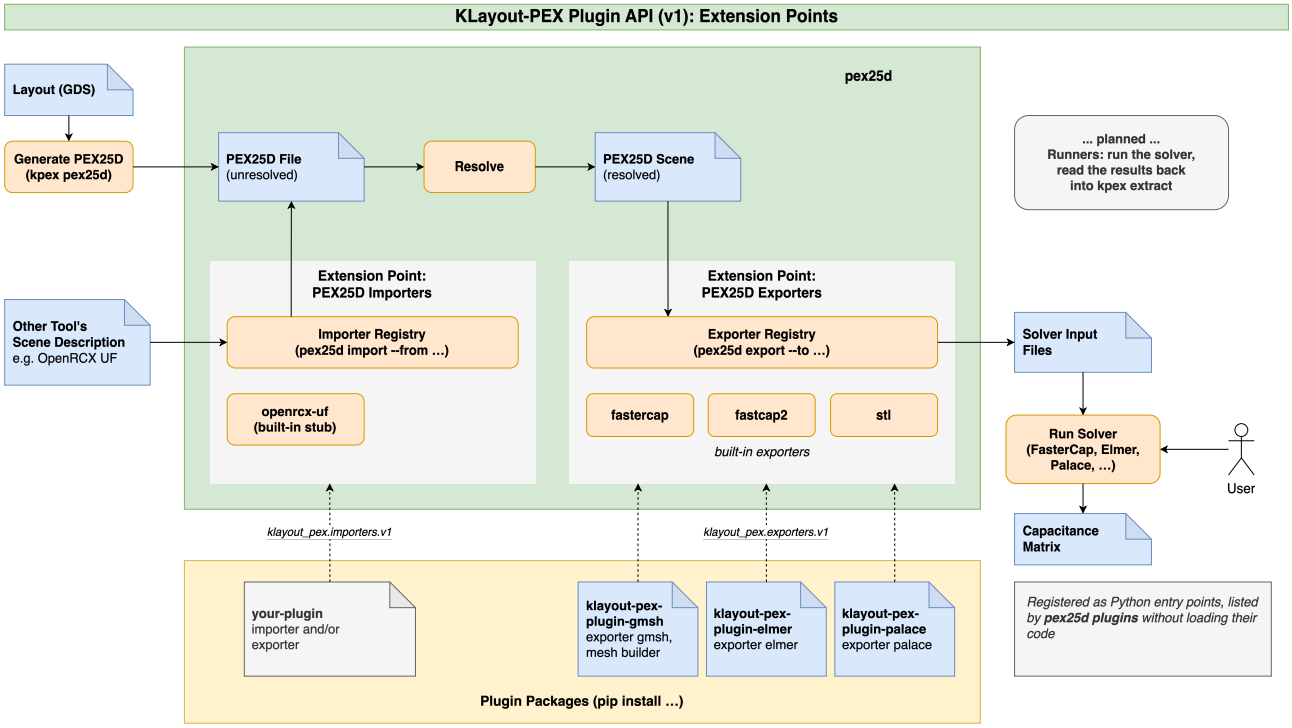


Figure 45: KPEX Plugin API: Extension Points

9.2.1 PEX25D Importers and Exporters

Currently, there are two extension points, both built around the PEX25D format (see Section 9.1):

Extension Point	PEX25D Importer	PEX25D Exporter
Description	Translate the scene description of another tool into a `PEX25DFile`	Write the input files of a solver from a resolved `PEX25DScene`
Input	Files of another tool, e.g. OpenRCX Universal Pattern Format	`PEX25DScene` protobuf
Entry Point Group	`klayout_pex.importers.v1`	`klayout_pex.exporters.v1`
Protocol	`PEX25DImporter`	`PEX25DSceneExporter`
CLI Invocation	`pex25d import --from NAME`	`pex25d export --to NAME`
Built-In Plugins	`openrcx-uf` (stub)	`fastercap`, `fastcap2`, `stl`
Maintained External Plugins	none yet	`elmer`, `palace`, `gmsh` (see Section 9.2.4.1)

Running the solver and reading its results back into `kpex extract` is not part of the plugin API yet, this is planned for a future version.

> **Caution**
>
> The plugin API `v1` is provisional: it is currently tested together with colleagues, and may still change until it is declared stable. Afterwards, breaking changes will get a new version `v2`, with

their own entry point groups.
Plugins should therefore pin `klayout-pex` to one minor release, e.g. `>=0.6.3,<0.7`.

9.2.2 Using Plugins

```
pip3 install --upgrade klayout-pex-plugin-palace

pex25d plugins     # overview: selector, package, version, description
pex25d exporters   # selectors only, one per line (for scripts)
pex25d importers
```

`pex25d plugins` only reads the package metadata, a plugin itself is loaded when it is used. A plugin that fails to load (e.g. because of a missing dependency) therefore does not affect the other ones.

Selecting a plugin:

- by its name, e.g. `--to palace`
- by `package:name`, if two packages provide the same name, e.g. `--to klayout-pex:stl`
- built-in plugins take precedence for plain names

Plugin specific options are passed with `--option NAME=VALUE` (since `klayout-pex` 0.6.3). The value is read as JSON if possible (`10`, `1e-9`, `true`), otherwise as text:

```
pex25d export cell.pex25d --to palace --out_dir deck \
  --option mesh_size_edge_um=0.2 --option order=2
```

Exit status	Meaning
2	Unknown plugin, invalid option, or the plugin rejected its input
3	The plugin does not implement the requested feature yet
4	Unexpected plugin failure (`--log_level debug` shows the traceback)

9.2.3 Writing a Plugin

A plugin is a Python package, which registers a factory function under one of the entry point groups. The factory is called without arguments and returns an object implementing the protocol. The protocols are defined in `klayout_pex.plugin_api.v1`, deriving from them is optional.

For the plugins maintained by us, we use these conventions:

- GitHub repository and PyPI package: `klayout-pex-plugin-<name>`
- Python package: `klayout_pex_plugins.<name>`
  - `site-packages/klayout_pex_plugins` is a namespace package shared by all plugins
  - it must not contain an `__init__.py`, otherwise all other plugins become invisible

The registration in `pyproject.toml`:

```toml
[project]
name = "klayout-pex-plugin-example"
dependencies = ["klayout-pex>=0.6.3,<0.7"]

[project.entry-points."klayout_pex.exporters.v1"]
nets = "klayout_pex_plugins.example:create_exporter"
```

General rules for plugins:

- import heavy dependencies (e.g. a mesh generator) only when the plugin is used, not when its module is imported
- check the names and values of the options, raise `ExportError` / `ImporterError` for invalid options or unsupported input
- raise `ExporterUnavailable` / `ImporterUnavailable` when a dependency is missing, e.g. native binaries or libraries
- raise `NotImplementedError` for features that are not implemented yet
- do not modify the scene or the options passed in

9.2.4 PEX25D Exporters

An exporter writes the input files of a solver from a resolved `PEX25DScene` (see Section 9.1.2), it does not run the solver:

```python
class PEX25DSceneExporter(Protocol):
    def export(self,
               scene: PEX25DScene,
               *,
               output_dir_path: str,
               prefix: str = '',
               options: Optional[Mapping[str, Any]] = None) -> List[str]:
        ...
```

- `scene`: the resolved scene (`pex25d export` resolves a `PEX25DFile` first)
- `output_dir_path`: the directory for the files, the exporter creates it if necessary
- `prefix`: prefix for the file names, an empty prefix means the exporter's own default
- `options`: exporter specific options, e.g. from `--option`
- returns the paths of the files written, the main input file first

A minimal exporter, which writes the net of each conductor into a text file:

```python
import os
from typing import Any, List, Mapping, Optional

from klayout_pex.plugin_api.v1 import ExportError, PEX25DSceneExporter


class NetsExporter(PEX25DSceneExporter):
    def export(self, scene, *, output_dir_path: str, prefix: str = '',
               options: Optional[Mapping[str, Any]] = None) -> List[str]:
        if options:
            raise ExportError(f"unknown options: {', '.join(options)}")
        os.makedirs(output_dir_path, exist_ok=True)
        path = os.path.join(output_dir_path, f"{prefix}nets.txt")
```

```
        with open(path, 'w') as file:
            for conductor in scene.conductors:
                file.write(f"{conductor.name} {conductor.net}\n")
        return [path]


def create_exporter() -> NetsExporter:
    return NetsExporter()
```

9.2.4.1 Example: FEM Solvers Elmer and Palace

As examples for external exporters, we provide plugins for two open source FEM field solvers, both used for capacitance extraction (electrostatics):

- klayout-pex-plugin-elmer: exporter `elmer` for Elmer FEM (`StatElecSolver`)
- klayout-pex-plugin-palace: exporter `palace` for Palace (`Electrostatic` problem)
- both use the volume mesh of klayout-pex-plugin-gmsh, which is also available as exporter `gmsh`

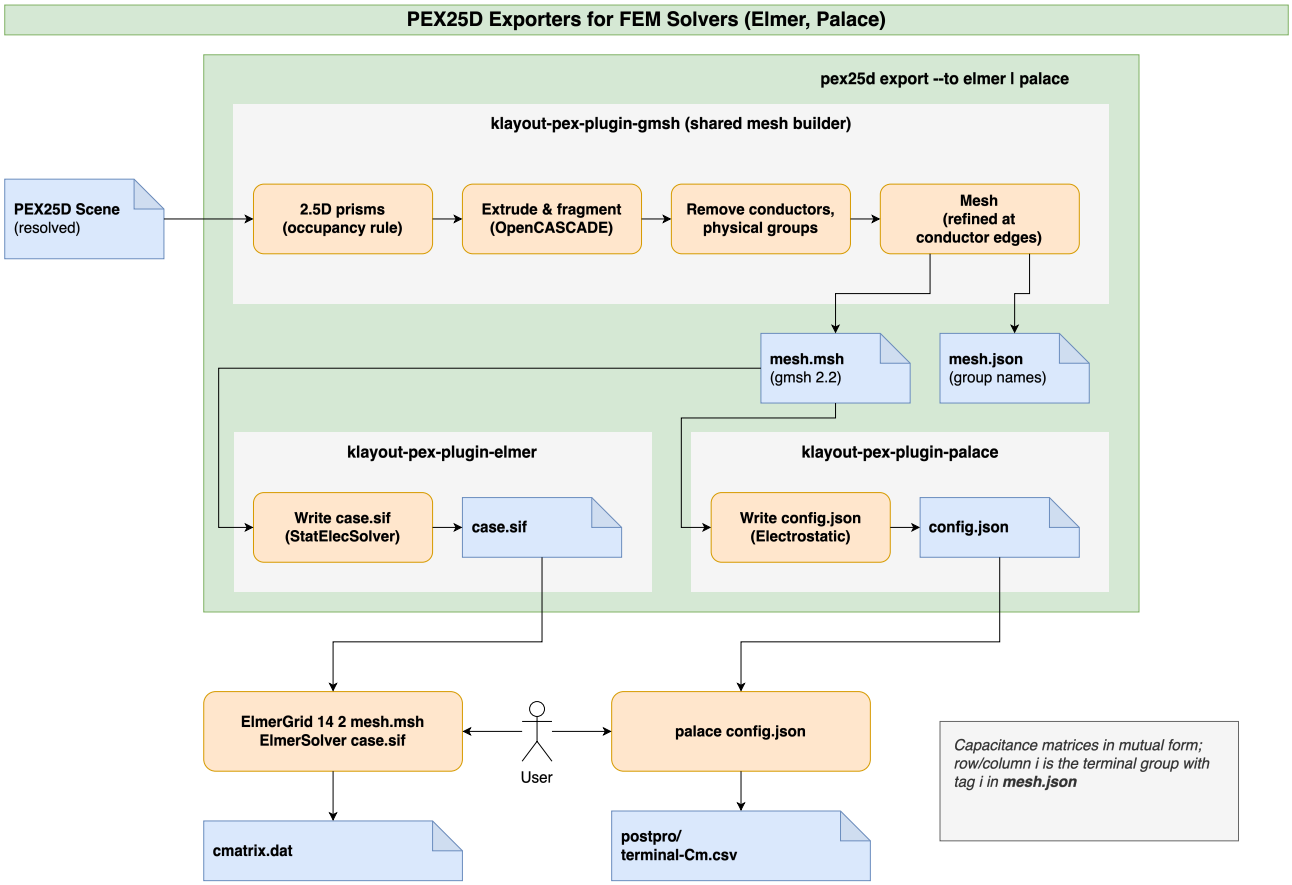


Figure 46: PEX25D Exporters for FEM Solvers

The mesh is built in these steps:

- 2.5D prisms: the domain is sliced at every z where a solid starts or ends
  - within each slab, conductors and the ground plane come first, then the dielectrics by wrap depth (see Section 9.1.5.4), and the background fills the rest
  - so the prisms fill the domain exactly once
- the prisms are extruded and fragmented using `OpenCASCADE`, so touching solids share their faces

- the conductors are removed, only their surfaces remain, as one physical group per net
- the elements are refined at the conductor edges, where the field concentrates
  - by default, down to the thickness of the thinnest conductor layer

Elmer FEM:

```
pip3 install --upgrade klayout-pex-plugin-elmer

pex25d export cell.pex25d --to elmer --out_dir elmer
cd elmer
ElmerGrid 14 2 mesh.msh -autoclean   # gmsh mesh -> Elmer mesh
ElmerSolver case.sif                 # writes cmatrix.dat
```

Palace:

```
pip3 install --upgrade klayout-pex-plugin-palace

pex25d export cell.pex25d --to palace --out_dir palace
cd palace
palace -np 4 config.json             # writes postpro/terminal-Cm.csv
```

Both solvers write the capacitance matrix in mutual form:

- off-diagonal cells contain the coupling capacitance between two terminals
- diagonal cells contain the capacitance to ground (zero, unless the outer boundary is grounded)
- row/column i belongs to the terminal group with tag i in `mesh.json` (nets, floating conductors, ground plane)

The options of the plugins (mesh sizes, element order, outer boundary condition, …) are listed in the `README` files of the repositories.

> **IIC-OSIC-TOOLS**
>
> The `IIC-OSIC-TOOLS` Docker image (see Section 1.5) contains Palace, Elmer FEM is not included yet. When running as `root`, call `palace -serial config.json`, as `mpirun` refuses to run as `root`.

For the test design `overlap_plates_100um_x_100um_m1_m2` (`ihp-sg13g2`, two plates of 100 µm × 100 µm in `Metal1` and `Metal2`, overlapping by 50 µm × 50 µm), the coupling capacitance between the plates is:

Solver	Mesh	$C_{LOWER,UPPER}$
Elmer FEM	default (94k tetrahedra)	181.3 fF
Palace, order 1	same mesh	181.3 fF
Palace, order 2	same mesh	179.4 fF
Elmer FEM	edges refined to 0.21 µm (~180k tetrahedra)	180.7 fF

A parallel plate estimate of the overlap alone gives 168 fF, the difference being the fringe capacitance.

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